



API Reference

Amazon CloudWatch



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Amazon CloudWatch: API Reference

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Welcome

Amazon CloudWatch enables you to publish, monitor, and manage various metrics, as well as configure alarm actions based on data from metrics. This guide provides detailed information about CloudWatch actions, data types, parameters, and errors. For more information about CloudWatch features, see [Amazon CloudWatch](#) and the *Amazon CloudWatch User Guide*.

For information about the metrics that other Amazon Web Services products send to CloudWatch, see the [Amazon CloudWatch Metrics and Dimensions Reference](#) in the *Amazon CloudWatch User Guide*.

Use the following links to get started using the CloudWatch Query API:

[Actions](#): An alphabetical list of all CloudWatch actions.

[Data Types](#): An alphabetical list of all CloudWatch data types.

[Common Parameters](#): Parameters that all Query actions can use.

[Common Error Types](#): Client and server errors that all actions can return.

[Regions and Endpoints](#): Supported regions and endpoints for all Amazon Web Services products.

Alternatively, you can use one of the [Amazon Web Services SDKs](#) to access CloudWatch using an API tailored to your programming language or platform.

Developers in the Amazon Web Services developer community also provide their own libraries, which you can find at the following Amazon Web Services developer centers:

[Java Developer Center](#)

[JavaScript Developer Center](#)

[Amazon Web Services Mobile Services](#)

[PHP Developer Center](#)

[Python Developer Center](#)

[Ruby Developer Center](#)

[Windows and .NET Developer Center](#)

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Actions

The following actions are supported:

- [AssociateDatasetKmsKey](#)
- [DeleteAlarmMuteRule](#)
- [DeleteAlarms](#)
- [DeleteAnomalyDetector](#)
- [DeleteDashboards](#)
- [DeleteInsightRules](#)
- [DeleteMetricStream](#)
- [DescribeAlarmContributors](#)
- [DescribeAlarmHistory](#)
- [DescribeAlarms](#)
- [DescribeAlarmsForMetric](#)
- [DescribeAnomalyDetectors](#)
- [DescribeInsightRules](#)
- [DisableAlarmActions](#)
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- [DisassociateDatasetKmsKey](#)
- [EnableAlarmActions](#)
- [EnableInsightRules](#)
- [GetAlarmMuteRule](#)
- [GetDashboard](#)
- [GetDataset](#)
- [GetInsightRuleReport](#)
- [GetMetricData](#)
- [GetMetricStatistics](#)
- [GetMetricStream](#)
- [GetMetricWidgetImage](#)
- [GetOTelEnrichment](#)

- [ListAlarmMuteRules](#)
- [ListDashboards](#)
- [ListManagedInsightRules](#)
- [ListMetrics](#)
- [ListMetricStreams](#)
- [ListTagsForResource](#)
- [PutAlarmMuteRule](#)
- [PutAnomalyDetector](#)
- [PutCompositeAlarm](#)
- [PutDashboard](#)
- [PutInsightRule](#)
- [PutLogAlarm](#)
- [PutManagedInsightRules](#)
- [PutMetricAlarm](#)
- [PutMetricData](#)
- [PutMetricStream](#)
- [SetAlarmState](#)
- [StartMetricStreams](#)
- [StartOTelEnrichment](#)
- [StopMetricStreams](#)
- [StopOTelEnrichment](#)
- [TagResource](#)
- [UntagResource](#)

AssociateDatasetKmsKey

Associates an AWS Key Management Service (AWS KMS) customer managed key with the specified dataset. After this operation completes, all data published to the dataset is encrypted at rest using the specified KMS key. Callers must have `kms:Decrypt` permission on the key to read the encrypted data.

Only the default dataset is supported. The default dataset is implicit for every account in every Region — you do not need to create it before calling this operation.

You can call `AssociateDatasetKmsKey` on a dataset that is already associated with a KMS key to replace the existing key with a different one. To replace a key, the caller must have `kms:Decrypt` permission on both the current key and the new key.

The KMS key that you specify must meet all of the following requirements:

- It must be a symmetric encryption KMS key (key spec `SYMMETRIC_DEFAULT`, key usage `ENCRYPT_DECRYPT`). Asymmetric keys, HMAC keys, and key material types other than `SYMMETRIC_DEFAULT` are not supported.
- It must be enabled and not pending deletion.
- Its key policy must grant the CloudWatch service principal (`cloudwatch.amazonaws.com`) these permissions: `kms:DescribeKey`, `kms:GenerateDataKey`, `kms:Encrypt`, `kms:Decrypt`, and `kms:ReEncrypt*`. Amazon CloudWatch requires these permissions to manage the data on your behalf.
- The calling principal must have `kms:Decrypt` permission on the key.
- It must be specified as a fully qualified key ARN. Key IDs, aliases, and alias ARNs are not accepted.
- It must be in the same AWS Region as the dataset.

Before completing the association, Amazon CloudWatch validates the key by performing a series of dry-run KMS operations. Service-principal checks run first to verify that the key policy grants the required access to Amazon CloudWatch. These checks include `kms:DescribeKey`, `kms:GenerateDataKey`, `kms:Encrypt`, `kms:Decrypt`, and `kms:ReEncrypt*`. After those succeed, a `kms:Decrypt` dry-run is run with the caller's credentials to verify that the calling principal can use the key. When you are replacing an existing key, the caller's `kms:Decrypt` dry-run is run on the current key first, and only then on the new key.

If any of these checks fails, the operation fails and the existing key association (if any) remains unchanged. Common failure causes include the key being disabled, the key policy not granting the required permissions to Amazon CloudWatch, or the caller lacking `kms:Decrypt` permission on the key.

For more information about using customer managed keys with Amazon CloudWatch, see [Encryption at rest with customer managed keys](#) in the *Amazon CloudWatch User Guide*.

Request Parameters

DatasetIdentifier

Specifies the identifier of the dataset that you want to associate the KMS key with. For the default dataset, you can specify either default or the full dataset Amazon Resource Name (ARN) in the format `arn:aws:cloudwatch:Region:account-id:dataset/default`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 2048.

Pattern: `(default|arn:[a-zA-Z0-9-]+:cloudwatch:[a-zA-Z0-9-]*:\d{12}:dataset/default)`

Required: Yes

KmsKeyArn

Specifies the Amazon Resource Name (ARN) of the customer managed KMS key to associate with the dataset. The key must be a symmetric encryption KMS key (SYMMETRIC_DEFAULT) in the same AWS Region as the dataset.

The ARN must be in the format `arn:aws:kms:Region:account-id:key/key-id`. Key IDs, aliases, and alias ARNs are not accepted.

For more information about KMS key ARNs, see [Key ARN](#) in the *AWS Key Management Service Developer Guide*.

Type: String

Length Constraints: Minimum length of 20. Maximum length of 2048.

Pattern: `arn:[a-zA-Z0-9-]+:kms:[a-zA-Z0-9-]+:\d{12}:key/[a-f0-9-]+`

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

KmsAccessDeniedException

The operation was denied because either the calling principal lacks the required AWS Key Management Service (AWS KMS) permission on the key, or the key policy does not grant Amazon CloudWatch the permissions it needs to use the key. Verify that the caller has `kms:Decrypt` permission on the key, and that the key policy grants the CloudWatch service principal the `kms:DescribeKey`, `kms:GenerateDataKey`, `kms:Encrypt`, `kms:Decrypt`, and `kms:ReEncrypt*` permissions described in [AssociateDatasetKmsKey](#).

HTTP Status Code: 400

KmsKeyDisabledException

The specified AWS Key Management Service (AWS KMS) key is disabled or pending deletion. Re-enable the key (or restore it, if it is pending deletion) and retry the operation.

HTTP Status Code: 400

KmsKeyNotFoundException

The specified AWS Key Management Service (AWS KMS) key could not be found. Verify that the key Amazon Resource Name (ARN) is correct, that the key exists, and that it is in the same AWS Region as the resource.

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteAlarmMuteRule

Deletes a specific alarm mute rule.

When you delete a mute rule, any alarms that are currently being muted by that rule are immediately unmuted. If those alarms are in an ALARM state, their configured actions will trigger.

This operation is idempotent. If you delete a mute rule that does not exist, the operation succeeds without returning an error.

Permissions

To delete a mute rule, you need the `cloudwatch:DeleteAlarmMuteRule` permission on the alarm mute rule resource.

Request Parameters

AlarmMuteRuleName

The name of the alarm mute rule to delete.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

Examples

Delete a mute rule

Delete a specific alarm mute rule. This operation does not return any output on success.

Sample Request

```
aws cloudwatch delete-alarm-mute-rule \
```

```
--alarm-mute-rule-name "DailyMaintenanceWindow"
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteAlarms

Deletes the specified alarms. You can delete up to 100 alarms in one operation. However, this total can include no more than one composite alarm. For example, you could delete 99 metric alarms and one composite alarms with one operation, but you can't delete two composite alarms with one operation. Log alarms cannot be batch deleted.

If you specify any incorrect alarm names, the alarms you specify with correct names are still deleted. Other syntax errors might result in no alarms being deleted. To confirm that alarms were deleted successfully, you can use the [DescribeAlarms](#) operation after using DeleteAlarms.

Note

It is possible to create a loop or cycle of composite alarms, where composite alarm A depends on composite alarm B, and composite alarm B also depends on composite alarm A. In this scenario, you can't delete any composite alarm that is part of the cycle because there is always still a composite alarm that depends on that alarm that you want to delete. To get out of such a situation, you must break the cycle by changing the rule of one of the composite alarms in the cycle to remove a dependency that creates the cycle. The simplest change to make to break a cycle is to change the `AlarmRule` of one of the alarms to `false`. Additionally, the evaluation of composite alarms stops if CloudWatch detects a cycle in the evaluation path.

Request Parameters

AlarmNames

The alarms to be deleted. Do not enclose the alarm names in quote marks.

Type: Array of strings

Array Members: Maximum number of 100 items.

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ResourceConflict

The operation could not be completed because the request conflicts with the current state of the alarm or its underlying scheduled query resource.

HTTP Status Code: 409

ResourceNotFound

The named resource does not exist.

message

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteAnomalyDetector

Deletes the specified anomaly detection model from your account. For more information about how to delete an anomaly detection model, see [Deleting an anomaly detection model](#) in the *CloudWatch User Guide*.

Request Parameters

Dimensions

This parameter has been deprecated.

The metric dimensions associated with the anomaly detection model to delete.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MetricMathAnomalyDetector

The metric math anomaly detector to be deleted.

When using `MetricMathAnomalyDetector`, you cannot include following parameters in the same operation:

- Dimensions,
- MetricName
- Namespace
- Stat
- the `SingleMetricAnomalyDetector` parameters of `DeleteAnomalyDetectorInput`

Instead, specify the metric math anomaly detector attributes as part of the `MetricMathAnomalyDetector` property.

Type: [MetricMathAnomalyDetector](#) object

Required: No

MetricName

This parameter has been deprecated.

The metric name associated with the anomaly detection model to delete.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

This parameter has been deprecated.

The namespace associated with the anomaly detection model to delete.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: No

SingleMetricAnomalyDetector

A single metric anomaly detector to be deleted.

When using `SingleMetricAnomalyDetector`, you cannot include the following parameters in the same operation:

- `Dimensions`,
- `MetricName`
- `Namespace`
- `Stat`
- the `MetricMathAnomalyDetector` parameters of `DeleteAnomalyDetectorInput`

Instead, specify the single metric anomaly detector attributes as part of the `SingleMetricAnomalyDetector` property.

Type: [SingleMetricAnomalyDetector](#) object

Required: No

Stat

This parameter has been deprecated.

The statistic associated with the anomaly detection model to delete.

Type: String

Length Constraints: Maximum length of 50.

Pattern: (SampleCount|Average|Sum|Minimum|Maximum|IQM|(p|tc|tm|ts|wm)
 (\d{1,2}(\.\d{0,10})?|100)|[ou]\d+(\.\d*)?)(_E|_L|_H)?|(TM|TC|TS|
 WM)\((((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):((\d{1,2})(\.
 \d{0,10})?|100(\.\d{0,10})?)%|((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):
 (((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%))?)\)| (TM|TC|TS|WM|PR)\((((\d
 +(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)):((\d+(\.\d{0,10})?|(\d
 +(\.\d{0,10})?[Ee][+-]?\d+)))?|((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee]
 [+-]?\d+)))?:((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))\)

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteDashboards

Deletes all dashboards that you specify. You can specify up to 100 dashboards to delete. If there is an error during this call, the operation attempts to delete as many dashboards as possible.

Request Parameters

DashboardNames

The dashboards to be deleted. This parameter is required.

Type: Array of strings

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
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DeleteInsightRules

Permanently deletes the specified Contributor Insights rules.

If you create a rule, delete it, and then re-create it with the same name, historical data from the first time the rule was created might not be available.

Request Parameters

RuleNames

An array of the rule names to delete. If you need to find out the names of your rules, use [DescribeInsightRules](#).

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

Response Elements

The following element is returned by the service.

Failures

An array listing the rules that could not be deleted. You cannot delete built-in rules.

Type: Array of [PartialFailure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteMetricStream

Permanently deletes the metric stream that you specify.

Request Parameters

Name

The name of the metric stream to delete.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeAlarmContributors

Returns the information of the current alarm contributors that are in ALARM state. This operation returns details about the individual time series that contribute to the alarm's state.

Request Parameters

AlarmName

The name of the alarm for which to retrieve contributor information.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

NextToken

The token returned by a previous call to indicate that there is more data available.

Type: String

Required: No

Response Elements

The following elements are returned by the service.

AlarmContributors

A list of alarm contributors that provide details about the individual time series contributing to the alarm's state.

Type: Array of [AlarmContributor](#) objects

NextToken

The token that marks the start of the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeAlarmHistory

Retrieves the history for the specified alarm. You can filter the results by date range or item type. If an alarm name is not specified, the histories for either all metric alarms or all composite alarms are returned.

CloudWatch retains the history of an alarm even if you delete the alarm.

To use this operation and return information about a composite alarm, you must be signed on with the `cloudwatch:DescribeAlarmHistory` permission that is scoped to `*`. You can't return information about composite alarms if your `cloudwatch:DescribeAlarmHistory` permission has a narrower scope.

Request Parameters

AlarmContributorId

The unique identifier of a specific alarm contributor to filter the alarm history results.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 16.

Required: No

AlarmName

The name of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

AlarmTypes

Use this parameter to specify whether you want the operation to return metric alarms, composite alarms, or log alarms. If you omit this parameter, only metric alarms are returned.

Type: Array of strings

Valid Values: `CompositeAlarm` | `MetricAlarm` | `LogAlarm`

Required: No

EndDate

The ending date to retrieve alarm history.

Type: Timestamp

Required: No

HistoryItemType

The type of alarm histories to retrieve.

Type: String

Valid Values: ConfigurationUpdate | StateUpdate | Action | AlarmContributorStateUpdate | AlarmContributorAction

Required: No

MaxRecords

The maximum number of alarm history records to retrieve.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 100.

Required: No

NextToken

The token returned by a previous call to indicate that there is more data available.

Type: String

Required: No

ScanBy

Specified whether to return the newest or oldest alarm history first. Specify `TimestampDescending` to have the newest event history returned first, and specify `TimestampAscending` to have the oldest history returned first.

Type: String

Valid Values: `TimestampDescending` | `TimestampAscending`

Required: No

StartDate

The starting date to retrieve alarm history.

Type: Timestamp

Required: No

Response Elements

The following elements are returned by the service.

AlarmHistoryItems

The alarm histories, in JSON format.

Type: Array of [AlarmHistoryItem](#) objects

NextToken

The token that marks the start of the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeAlarms

Retrieves the specified alarms. You can filter the results by specifying a prefix for the alarm name, the alarm state, or a prefix for any action.

To use this operation and return information about composite alarms, you must be signed on with the `cloudwatch:DescribeAlarms` permission that is scoped to `*`. You can't return information about composite alarms if your `cloudwatch:DescribeAlarms` permission has a narrower scope.

Request Parameters

ActionPrefix

Use this parameter to filter the results of the operation to only those alarms that use a certain alarm action. For example, you could specify the ARN of an SNS topic to find all alarms that send notifications to that topic.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmNamePrefix

An alarm name prefix. If you specify this parameter, you receive information about all alarms that have names that start with this prefix.

If this parameter is specified, you cannot specify `AlarmNames`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

AlarmNames

The names of the alarms to retrieve information about.

Type: Array of strings

Array Members: Maximum number of 100 items.

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

AlarmTypes

Use this parameter to specify whether you want the operation to return metric alarms, composite alarms, or log alarms. If you omit this parameter, only metric alarms are returned, even if composite alarms or log alarms exist in the account.

For example, if you omit this parameter or specify `MetricAlarms`, the operation returns only a list of metric alarms. It does not return any composite alarms or log alarms, even if they exist in the account.

If you specify `CompositeAlarms`, the operation returns only a list of composite alarms, and does not return any metric alarms or log alarms.

If you specify `LogAlarms`, the operation returns only a list of log alarms, and does not return any metric alarms or composite alarms.

Type: Array of strings

Valid Values: `CompositeAlarm` | `MetricAlarm` | `LogAlarm`

Required: No

ChildrenOfAlarmName

If you use this parameter and specify the name of a composite alarm, the operation returns information about the "children" alarms of the alarm you specify. These are the metric alarms and composite alarms referenced in the `AlarmRule` field of the composite alarm that you specify in `ChildrenOfAlarmName`. Information about the composite alarm that you name in `ChildrenOfAlarmName` is not returned.

If you specify `ChildrenOfAlarmName`, you cannot specify any other parameters in the request except for `MaxRecords` and `NextToken`. If you do so, you receive a validation error.

Note

Only the `AlarmName`, `ARN`, `StateValue` (`OK`/`ALARM`/`INSUFFICIENT_DATA`), and `StateUpdatedTimestamp` information are returned by this operation when you use this parameter. To get complete information about these alarms, perform another

DescribeAlarms operation and specify the parent alarm names in the AlarmNames parameter.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

MaxRecords

The maximum number of alarm descriptions to retrieve.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 100.

Required: No

NextToken

The token returned by a previous call to indicate that there is more data available.

Type: String

Required: No

ParentsOfAlarmName

If you use this parameter and specify the name of a metric or composite alarm, the operation returns information about the "parent" alarms of the alarm you specify. These are the composite alarms that have AlarmRule parameters that reference the alarm named in ParentsOfAlarmName. Information about the alarm that you specify in ParentsOfAlarmName is not returned.

If you specify ParentsOfAlarmName, you cannot specify any other parameters in the request except for MaxRecords and NextToken. If you do so, you receive a validation error.

Note

Only the Alarm Name and ARN are returned by this operation when you use this parameter. To get complete information about these alarms, perform another

DescribeAlarms operation and specify the parent alarm names in the AlarmNames parameter.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

StateValue

Specify this parameter to receive information only about alarms that are currently in the state that you specify.

Type: String

Valid Values: OK | ALARM | INSUFFICIENT_DATA

Required: No

Response Elements

The following elements are returned by the service.

CompositeAlarms

The information about any composite alarms returned by the operation.

Type: Array of [CompositeAlarm](#) objects

LogAlarms

The information about any log alarms returned by the operation.

Type: Array of [LogAlarm](#) objects

MetricAlarms

The information about any metric alarms returned by the operation.

Type: Array of [MetricAlarm](#) objects

NextToken

The token that marks the start of the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeAlarmsForMetric

Retrieves the alarms for the specified metric. To filter the results, specify a statistic, period, or unit.

This operation retrieves only standard alarms that are based on the specified metric. It does not return alarms based on math expressions that use the specified metric, or composite alarms that use the specified metric.

Request Parameters

Dimensions

The dimensions associated with the metric. If the metric has any associated dimensions, you must specify them in order for the call to succeed.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

ExtendedStatistic

The percentile statistic for the metric. Specify a value between p0.0 and p100.

Type: String

Required: No

MetricName

The name of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Namespace

The namespace of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*.*`

Required: Yes

Period

The period, in seconds, over which the statistic is applied.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

Statistic

The statistic for the metric, other than percentiles. For percentile statistics, use `ExtendedStatistics`.

Type: String

Valid Values: `SampleCount` | `Average` | `Sum` | `Minimum` | `Maximum`

Required: No

Unit

The unit for the metric.

Type: String

Valid Values: `Seconds` | `Microseconds` | `Milliseconds` | `Bytes` | `Kilobytes` | `Megabytes` | `Gigabytes` | `Terabytes` | `Bits` | `Kilobits` | `Megabits` | `Gigabits` | `Terabits` | `Percent` | `Count` | `Bytes/Second` | `Kilobytes/Second` | `Megabytes/Second` | `Gigabytes/Second` | `Terabytes/Second` | `Bits/Second` | `Kilobits/Second` | `Megabits/Second` | `Gigabits/Second` | `Terabits/Second` | `Count/Second` | `None`

Required: No

Response Elements

The following element is returned by the service.

MetricAlarms

The information for each alarm with the specified metric.

Type: Array of [MetricAlarm](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeAnomalyDetectors

Lists the anomaly detection models that you have created in your account. For single metric anomaly detectors, you can list all of the models in your account or filter the results to only the models that are related to a certain namespace, metric name, or metric dimension. For metric math anomaly detectors, you can list them by adding `METRIC_MATH` to the `AnomalyDetectorTypes` array. This will return all metric math anomaly detectors in your account.

Request Parameters

AnomalyDetectorTypes

The anomaly detector types to request when using `DescribeAnomalyDetectorsInput`. If empty, defaults to `SINGLE_METRIC`.

Type: Array of strings

Array Members: Maximum number of 2 items.

Valid Values: `SINGLE_METRIC` | `METRIC_MATH`

Required: No

Dimensions

Limits the results to only the anomaly detection models that are associated with the specified metric dimensions. If there are multiple metrics that have these dimensions and have anomaly detection models associated, they're all returned.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MaxResults

The maximum number of results to return in one operation. The maximum value that you can specify is 100.

To retrieve the remaining results, make another call with the returned `NextToken` value.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

MetricName

Limits the results to only the anomaly detection models that are associated with the specified metric name. If there are multiple metrics with this name in different namespaces that have anomaly detection models, they're all returned.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

Limits the results to only the anomaly detection models that are associated with the specified namespace.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: No

NextToken

Use the token returned by the previous operation to request the next page of results.

Type: String

Required: No

Response Elements

The following elements are returned by the service.

AnomalyDetectors

The list of anomaly detection models returned by the operation.

Type: Array of [AnomalyDetector](#) objects

NextToken

A token that you can use in a subsequent operation to retrieve the next set of results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

Examples

Example

The following example lists all the anomaly detectors for metrics with the name CPUUtilization.

Sample Request

```
{
  "MetricName": "CPUUtilization"
}
```

Sample Response

```
{
  "AnomalyDetectors": [
    {
      "Namespace": "AWS/EC2",
      "MetricName": "CPUUtilization",
      "Dimensions": [
        {
          "Name": "dimension1",
          "Value": "value1"
        },
        {
          "Name": "dimension2",
          "Value": "value2"
        }
      ],
      "Stat": "Average",
      "Configuration": {
        "ExcludedTimeRanges": [
        ]
      }
    },
    {
      "Namespace": "AWS/EC2",
      "MetricName": "CPUUtilization",
      "Dimensions": [
        {
          "Name": "dimension1",
```

```
        "Value": "value1"
      }
    ],
    "Stat": "SampleCount",
    "Configuration": {
      "ExcludedTimeRanges": [

      ]
    }
  },
  {
    "Namespace": "APITest1",
    "MetricName": "Metric1",
    "Dimensions": [
      {
        "Name": "dimension1",
        "Value": "value1"
      }
    ],
    "Stat": "SampleCount",
    "Configuration": {
      "ExcludedTimeRanges": [

      ]
    }
  },
  {
    "Namespace": "CustomNamespace",
    "MetricName": "CPUUtilization",
    "Dimensions": [
      {
        "Name": "dimension1",
        "Value": "value1"
      },
      {
        "Name": "dimension2",
        "Value": "value2"
      }
    ],
    "Stat": "Maximum",
    "Configuration": {
      "ExcludedTimeRanges": [

      ]
    }
  }
]
```

```
}  
  }  
}  
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeInsightRules

Returns a list of all the Contributor Insights rules in your account.

For more information about Contributor Insights, see [Using Contributor Insights to Analyze High-Cardinality Data](#).

Request Parameters

MaxResults

The maximum number of results to return in one operation. If you omit this parameter, the default of 500 is used.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 500.

Required: No

NextToken

Include this value, if it was returned by the previous operation, to get the next set of rules.

Type: String

Required: No

Response Elements

The following elements are returned by the service.

InsightRules

The rules returned by the operation.

Type: Array of [InsightRule](#) objects

NextToken

If this parameter is present, it is a token that marks the start of the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DisableAlarmActions

Disables the actions for the specified alarms. When an alarm's actions are disabled, the alarm actions do not execute when the alarm state changes.

Request Parameters

AlarmNames

The names of the alarms.

Type: Array of strings

Array Members: Maximum number of 100 items.

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)

- [AWS SDK for Ruby V3](#)

DisableInsightRules

Disables the specified Contributor Insights rules. When rules are disabled, they do not analyze log groups and do not incur costs.

Request Parameters

RuleNames

An array of the rule names to disable. If you need to find out the names of your rules, use [DescribeInsightRules](#).

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

Response Elements

The following element is returned by the service.

Failures

An array listing the rules that could not be disabled. You cannot disable built-in rules.

Type: Array of [PartialFailure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DisassociateDatasetKmsKey

Removes the customer managed AWS Key Management Service (AWS KMS) key association from the specified dataset. After this operation completes, data that you publish to the dataset is encrypted at rest using an AWS owned key managed by Amazon CloudWatch.

Only the default dataset is supported. To call this operation, the dataset must currently have a customer managed KMS key associated with it. If the dataset has no associated KMS key, the operation fails with `ResourceNotFoundException`.

Amazon CloudWatch performs a dry-run `kms:Decrypt` call on the key as part of this operation. This verifies that the caller is authorized to use the currently associated key. The caller must have `kms:Decrypt` permission on the currently associated key, and the key must be enabled and accessible. If the key has been disabled or scheduled for deletion, you must first re-enable or restore it before you can disassociate it from the dataset.

Important

Disassociating a KMS key from a dataset does not immediately remove the `kms:Decrypt` requirement on data plane operations. For up to three hours after disassociation, callers must continue to have `kms:Decrypt` permission on the previously associated key. Some data may still be encrypted with that key during this window. After this enforcement window elapses, the `kms:Decrypt` requirement is lifted.

For more information about using customer managed keys with Amazon CloudWatch, see [Encryption at rest with customer managed keys](#) in the *Amazon CloudWatch User Guide*.

Request Parameters

DatasetIdentifier

Specifies the identifier of the dataset from which to remove the KMS key association. For the default dataset, you can specify either `default` or the full dataset Amazon Resource Name (ARN) in the format `arn:aws:cloudwatch:Region:account-id:dataset/default`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 2048.

Pattern: (default|arn:[a-zA-Z0-9-]+:cloudwatch:[a-zA-Z0-9-]*:
\d{12}:dataset/default)

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

EnableAlarmActions

Enables the actions for the specified alarms.

Request Parameters

AlarmNames

The names of the alarms.

Type: Array of strings

Array Members: Maximum number of 100 items.

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

EnableInsightRules

Enables the specified Contributor Insights rules. When rules are enabled, they immediately begin analyzing log data.

Request Parameters

RuleNames

An array of the rule names to enable. If you need to find out the names of your rules, use [DescribeInsightRules](#).

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

Response Elements

The following element is returned by the service.

Failures

An array listing the rules that could not be enabled. You cannot disable or enable built-in rules.

Type: Array of [PartialFailure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

LimitExceededException

The operation exceeded one or more limits.

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetAlarmMuteRule

Retrieves details for a specific alarm mute rule.

This operation returns complete information about the mute rule, including its configuration, status, targeted alarms, and metadata.

The returned status indicates the current state of the mute rule:

- **SCHEDULED:** The mute rule is configured and will become active in the future
- **ACTIVE:** The mute rule is currently muting alarm actions
- **EXPIRED:** The mute rule has passed its expiration date and will no longer become active

Permissions

To retrieve details for a mute rule, you need the `cloudwatch:GetAlarmMuteRule` permission on the alarm mute rule resource.

Request Parameters

AlarmMuteRuleName

The name of the alarm mute rule to retrieve.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Response Elements

The following elements are returned by the service.

AlarmMuteRuleArn

The Amazon Resource Name (ARN) of the alarm mute rule.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Description

The description of the alarm mute rule.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

ExpireDate

The date and time when the mute rule expires and is no longer evaluated.

Type: Timestamp

LastUpdatedTimestamp

The date and time when the mute rule was last updated.

Type: Timestamp

MuteTargets

Specifies which alarms this rule applies to.

Type: [MuteTargets](#) object

MuteType

Indicates whether the mute rule is one-time or recurring. Valid values are ONE_TIME or RECURRING.

Type: String

Name

The name of the alarm mute rule.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Rule

The configuration that defines when and how long alarms are muted.

Type: [Rule](#) object

StartDate

The date and time when the mute rule becomes active. If not set, the rule is active immediately.

Type: Timestamp

Status

The current status of the alarm mute rule. Valid values are SCHEDULED, ACTIVE, or EXPIRED.

Type: String

Valid Values: SCHEDULED | ACTIVE | EXPIRED

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

Examples

Get details for a mute rule

Retrieve complete details for a specific alarm mute rule.

Sample Request

```
aws cloudwatch get-alarm-mute-rule \  
--alarm-mute-rule-name "DailyMaintenanceWindow"
```

Sample Response

```
{
```

```
{
  "Name": "DailyMaintenanceWindow",
  "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-east-1:123456789012:alarm-mute-rule:DailyMaintenanceWindow",
  "Description": "Mute alarms during daily maintenance",
  "Rule": {
    "Schedule": {
      "Expression": "cron(0 2 * * ?)",
      "Duration": "PT2H",
      "Timezone": "UTC"
    }
  },
  "MuteTargets": {
    "AlarmNames": [
      "WebServerCPUAlarm",
      "DatabaseConnectionAlarm"
    ]
  },
  "Status": "SCHEDULED",
  "LastUpdatedTimestamp": "2026-01-15T10:30:00Z"
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetDashboard

Displays the details of the dashboard that you specify.

To copy an existing dashboard, use `GetDashboard`, and then use the data returned within `DashboardBody` as the template for the new dashboard when you call `PutDashboard` to create the copy.

Request Parameters

DashboardName

The name of the dashboard to be described.

Type: String

Required: Yes

Response Elements

The following elements are returned by the service.

DashboardArn

The Amazon Resource Name (ARN) of the dashboard.

Type: String

DashboardBody

The detailed information about the dashboard, including what widgets are included and their location on the dashboard. For more information about the `DashboardBody` syntax, see [Dashboard Body Structure and Syntax](#).

Type: String

DashboardName

The name of the dashboard.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

ResourceNotFound

The specified dashboard does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetDataset

Returns information about the specified dataset. This includes its identifier, Amazon Resource Name (ARN), and any customer managed AWS Key Management Service (AWS KMS) key that is currently associated with it.

Only the default dataset is supported. The default dataset is implicit for every account in every Region — you can call `GetDataset` for it without first creating it. If no customer managed KMS key has been associated with the dataset, the response omits the `KmsKeyArn` field, indicating that data is encrypted at rest using an AWS owned key managed by Amazon CloudWatch.

To associate a customer managed KMS key with a dataset, use [AssociateDatasetKmsKey](#). To remove the association, use [DisassociateDatasetKmsKey](#).

Request Parameters

DatasetIdentifier

Specifies the identifier of the dataset to retrieve. For the default dataset, you can specify either `default` or the full dataset Amazon Resource Name (ARN) in the format `arn:aws:cloudwatch:Region:account-id:dataset/default`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 2048.

Pattern: `(default|arn:[a-zA-Z0-9-]+:cloudwatch:[a-zA-Z0-9-]*:\d{12}:dataset/default)`

Required: Yes

Response Elements

The following elements are returned by the service.

Arn

Returns the Amazon Resource Name (ARN) of the dataset, in the format `arn:aws:cloudwatch:Region:account-id:dataset/dataset-id`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 2048.

Pattern: `arn:[a-zA-Z0-9-]+:cloudwatch:[a-zA-Z0-9-]*:\d{12}:dataset/default`

DatasetId

Returns the identifier of the dataset.

Type: String

Length Constraints: Fixed length of 7.

Pattern: `default`

KmsKeyArn

Returns the Amazon Resource Name (ARN) of the customer managed AWS KMS key that is currently associated with the dataset, if any. If the dataset is not associated with a customer managed KMS key, this field is not included in the response and the dataset is encrypted at rest using an AWS owned key.

Type: String

Length Constraints: Minimum length of 20. Maximum length of 2048.

Pattern: `arn:[a-zA-Z0-9-]+:kms:[a-zA-Z0-9-]+:\d{12}:key/[a-f0-9-]+`

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetInsightRuleReport

This operation returns the time series data collected by a Contributor Insights rule. The data includes the identity and number of contributors to the log group.

You can also optionally return one or more statistics about each data point in the time series. These statistics can include the following:

- `UniqueContributors` -- the number of unique contributors for each data point.
- `MaxContributorValue` -- the value of the top contributor for each data point. The identity of the contributor might change for each data point in the graph.

If this rule aggregates by `COUNT`, the top contributor for each data point is the contributor with the most occurrences in that period. If the rule aggregates by `SUM`, the top contributor is the contributor with the highest sum in the log field specified by the rule's `Value`, during that period.

- `SampleCount` -- the number of data points matched by the rule.
- `Sum` -- the sum of the values from all contributors during the time period represented by that data point.
- `Minimum` -- the minimum value from a single observation during the time period represented by that data point.
- `Maximum` -- the maximum value from a single observation during the time period represented by that data point.
- `Average` -- the average value from all contributors during the time period represented by that data point.

Request Parameters

EndTime

The end time of the data to use in the report. When used in a raw HTTP Query API, it is formatted as `yyyy-MM-dd'T'HH:mm:ss`. For example, `2019-07-01T23:59:59`.

Type: Timestamp

Required: Yes

MaxContributorCount

The maximum number of contributors to include in the report. The range is 1 to 100. If you omit this, the default of 10 is used.

Type: Integer

Required: No

Metrics

Specifies which metrics to use for aggregation of contributor values for the report. You can specify one or more of the following metrics:

- `UniqueContributors` -- the number of unique contributors for each data point.
- `MaxContributorValue` -- the value of the top contributor for each data point. The identity of the contributor might change for each data point in the graph.

If this rule aggregates by `COUNT`, the top contributor for each data point is the contributor with the most occurrences in that period. If the rule aggregates by `SUM`, the top contributor is the contributor with the highest sum in the log field specified by the rule's `Value`, during that period.

- `SampleCount` -- the number of data points matched by the rule.
- `Sum` -- the sum of the values from all contributors during the time period represented by that data point.
- `Minimum` -- the minimum value from a single observation during the time period represented by that data point.
- `Maximum` -- the maximum value from a single observation during the time period represented by that data point.
- `Average` -- the average value from all contributors during the time period represented by that data point.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 32.

Pattern: `[\x20-\x7E]+`

Required: No

OrderBy

Determines what statistic to use to rank the contributors. Valid values are Sum and Maximum.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 32.

Pattern: `[\x20-\x7E]+`

Required: No

Period

The period, in seconds, to use for the statistics in the `InsightRuleMetricDatapoint` results.

Type: Integer

Valid Range: Minimum value of 1.

Required: Yes

RuleName

The name of the rule that you want to see data from.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

StartTime

The start time of the data to use in the report. When used in a raw HTTP Query API, it is formatted as `yyyy-MM-dd'T'HH:mm:ss`. For example, `2019-07-01T23:59:59`.

Type: Timestamp

Required: Yes

Response Elements

The following elements are returned by the service.

AggregateValue

The sum of the values from all individual contributors that match the rule.

Type: Double

AggregationStatistic

Specifies whether this rule aggregates contributor data by COUNT or SUM.

Type: String

ApproximateUniqueCount

An approximate count of the unique contributors found by this rule in this time period.

Type: Long

Contributors

An array of the unique contributors found by this rule in this time period. If the rule contains multiple keys, each combination of values for the keys counts as a unique contributor.

Type: Array of [InsightRuleContributor](#) objects

KeyLabels

An array of the strings used as the keys for this rule. The keys are the dimensions used to classify contributors. If the rule contains more than one key, then each unique combination of values for the keys is counted as a unique contributor.

Type: Array of strings

MetricDatapoints

A time series of metric data points that matches the time period in the rule request.

Type: Array of [InsightRuleMetricDatapoint](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetMetricData

You can use the `GetMetricData` API to retrieve CloudWatch metric values. The operation can also include a CloudWatch Metrics Insights query, and one or more metric math functions.

A `GetMetricData` operation that does not include a query can retrieve as many as 500 different metrics in a single request, with a total of as many as 100,800 data points. You can also optionally perform metric math expressions on the values of the returned statistics, to create new time series that represent new insights into your data. For example, using Lambda metrics, you could divide the Errors metric by the Invocations metric to get an error rate time series. For more information about metric math expressions, see [Metric Math Syntax and Functions](#) in the *Amazon CloudWatch User Guide*.

If you include a Metrics Insights query, each `GetMetricData` operation can include only one query. But the same `GetMetricData` operation can also retrieve other metrics. Metrics Insights queries can query only the most recent three hours of metric data. For more information about Metrics Insights, see [Query your metrics with CloudWatch Metrics Insights](#).

Calls to the `GetMetricData` API have a different pricing structure than calls to `GetMetricStatistics`. For more information about pricing, see [Amazon CloudWatch Pricing](#).

Amazon CloudWatch retains metric data as follows:

- Data points with a period of less than 60 seconds are available for 3 hours. These data points are high-resolution metrics and are available only for custom metrics that have been defined with a `StorageResolution` of 1.
- Data points with a period of 60 seconds (1-minute) are available for 15 days.
- Data points with a period of 300 seconds (5-minute) are available for 63 days.
- Data points with a period of 3600 seconds (1 hour) are available for 455 days (15 months).

Data points that are initially published with a shorter period are aggregated together for long-term storage. For example, if you collect data using a period of 1 minute, the data remains available for 15 days with 1-minute resolution. After 15 days, this data is still available, but is aggregated and retrievable only with a resolution of 5 minutes. After 63 days, the data is further aggregated and is available with a resolution of 1 hour.

If you omit `Unit` in your request, all data that was collected with any unit is returned, along with the corresponding units that were specified when the data was reported to CloudWatch. If

you specify a unit, the operation returns only data that was collected with that unit specified. If you specify a unit that does not match the data collected, the results of the operation are null. CloudWatch does not perform unit conversions.

Using Metrics Insights queries with metric math

You can't mix a Metric Insights query and metric math syntax in the same expression, but you can reference results from a Metrics Insights query within other Metric math expressions. A Metrics Insights query without a **GROUP BY** clause returns a single time-series (TS), and can be used as input for a metric math expression that expects a single time series. A Metrics Insights query with a **GROUP BY** clause returns an array of time-series (TS[]), and can be used as input for a metric math expression that expects an array of time series.

Request Parameters

EndTime

The time stamp indicating the latest data to be returned.

The value specified is exclusive; results include data points up to the specified time stamp.

For better performance, specify `StartTime` and `EndTime` values that align with the value of the metric's `Period` and sync up with the beginning and end of an hour. For example, if the `Period` of a metric is 5 minutes, specifying 12:05 or 12:30 as `EndTime` can get a faster response from CloudWatch than setting 12:07 or 12:29 as the `EndTime`.

Type: Timestamp

Required: Yes

LabelOptions

This structure includes the `Timezone` parameter, which you can use to specify your time zone so that the labels of returned data display the correct time for your time zone.

Type: [LabelOptions](#) object

Required: No

MaxDatapoints

The maximum number of data points the request should return before paginating. If you omit this, the default of 100,800 is used.

Type: Integer

Required: No

MetricDataQueries

The metric queries to be returned. A single `GetMetricData` call can include as many as 500 `MetricDataQuery` structures. Each of these structures can specify either a metric to retrieve, a Metrics Insights query, or a math expression to perform on retrieved data.

Type: Array of [MetricDataQuery](#) objects

Required: Yes

NextToken

Include this value, if it was returned by the previous `GetMetricData` operation, to get the next set of data points.

Type: String

Required: No

ScanBy

The order in which data points should be returned. `TimestampDescending` returns the newest data first and paginates when the `MaxDatapoints` limit is reached. `TimestampAscending` returns the oldest data first and paginates when the `MaxDatapoints` limit is reached.

If you omit this parameter, the default of `TimestampDescending` is used.

Type: String

Valid Values: `TimestampDescending` | `TimestampAscending`

Required: No

StartTime

The time stamp indicating the earliest data to be returned.

The value specified is inclusive; results include data points with the specified time stamp.

CloudWatch rounds the specified time stamp as follows:

- Start time less than 15 days ago - Round down to the nearest whole minute. For example, 12:32:34 is rounded down to 12:32:00.
- Start time between 15 and 63 days ago - Round down to the nearest 5-minute clock interval. For example, 12:32:34 is rounded down to 12:30:00.
- Start time greater than 63 days ago - Round down to the nearest 1-hour clock interval. For example, 12:32:34 is rounded down to 12:00:00.

If you set `Period` to 5, 10, 20, or 30, the start time of your request is rounded down to the nearest time that corresponds to even 5-, 10-, 20-, or 30-second divisions of a minute. For example, if you make a query at (HH:mm:ss) 01:05:23 for the previous 10-second period, the start time of your request is rounded down and you receive data from 01:05:10 to 01:05:20. If you make a query at 15:07:17 for the previous 5 minutes of data, using a period of 5 seconds, you receive data timestamped between 15:02:15 and 15:07:15.

For better performance, specify `StartTime` and `EndTime` values that align with the value of the metric's `Period` and sync up with the beginning and end of an hour. For example, if the `Period` of a metric is 5 minutes, specifying 12:05 or 12:30 as `StartTime` can get a faster response from CloudWatch than setting 12:07 or 12:29 as the `StartTime`.

Type: Timestamp

Required: Yes

Response Elements

The following elements are returned by the service.

Messages

Contains a message about this `GetMetricData` operation, if the operation results in such a message. An example of a message that might be returned is `Maximum number of allowed metrics exceeded`. If there is a message, as much of the operation as possible is still executed.

A message appears here only if it is related to the global `GetMetricData` operation. Any message about a specific metric returned by the operation appears in the `MetricDataResult` object returned for that metric.

Type: Array of [MessageData](#) objects

MetricDataResults

The metrics that are returned, including the metric name, namespace, and dimensions.

Type: Array of [MetricDataResult](#) objects

NextToken

A token that marks the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

Examples

Example

The following example requests a Metrics Insights query for aggregated CPUUtilization, and a metric.

Sample Request

```
{
  "StartTime": 1637061900,
  "EndTime": 1637074500,
  "MetricDataQueries": [
    {
      "Expression": "SELECT AVG(CPUUtilization) FROM SCHEMA(\"AWS/EC2\",
InstanceId)",
      "Id": "q1",
```

```

        "Period": 300,
        "Label": "Cluster CpuUtilization"
    },
    {
        "Id": "m1",
        "Label": "Unhealthy Behind Load Balancer",
        "MetricStat": {
            "Metric": {
                "Namespace": "AWS/ApplicationELB",
                "MetricName": "UnHealthyHostCount",
                "Dimensions": [
                    {
                        "Name": "TargetGroup",
                        "Value": "targetgroup/EC2Co-Defau-
EXAMPLEWNAD/89cc68152b367e5f"
                    },
                    {
                        "Name": "LoadBalancer",
                        "Value": "app/EC2Co-EcsE1-EXAMPLE69Q/fdd2210e799e4376"
                    }
                ]
            },
            "Period": 300,
            "Stat": "Average"
        }
    }
]
}

```

Sample Response

```

{
    "Messages": [],
    "MetricDataResults": [
        {
            "Id": "m1",
            "Label": "Unhealthy Behind Load Balancer",
            "StatusCode": "Complete",
            "Timestamps": [
                1637074200,
                1637073900,
                1637073600
            ],

```



```
        "Values": [
            0,
            0,
            0
        ],
    },
    {
        "Id": "q1",
        "Label": "Cluster CpuUtilization",
        "StatusCode": "Complete",
        "Timestamps": [
            1637074245,
            1637073945,
            1637073645
        ],
        "Values": [
            1.2158469945359334,
            0.8678863271635757,
            0.7201860957623283
        ]
    }
]
```

Example

The following example includes a Metrics Insights query for that is given the ID `error_rate`. The returned results of the query are then used in the metric math expression to return availability.

Sample Request

```
{
  "StartTime": 1518867432,
  "EndTime": 1518868032,
  "MetricQueries": [
    {
      "Id": "availability",
      "Expression": "(1 - error_rate) * 100",
      "Label": "Availability"
    },
    {
      "Id": " error_rate",
```

```
        "Expression": "SELECT AVG(ErrorRate) FROM MyService",
        "Period": 300,
        "ReturnData": false
    }
]
}
```

Example

The following example requests three separate metrics across two namespaces. The labels of the first two metrics use dynamic labels to display the peak value of CPUUtilization during the time shown on the graph, and also the time that the peak value was recorded. The Timezone setting specifies that the times shown in those dynamic labels reflect the United States Eastern time zone, which is 4 hours behind UTC.

Sample Request

```
{
  "StartTime": 1518867432,
  "EndTime": 1518868232,
  "LabelOptions": {
    "Timezone" : "-0400"
  },
  "MetricDataQueries": [
    {
      "Id": "m1",
      "Label": "CPUUtilization, peak of ${MAX} was at ${MAX_TIME}",
      "MetricStat": {
        "Metric": {
          "Namespace": "AWS/EC2",
          "MetricName": "CPUUtilization",
          "Dimensions": [
            {
              "Name": "InstanceId",
              "Value": "i-1234567890abcdef0"
            }
          ]
        },
        "Period": 300,
        "Stat": "Average"
      }
    },
    {

```

```
    "Id": "m2",
    "Label": "CPUUtilization, peak of ${MAX} was at ${MAX_TIME}",
    "MetricStat": {
      "Metric": {
        "Namespace": "AWS/EC2",
        "MetricName": "CPUUtilization",
        "Dimensions": [
          {
            "Name": "InstanceId",
            "Value": "i-11111111111111111"
          }
        ]
      },
      "Period": 300,
      "Stat": "Average"
    }
  },
  {
    "Id": "m3",
    "MetricStat": {
      "Metric": {
        "Namespace": "AWS/ELB",
        "MetricName": "HealthyHostCount",
        "Dimensions": [
          {
            "Name": "LoadBalancerName",
            "Value": "my-lb-B"
          }
        ]
      },
      "Period": 300,
      "Stat": "Sum",
      "Unit": "None"
    }
  }
]
```

Sample Response

```
{
  "MetricDataResults": [
    {
```

```
"Id": "m1",
"StatusCode": "Complete",
"Label": "CPUUtilization, peak of 31.5 was at 1-22 13:05",
"Timestamps": [
    1518868032,
    1518867732,
    1518867432
],
"Values": [
    15000,
    14000,
    16000
]
},
{
    "Id": "m2",
    "StatusCode": "Complete",
    "Label": "CPUUtilization, peak of 63.2 was at 1-22 13:20",
    "Timestamps": [
        1518868032,
        1518867732,
        1518867432
    ],
    "Values": [
        15,
        14,
        16
    ]
},
{
    "Id": "m3",
    "StatusCode": "Complete",
    "Label": "AWS/EC2 HealthyHostCount",
    "Timestamps": [
        1518868032,
        1518867732,
        1518867432
    ],
    "Values": [
        15,
        14,
        16
    ]
}
```

```
]
}
```

Example

The following example retrieves the `NetworkOut` metric for two Auto Scaling groups, and uses them in an expression. These two metrics are called `m1` and `m2`, and the expression calculates `e1` as the results of `m2/m1`. The raw values and time stamps of the `NetworkOut` metrics are not returned.

Sample Request

```
{
  "StartTime": 1518867432,
  "EndTime": 1518868232,
  "MetricQueries": [
    {
      "Id": "e1",
      "Expression": "m2 / m1",
      "Label": "my-asg-B / my-asg-A"
    },
    {
      "Id": "m1",
      "MetricStat": {
        "Metric": {
          "Namespace": "AWS/EC2",
          "MetricName": "NetworkOut",
          "Dimensions": [
            {
              "Name": "AutoScalingGroupName",
              "Value": "my-asg-A"
            }
          ]
        },
        "Period": 300,
        "Stat": "SampleCount",
        "Unit": "Bytes"
      },
      "ReturnData": false
    },
    {
      "Id": "m2",
      "MetricStat": {
```

```
    "Metric": {
      "Namespace": "AWS/EC2",
      "MetricName": "NetworkOut",
      "Dimensions": [
        {
          "Name": "AutoScalingGroupName",
          "Value": "my-asg-B"
        }
      ]
    },
    "Period": 300,
    "Stat": "SampleCount",
    "Unit": "Bytes"
  },
  "ReturnData": false
}
]
```

Sample Response

```
{
  "MetricDataResults": [
    {
      "Id": "m1",
      "StatusCode": "Complete"
    },
    {
      "Id": "m2",
      "StatusCode": "Complete"
    },
    {
      "Id": "e1",
      "StatusCode": "Complete",
      "Label": "my-asg-B / my-asg-A",
      "Timestamps": [
        1518868032,
        1518867732,
        1518867432
      ],
      "Values": [
        100,
        100,
```

```
        100
      ]
    }
  ]
}
```

Example

In the following example, two levels of metric math expressions are used, with the result of one expression used as an input to the next expression:

Sample Request

```
{
  "StartTime": 1518867432,
  "EndTime": 1518868232,
  "MetricDataQueries": [
    {
      "Id": "e1",
      "Expression": "e2 + m3",
      "Label": "my-asg-A * my-asg-B + my-asg-C"
    },
    {
      "Id": "e2",
      "Expression": "m1 * m2",
      "Label": "my-asg-A * my-asg-B"
    },
    {
      "Id": "m1",
      "MetricStat": {
        "Metric": {
          "Namespace": "AWS/EC2",
          "MetricName": "NetworkOut",
          "Dimensions": [
            {
              "Name": "AutoScalingGroupName",
              "Value": "my-asg-A"
            }
          ]
        },
        "Period": 300,
        "Stat": "SampleCount",
        "Unit": "Bytes"
      }
    }
  ]
}
```

```
    },
    "ReturnData": false
  },
  {
    "Id": "m2",
    "MetricStat": {
      "Metric": {
        "Namespace": "AWS/EC2",
        "MetricName": "NetworkOut",
        "Dimensions": [
          {
            "Name": "AutoScalingGroupName",
            "Value": "my-asg-B"
          }
        ]
      },
      "Period": 300,
      "Stat": "SampleCount",
      "Unit": "Bytes"
    },
    "ReturnData": false
  },
  {
    "Id": "m3",
    "MetricStat": {
      "Metric": {
        "Namespace": "AWS/EC2",
        "MetricName": "NetworkOut",
        "Dimensions": [
          {
            "Name": "AutoScalingGroupName",
            "Value": "my-asg-C"
          }
        ]
      },
      "Period": 300,
      "Stat": "SampleCount",
      "Unit": "Bytes"
    },
    "ReturnData": false
  }
]
```


Sample Response

```
{
  "MetricDataResults": [
    {
      "Id": "m1",
      "StatusCode": "Complete"
    },
    {
      "Id": "m2",
      "StatusCode": "Complete"
    },
    {
      "Id": "m3",
      "StatusCode": "Complete"
    },
    {
      "Id": "e1",
      "StatusCode": "Complete",
      "Label": "my-asg-A * my-asg-B + my-asg-C",
      "Timestamps": [
        1518868032,
        1518867732,
        1518867432
      ],
      "Values": [
        200,
        200,
        200
      ]
    },
    {
      "Id": "e2",
      "StatusCode": "Complete",
      "Label": "my-asg-A * my-asg-B",
      "Timestamps": [
        1518868032,
        1518867732,
        1518867432
      ],
      "Values": [
        100,
        100,
        100
      ]
    }
  ]
}
```

```
]
}
]
}
```

Example

In the following example, custom metrics are searched and assigned IDs that contain either "error" or "request", even if the original metric names did not contain those words. Then an error rate is calculated using the `METRICS("string")` function on the assigned IDs.

Sample Request

```
{
  "StartTime": 1518867432,
  "EndTime": 1518868432,
  "MetricDataQueries": [
    {
      "Id": "errorRate",
      "Label": "Error Rate",
      "Expression": "errors/requests"
    },
    {
      "Id": "errorRatePercent",
      "Label": "% Error Rate",
      "Expression": "errorRate*100"
    },
    {
      "Id": "requests",
      "Expression": "SUM(METRICS('request'))",
      "ReturnData": false
    },
    {
      "Id": "errors",
      "Expression": "SUM(METRICS('error'))",
      "ReturnData": false
    },
    {
      "Id": "error1",
      "MetricStat": {
        "Metric": {
          "Namespace": "MyService",
          "MetricName": "BadRequests",
```

```
        "Dimensions": [
            {
                "Name": "Component",
                "Value": "component-1"
            }
        ],
    },
    "Period": 60,
    "Stat": "Sum"
},
"ReturnData": false
},
{
    "Id": "error2",
    "MetricStat": {
        "Metric": {
            "Namespace": "MyService",
            "MetricName": "ConnectionErrors",
            "Dimensions": [
                {
                    "Name": "Component",
                    "Value": "component-1"
                }
            ]
        },
        "Period": 60,
        "Stat": "Sum"
    },
    "ReturnData": false
},
{
    "Id": "request1",
    "MetricStat": {
        "Metric": {
            "Namespace": "MyService",
            "MetricName": "InternalRequests",
            "Dimensions": [
                {
                    "Name": "Component",
                    "Value": "component-1"
                }
            ]
        },
        "Period": 60,
```

```
    "Stat": "Sum"
  },
  "ReturnData": false
},
{
  "Id": "request2",
  "MetricStat": {
    "Metric": {
      "Namespace": "MyService",
      "MetricName": "ExternalRequests",
      "Dimensions": [
        {
          "Name": "Component",
          "Value": "component-1"
        }
      ]
    },
    "Period": 60,
    "Stat": "Sum"
  },
  "ReturnData": false
}
]
```

Sample Response

```
{
  "MetricDataResults": [
    {
      "Id": "errorRate",
      "Label": "Error Rate",
      "StatusCode": "Complete",
      "Timestamps": [
        1518868032,
        1518867732,
        1518867432
      ],
      "Values": [
        0.1,
        0.5,
        0.3
      ]
    }
  ]
}
```

```
    },  
    {  
      "Id": "errorRatePercent",  
      "Label": "% Error Rate",  
      "StatusCode": "Complete",  
      "Timestamps": [  
        1518868032,  
        1518867732,  
        1518867432  
      ],  
      "Values": [  
        10,  
        50,  
        30  
      ]  
    }  
  ]  
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetMetricStatistics

Gets statistics for the specified metric.

The maximum number of data points returned from a single call is 1,440. If you request more than 1,440 data points, CloudWatch returns an error. To reduce the number of data points, you can narrow the specified time range and make multiple requests across adjacent time ranges, or you can increase the specified period. Data points are not returned in chronological order.

CloudWatch aggregates data points based on the length of the period that you specify. For example, if you request statistics with a one-hour period, CloudWatch aggregates all data points with time stamps that fall within each one-hour period. Therefore, the number of values aggregated by CloudWatch is larger than the number of data points returned.

CloudWatch needs raw data points to calculate percentile statistics. If you publish data using a statistic set instead, you can only retrieve percentile statistics for this data if one of the following conditions is true:

- The SampleCount value of the statistic set is 1.
- The Min and the Max values of the statistic set are equal.

Percentile statistics are not available for metrics when any of the metric values are negative numbers.

Amazon CloudWatch retains metric data as follows:

- Data points with a period of less than 60 seconds are available for 3 hours. These data points are high-resolution metrics and are available only for custom metrics that have been defined with a StorageResolution of 1.
- Data points with a period of 60 seconds (1-minute) are available for 15 days.
- Data points with a period of 300 seconds (5-minute) are available for 63 days.
- Data points with a period of 3600 seconds (1 hour) are available for 455 days (15 months).

Data points that are initially published with a shorter period are aggregated together for long-term storage. For example, if you collect data using a period of 1 minute, the data remains available for 15 days with 1-minute resolution. After 15 days, this data is still available, but is aggregated and

retrievable only with a resolution of 5 minutes. After 63 days, the data is further aggregated and is available with a resolution of 1 hour.

CloudWatch started retaining 5-minute and 1-hour metric data as of July 9, 2016.

For information about metrics and dimensions supported by AWS services, see the [Amazon CloudWatch Metrics and Dimensions Reference](#) in the *Amazon CloudWatch User Guide*.

Request Parameters

Dimensions

The dimensions. If the metric contains multiple dimensions, you must include a value for each dimension. CloudWatch treats each unique combination of dimensions as a separate metric. If a specific combination of dimensions was not published, you can't retrieve statistics for it. You must specify the same dimensions that were used when the metrics were created. For an example, see [Dimension Combinations](#) in the *Amazon CloudWatch User Guide*. For more information about specifying dimensions, see [Publishing Metrics](#) in the *Amazon CloudWatch User Guide*.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

EndTime

The time stamp that determines the last data point to return.

The value specified is exclusive; results include data points up to the specified time stamp. In a raw HTTP query, the time stamp must be in ISO 8601 UTC format (for example, 2016-10-10T23:00:00Z).

Type: Timestamp

Required: Yes

ExtendedStatistics

The percentile statistics. Specify values between p0.0 and p100. When calling `GetMetricStatistics`, you must specify either `Statistics` or `ExtendedStatistics`, but

not both. Percentile statistics are not available for metrics when any of the metric values are negative numbers.

Type: Array of strings

Array Members: Minimum number of 1 item. Maximum number of 10 items.

Required: No

MetricName

The name of the metric, with or without spaces.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Namespace

The namespace of the metric, with or without spaces.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: Yes

Period

The granularity, in seconds, of the returned data points. For metrics with regular resolution, a period can be as short as one minute (60 seconds) and must be a multiple of 60. For high-resolution metrics that are collected at intervals of less than one minute, the period can be 1, 5, 10, 20, 30, 60, or any multiple of 60. High-resolution metrics are those metrics stored by a `PutMetricData` call that includes a `StorageResolution` of 1 second.

If the `StartTime` parameter specifies a time stamp that is greater than 3 hours ago, you must specify the period as follows or no data points in that time range is returned:

- Start time between 3 hours and 15 days ago - Use a multiple of 60 seconds (1 minute).
- Start time between 15 and 63 days ago - Use a multiple of 300 seconds (5 minutes).

- Start time greater than 63 days ago - Use a multiple of 3600 seconds (1 hour).

Type: Integer

Valid Range: Minimum value of 1.

Required: Yes

StartTime

The time stamp that determines the first data point to return. Start times are evaluated relative to the time that CloudWatch receives the request.

The value specified is inclusive; results include data points with the specified time stamp. In a raw HTTP query, the time stamp must be in ISO 8601 UTC format (for example, 2016-10-03T23:00:00Z).

CloudWatch rounds the specified time stamp as follows:

- Start time less than 15 days ago - Round down to the nearest whole minute. For example, 12:32:34 is rounded down to 12:32:00.
- Start time between 15 and 63 days ago - Round down to the nearest 5-minute clock interval. For example, 12:32:34 is rounded down to 12:30:00.
- Start time greater than 63 days ago - Round down to the nearest 1-hour clock interval. For example, 12:32:34 is rounded down to 12:00:00.

If you set `Period` to 5, 10, 20, or 30, the start time of your request is rounded down to the nearest time that corresponds to even 5-, 10-, 20-, or 30-second divisions of a minute. For example, if you make a query at (HH:mm:ss) 01:05:23 for the previous 10-second period, the start time of your request is rounded down and you receive data from 01:05:10 to 01:05:20. If you make a query at 15:07:17 for the previous 5 minutes of data, using a period of 5 seconds, you receive data timestamped between 15:02:15 and 15:07:15.

Type: Timestamp

Required: Yes

Statistics

The metric statistics, other than percentile. For percentile statistics, use `ExtendedStatistics`. When calling `GetMetricStatistics`, you must specify either `Statistics` or `ExtendedStatistics`, but not both.

Type: Array of strings

Array Members: Minimum number of 1 item. Maximum number of 5 items.

Valid Values: SampleCount | Average | Sum | Minimum | Maximum

Required: No

Unit

The unit for a given metric. If you omit Unit, all data that was collected with any unit is returned, along with the corresponding units that were specified when the data was reported to CloudWatch. If you specify a unit, the operation returns only data that was collected with that unit specified. If you specify a unit that does not match the data collected, the results of the operation are null. CloudWatch does not perform unit conversions.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

Response Elements

The following elements are returned by the service.

Datapoints

The data points for the specified metric.

Type: Array of [Datapoint](#) objects

Label

A label for the specified metric.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)

- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetMetricStream

Returns information about the metric stream that you specify.

Request Parameters

Name

The name of the metric stream to retrieve information about.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Response Elements

The following elements are returned by the service.

Arn

The ARN of the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

CreationDate

The date that the metric stream was created.

Type: Timestamp

ExcludeFilters

If this array of metric namespaces is present, then these namespaces are the only metric namespaces that are not streamed by this metric stream. In this case, all other metric namespaces in the account are streamed by this metric stream.

Type: Array of [MetricStreamFilter](#) objects

FirehoseArn

The ARN of the Amazon Kinesis Data Firehose delivery stream that is used by this metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

IncludeFilters

If this array of metric namespaces is present, then these namespaces are the only metric namespaces that are streamed by this metric stream.

Type: Array of [MetricStreamFilter](#) objects

IncludeLinkedAccountsMetrics

If this is `true` and this metric stream is in a monitoring account, then the stream includes metrics from source accounts that the monitoring account is linked to.

Type: Boolean

LastUpdateDate

The date of the most recent update to the metric stream's configuration.

Type: Timestamp

Name

The name of the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

OutputFormat

The output format for the stream. Valid values are `json`, `opentelemetry1.0`, and `opentelemetry0.7`. For more information about metric stream output formats, see [Metric streams output formats](#).

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Valid Values: `json` | `opentelemetry0.7` | `opentelemetry1.0`

RoleArn

The ARN of the IAM role that is used by this metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

State

The state of the metric stream. The possible values are `running` and `stopped`.

Type: String

StatisticsConfigurations

Each entry in this array displays information about one or more metrics that include additional statistics in the metric stream. For more information about the additional statistics, see [CloudWatch statistics definitions](#).

Type: Array of [MetricStreamStatisticsConfiguration](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetMetricWidgetImage

You can use the `GetMetricWidgetImage` API to retrieve a snapshot graph of one or more Amazon CloudWatch metrics as a bitmap image. You can then embed this image into your services and products, such as wiki pages, reports, and documents. You could also retrieve images regularly, such as every minute, and create your own custom live dashboard.

The graph you retrieve can include all CloudWatch metric graph features, including metric math and horizontal and vertical annotations.

There is a limit of 20 transactions per second for this API. Each `GetMetricWidgetImage` action has the following limits:

- As many as 100 metrics in the graph.
- Up to 100 KB uncompressed payload.

Request Parameters

MetricWidget

A JSON string that defines the bitmap graph to be retrieved. The string includes the metrics to include in the graph, statistics, annotations, title, axis limits, and so on. You can include only one `MetricWidget` parameter in each `GetMetricWidgetImage` call.

For more information about the syntax of `MetricWidget` see [GetMetricWidgetImage: Metric Widget Structure and Syntax](#).

If any metric on the graph could not load all the requested data points, an orange triangle with an exclamation point appears next to the graph legend.

Type: String

Required: Yes

OutputFormat

The format of the resulting image. Only PNG images are supported.

The default is `png`. If you specify `png`, the API returns an HTTP response with the content-type set to `text/xml`. The image data is in a `MetricWidgetImage` field. For example:

```
<GetMetricWidgetImageResponse xmlns=<URLstring>>

  <GetMetricWidgetImageResult>

    <MetricWidgetImage>

      iVBORw0KGgoAAAANSUhEUgAAAlgAAAGQEAYAAAAip...

    </MetricWidgetImage>

  </GetMetricWidgetImageResult>

  <ResponseMetadata>

    <RequestId>6f0d4192-4d42-11e8-82c1-f539a07e0e3b</RequestId>

  </ResponseMetadata>

</GetMetricWidgetImageResponse>
```

The `image/png` setting is intended only for custom HTTP requests. For most use cases, and all actions using an AWS SDK, you should use `png`. If you specify `image/png`, the HTTP response has a content-type set to `image/png`, and the body of the response is a PNG image.

Type: String

Required: No

Response Elements

The following element is returned by the service.

MetricWidgetImage

The image of the graph, in the output format specified. The output is base64-encoded.

Type: Base64-encoded binary data object

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

Examples

Example

The following is an example of a `GetMetricWidgetImage` call. This example displays a graph showing an image of the Average statistic for the `CPUUtilization` metric for two Amazon EC2 instances, with both horizontal and vertical annotations.

```
{
  "OutputFormat": "png",
  "MetricWidget": "{\\\"width\\\":600,\\\"height\\\":395,\\\"metrics\\\":[[\\\"AWS/EC2\\\",
\\\"CPUUtilization\\\",\\\"InstanceId\\\",\\\"i-1234567890abcdef0\\\",{\\\"stat\\\":\\\"Average\\\"}],
[\\\"AWS/EC2\\\",\\\"CPUUtilization\\\",\\\"InstanceId\\\",\\\"i-0987654321abcdef0\\\",{\\\"stat\\\":
\\\"Average\\\"}]]],\\\"period\\\":300,\\\"start\\\":\\\"-P30D\\\",\\\"end\\\":\\\"PT0H\\\",\\\"stacked\\\":false,
\\\"yAxis\\\":{\\\"left\\\":{\\\"min\\\":0.1,\\\"max\\\":1},\\\"right\\\":{\\\"min\\\":0}},\\\"title\\\":\\\"CPU
for Two Instances\\\",\\\"annotations\\\":{\\\"horizontal\\\":[{\\\"color\\\":\\\"#ff6961\\\",\\\"label
\\\":\\\"Trouble threshold start\\\",\\\"fill\\\":\\\"above\\\",\\\"value\\\":0.5}],\\\"vertical\\\":
[{\\\"visible\\\":true,\\\"color\\\":\\\"#9467bd\\\",\\\"label\\\":\\\"Bug fix deployed\\\",\\\"value\\\":
\\\"2018-08-28T15:25:26Z\\\",\\\"fill\\\":\\\"after\\\"}]}}}"
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetOTelEnrichment

Returns the current status of vended metric enrichment for the account, including whether CloudWatch vended metrics are enriched with resource ARN and resource tag labels and queryable using PromQL. For the list of supported resources, see [Supported AWS infrastructure metrics](#).

Response Elements

The following element is returned by the service.

Status

The status of OTel enrichment for the account. Valid values are Running (enrichment is enabled) and Stopped (enrichment is disabled).

Type: String

Valid Values: Running | Stopped

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)

- [AWS SDK for Ruby V3](#)

ListAlarmMuteRules

Lists alarm mute rules in your AWS account and region.

You can filter the results by alarm name to find all mute rules targeting a specific alarm, or by status to find rules that are scheduled, active, or expired.

This operation supports pagination for accounts with many mute rules. Use the `MaxRecords` and `NextToken` parameters to retrieve results in multiple calls.

Permissions

To list mute rules, you need the `cloudwatch:ListAlarmMuteRules` permission.

Request Parameters

AlarmName

Filter results to show only mute rules that target the specified alarm name.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

MaxRecords

The maximum number of mute rules to return in one call. The default is 50.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 100.

Required: No

NextToken

The token returned from a previous call to indicate where to continue retrieving results.

Type: String

Required: No

Statuses

Filter results to show only mute rules with the specified statuses. Valid values are SCHEDULED, ACTIVE, or EXPIRED.

Type: Array of strings

Valid Values: SCHEDULED | ACTIVE | EXPIRED

Required: No

Response Elements

The following elements are returned by the service.

AlarmMuteRuleSummaries

A list of alarm mute rule summaries.

Type: Array of [AlarmMuteRuleSummary](#) objects

NextToken

The token to use when requesting the next set of results. If this field is absent, there are no more results to retrieve.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

Examples

List all mute rules

List all alarm mute rules in your account.

Sample Request

```
aws cloudwatch list-alarm-mute-rules
```

Sample Response

```
{
  "AlarmMuteRuleSummaries": [
    {
      "Name": "DailyMaintenanceWindow",
      "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-east-1:123456789012:alarm-mute-rule:DailyMaintenanceWindow",
      "Status": "SCHEDULED",
      "MuteType": "RECURRING",
      "LastUpdatedTimestamp": "2026-01-15T10:30:00Z"
    },
    {
      "Name": "ProductionDeployment-2026-01-20",
      "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-east-1:123456789012:alarm-mute-rule:ProductionDeployment-2026-01-20",
      "Status": "ACTIVE",
      "MuteType": "ONE_TIME",
      "LastUpdatedTimestamp": "2026-01-20T13:00:00Z"
    },
    {
      "Name": "WeeklyBackupWindow",
      "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-west-2:123456789012:alarm-mute-rule:WeeklyBackupWindow",
      "Status": "SCHEDULED",
      "MuteType": "RECURRING",

```



```
        "ExpireDate": "2026-12-31T23:59:59Z",
        "LastUpdatedTimestamp": "2026-01-05T12:00:00Z"
    }
  ]
}
```

List mute rules targeting a specific alarm

List all mute rules that target a specific alarm.

Sample Request

```
aws cloudwatch list-alarm-mute-rules \
  --alarm-name "WebServerCPULAlarm"
```

Sample Response

```
{
  "AlarmMuteRuleSummaries": [
    {
      "Name": "DailyMaintenanceWindow",
      "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-east-1:123456789012:alarm-mute-rule:DailyMaintenanceWindow",
      "Status": "SCHEDULED",
      "MuteType": "RECURRING",
      "LastUpdatedTimestamp": "2026-01-15T10:30:00Z"
    },
    {
      "Name": "EmergencyMuteRule",
      "AlarmMuteRuleArn": "arn:aws:cloudwatch:us-east-1:123456789012:alarm-mute-rule:EmergencyMuteRule",
      "Status": "ACTIVE",
      "MuteType": "ONE_TIME",
      "LastUpdatedTimestamp": "2026-01-21T14:00:00Z"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListDashboards

Returns a list of the dashboards for your account. If you include `DashboardNamePrefix`, only those dashboards with names starting with the prefix are listed. Otherwise, all dashboards in your account are listed.

`ListDashboards` returns up to 1000 results on one page. If there are more than 1000 dashboards, you can call `ListDashboards` again and include the value you received for `NextToken` in the first call, to receive the next 1000 results.

Request Parameters

DashboardNamePrefix

If you specify this parameter, only the dashboards with names starting with the specified string are listed. The maximum length is 255, and valid characters are A-Z, a-z, 0-9, ".", "-", and "_".

Type: String

Required: No

NextToken

The token returned by a previous call to indicate that there is more data available.

Type: String

Required: No

Response Elements

The following elements are returned by the service.

DashboardEntries

The list of matching dashboards.

Type: Array of [DashboardEntry](#) objects

NextToken

The token that marks the start of the next batch of returned results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListManagedInsightRules

Returns a list that contains the number of managed Contributor Insights rules in your account.

Request Parameters

MaxResults

The maximum number of results to return in one operation. If you omit this parameter, the default number is used. The default number is 100.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 500.

Required: No

NextToken

Include this value to get the next set of rules if the value was returned by the previous operation.

Type: String

Required: No

ResourceARN

The ARN of an AWS resource that has managed Contributor Insights rules.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

Response Elements

The following elements are returned by the service.

ManagedRules

The managed rules that are available for the specified AWS resource.

Type: Array of [ManagedRuleDescription](#) objects

NextToken

Include this value to get the next set of rules if the value was returned by the previous operation.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)

- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListMetrics

List the specified metrics. You can use the returned metrics with [GetMetricData](#) or [GetMetricStatistics](#) to get statistical data.

Up to 500 results are returned for any one call. To retrieve additional results, use the returned token with subsequent calls.

After you create a metric, allow up to 15 minutes for the metric to appear. To see metric statistics sooner, use [GetMetricData](#) or [GetMetricStatistics](#).

If you are using CloudWatch cross-account observability, you can use this operation in a monitoring account and view metrics from the linked source accounts. For more information, see [CloudWatch cross-account observability](#).

ListMetrics doesn't return information about metrics if those metrics haven't reported data in the past two weeks. To retrieve those metrics, use [GetMetricData](#) or [GetMetricStatistics](#).

Request Parameters

Dimensions

The dimensions to filter against. Only the dimension with names that match exactly will be returned. If you specify one dimension name and a metric has that dimension and also other dimensions, it will be returned.

Type: Array of [DimensionFilter](#) objects

Array Members: Maximum number of 10 items.

Required: No

IncludeLinkedAccounts

If you are using this operation in a monitoring account, specify `true` to include metrics from source accounts in the returned data.

The default is `false`.

Type: Boolean

Required: No

MetricName

The name of the metric to filter against. Only the metrics with names that match exactly will be returned.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

The metric namespace to filter against. Only the namespace that matches exactly will be returned.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: No

NextToken

The token returned by a previous call to indicate that there is more data available.

Type: String

Required: No

OwningAccount

When you use this operation in a monitoring account, use this field to return metrics only from one source account. To do so, specify that source account ID in this field, and also specify `true` for `IncludeLinkedAccounts`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

RecentlyActive

To filter the results to show only metrics that have had data points published in the past three hours, specify this parameter with a value of PT3H. This is the only valid value for this parameter.

The results that are returned are an approximation of the value you specify. There is a low probability that the returned results include metrics with last published data as much as 50 minutes more than the specified time interval.

Type: String

Valid Values: PT3H

Required: No

Response Elements

The following elements are returned by the service.

Metrics

The metrics that match your request.

Type: Array of [Metric](#) objects

NextToken

The token that marks the start of the next batch of returned results.

Type: String

OwningAccounts

If you are using this operation in a monitoring account, this array contains the account IDs of the source accounts where the metrics in the returned data are from.

This field is a 1:1 mapping between each metric that is returned and the ID of the owning account.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 255.

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

Examples

List metrics in a specified namespace from all source accounts and from the monitoring account

The following example lists metrics in the AWS/EC2 namespace from the monitoring account itself and all source accounts.

Sample Request

```
{
  "IncludeLinkedAccounts": true,
  "Namespace" : "AWS/EC2"
}
```

List metrics from a namespace in just one source account

The following example lists metrics in the AWS/EC2 namespace from only the source account with the ID 111111111111.

Sample Request

```
{
  "IncludeLinkedAccounts": "true",
  "OwningAccount" : "111111111111",
  "Namespace" : "AWS/EC2"
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListMetricStreams

Returns a list of metric streams in this account.

Request Parameters

MaxResults

The maximum number of results to return in one operation.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 500.

Required: No

NextToken

Include this value, if it was returned by the previous call, to get the next set of metric streams.

Type: String

Required: No

Response Elements

The following elements are returned by the service.

Entries

The array of metric stream information.

Type: Array of [MetricStreamEntry](#) objects

NextToken

The token that marks the start of the next batch of returned results. You can use this token in a subsequent operation to get the next batch of results.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidNextToken

The next token specified is invalid.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListTagsForResource

Displays the tags associated with a CloudWatch resource. Currently, alarms, dashboards, metric streams and Contributor Insights rules support tagging.

Request Parameters

ResourceARN

The ARN of the CloudWatch resource that you want to view tags for.

The ARN format of an alarm is `arn:aws:cloudwatch:Region:account-id:alarm:alarm-name`

The ARN format of a Contributor Insights rule is `arn:aws:cloudwatch:Region:account-id:insight-rule/insight-rule-name`

The ARN format of a dashboard is `arn:aws:cloudwatch::account-id:dashboard/dashboard-name`

The ARN format of a metric stream is `arn:aws:cloudwatch:Region:account-id:metric-stream/metric-stream-name`

For more information about ARN format, see [Resource Types Defined by Amazon CloudWatch](#) in the *Amazon Web Services General Reference*.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

Response Elements

The following element is returned by the service.

Tags

The list of tag keys and values associated with the resource you specified.

Type: Array of [Tag](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutAlarmMuteRule

Creates or updates an alarm mute rule.

Alarm mute rules automatically mute alarm actions during predefined time windows. When a mute rule is active, targeted alarms continue to evaluate metrics and transition between states, but their configured actions (such as Amazon SNS notifications or Auto Scaling actions) are muted.

You can create mute rules with recurring schedules using `cron` expressions or one-time mute windows using `at` expressions. Each mute rule can target up to 100 specific alarms by name.

If you specify a rule name that already exists, this operation updates the existing rule with the new configuration.

Permissions

To create or update a mute rule, you must have the `cloudwatch:PutAlarmMuteRule` permission on two types of resources: the alarm mute rule resource itself, and each alarm that the rule targets.

For example, If you want to allow a user to create mute rules that target only specific alarms named "WebServerCPULAlarm" and "DatabaseConnectionAlarm", you would create an IAM policy with one statement granting `cloudwatch:PutAlarmMuteRule` on the alarm mute rule resource (`arn:aws:cloudwatch:[REGION]:123456789012:alarm-mute-rule:*`), and another statement granting `cloudwatch:PutAlarmMuteRule` on the targeted alarm resources (`arn:aws:cloudwatch:[REGION]:123456789012:alarm:WebServerCPULAlarm` and `arn:aws:cloudwatch:[REGION]:123456789012:alarm:DatabaseConnectionAlarm`).

You can also use IAM policy conditions to allow targeting alarms based on resource tags. For example, you can restrict users to create/update mute rules to only target alarms that have a specific tag key-value pair, such as `Team=TeamA`.

Request Parameters

Description

A description of the alarm mute rule that helps you identify its purpose.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

ExpireDate

The date and time when the mute rule expires and is no longer evaluated, specified as a timestamp in ISO 8601 format (for example, 2026-12-31T23:59:59Z). After this time, the rule status becomes EXPIRED and will no longer mute the targeted alarms.

Type: Timestamp

Required: No

MuteTargets

Specifies which alarms this rule applies to.

Type: [MuteTargets](#) object

Required: No

Name

The name of the alarm mute rule. This name must be unique within your AWS account and region.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Rule

The configuration that defines when and how long alarms should be muted.

Type: [Rule](#) object

Required: Yes

StartDate

The date and time after which the mute rule takes effect, specified as a timestamp in ISO 8601 format (for example, 2026-04-15T08:00:00Z). If not specified, the mute rule takes effect immediately upon creation and the mutes are applied as per the schedule expression.

Type: Timestamp

Required: No

Tags

A list of key-value pairs to associate with the alarm mute rule. You can use tags to categorize and manage your mute rules.

Type: Array of [Tag](#) objects

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

LimitExceeded

The quota for alarms for this customer has already been reached.

message

HTTP Status Code: 400

Examples

Create a recurring daily mute rule

Create a mute rule that mutes specific alarms every day from 2:00 AM to 4:00 AM UTC.

Sample Request

```
aws cloudwatch put-alarm-mute-rule \  
  --name "DailyMaintenanceWindow" \  
  --description "Mute alarms during daily maintenance" \  
  --rule '{  
    "Schedule": {  
      "Expression": "cron(0 2 * * ?)",  
      "Duration": "PT2H",  
      "Timezone": "UTC"  
    }  
  }
```

```
}' \  
--mute-targets '{  
    "AlarmNames": ["WebServerCPULAlarm", "DatabaseConnectionAlarm"]  
}'
```

Create a one-time mute rule

Create a mute rule for a one-time deployment window.

Sample Request

```
aws cloudwatch put-alarm-mute-rule \  
--name "ProductionDeployment-2026-01-20" \  
--description "Mute alarms during production deployment" \  
--rule '{  
    "Schedule": {  
        "Expression": "at(2026-01-20T14:00)",  
        "Duration": "PT1H",  
        "Timezone": "America/New_York"  
    }  
}' \  
--mute-targets '{  
    "AlarmNames": ["APILatencyAlarm", "ErrorRateAlarm"]  
}'
```

Create a weekly mute rule with tags

Create a mute rule that mutes specific alarms every Saturday for 4 hours.

Sample Request

```
aws cloudwatch put-alarm-mute-rule \  
--name "WeeklyBackupWindow" \  
--description "Mute alarms during weekly backup" \  
--rule '{  
    "Schedule": {  
        "Expression": "cron(0 0 ? * SAT)",  
        "Duration": "PT4H",
```

```
        "Timezone": "America/Los_Angeles"
    }
}' \
--mute-targets '{
    "AlarmNames": ["BackupAlarm", "StorageAlarm"]
}' \
--tags '[
    {"Key": "Environment", "Value": "Production"},
    {"Key": "Team", "Value": "Operations"}
]'
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutAnomalyDetector

Creates an anomaly detection model for a CloudWatch metric. You can use the model to display a band of expected normal values when the metric is graphed.

If you have enabled unified cross-account observability, and this account is a monitoring account, the metric can be in the same account or a source account. You can specify the account ID in the object you specify in the `SingleMetricAnomalyDetector` parameter.

For more information, see [CloudWatch Anomaly Detection](#).

Request Parameters

Configuration

The configuration specifies details about how the anomaly detection model is to be trained, including time ranges to exclude when training and updating the model. You can specify as many as 10 time ranges.

The configuration can also include the time zone to use for the metric.

Type: [AnomalyDetectorConfiguration](#) object

Required: No

Dimensions

This parameter has been deprecated.

The metric dimensions to create the anomaly detection model for.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MetricCharacteristics

Use this object to include parameters to provide information about your metric to CloudWatch to help it build more accurate anomaly detection models. Currently, it includes the `PeriodicSpikes` parameter.

Type: [MetricCharacteristics](#) object

Required: No

MetricMathAnomalyDetector

The metric math anomaly detector to be created.

When using `MetricMathAnomalyDetector`, you cannot include the following parameters in the same operation:

- `Dimensions`
- `MetricName`
- `Namespace`
- `Stat`
- the `SingleMetricAnomalyDetector` parameters of `PutAnomalyDetectorInput`

Instead, specify the metric math anomaly detector attributes as part of the property `MetricMathAnomalyDetector`.

Type: [MetricMathAnomalyDetector](#) object

Required: No

MetricName

This parameter has been deprecated.

The name of the metric to create the anomaly detection model for.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

This parameter has been deprecated.

The namespace of the metric to create the anomaly detection model for.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `^[^:]*`

Required: No

SingleMetricAnomalyDetector

A single metric anomaly detector to be created.

When using `SingleMetricAnomalyDetector`, you cannot include the following parameters in the same operation:

- `Dimensions`
- `MetricName`
- `Namespace`
- `Stat`
- the `MetricMathAnomalyDetector` parameters of `PutAnomalyDetectorInput`

Instead, specify the single metric anomaly detector attributes as part of the property `SingleMetricAnomalyDetector`.

Type: [SingleMetricAnomalyDetector](#) object

Required: No

Stat

This parameter has been deprecated.

The statistic to use for the metric and the anomaly detection model.

Type: String

Length Constraints: Maximum length of 50.

Pattern: `(SampleCount|Average|Sum|Minimum|Maximum|IQM|(p|tc|tm|ts|wm)(\d{1,2}(\.\d{0,10})?|100)|[ou]\d+(\.\d*)?)(_E|_L|_H)?|(TM|TC|TS|WM)\((((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%|((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):(((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%))?)\)| (TM|TC|TS|WM|PR)\((((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)):(\d+(\.\d{0,10})?|(\d`

```
+((\.\d{0,10})?[Ee][+-]?\d+))?)|((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))?:((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))\)
```

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

LimitExceededException

The operation exceeded one or more limits.

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutCompositeAlarm

Creates or updates a *composite alarm*. When you create a composite alarm, you specify a rule expression for the alarm that takes into account the alarm states of other alarms that you have created. The composite alarm goes into ALARM state only if all conditions of the rule are met.

The alarms specified in a composite alarm's rule expression can include metric alarms and other composite alarms. The rule expression of a composite alarm can include as many as 100 underlying alarms. Any single alarm can be included in the rule expressions of as many as 150 composite alarms.

Using composite alarms can reduce alarm noise. You can create multiple metric alarms, and also create a composite alarm and set up alerts only for the composite alarm. For example, you could create a composite alarm that goes into ALARM state only when more than one of the underlying metric alarms are in ALARM state.

Composite alarms can take the following actions:

- Notify Amazon SNS topics.
- Invoke Lambda functions.
- Create OpsItems in Systems Manager Ops Center.
- Create incidents in Systems Manager Incident Manager.

Note

It is possible to create a loop or cycle of composite alarms, where composite alarm A depends on composite alarm B, and composite alarm B also depends on composite alarm A. In this scenario, you can't delete any composite alarm that is part of the cycle because there is always still a composite alarm that depends on that alarm that you want to delete. To get out of such a situation, you must break the cycle by changing the rule of one of the composite alarms in the cycle to remove a dependency that creates the cycle. The simplest change to make to break a cycle is to change the `AlarmRule` of one of the alarms to `false`.

Additionally, the evaluation of composite alarms stops if CloudWatch detects a cycle in the evaluation path.

When this operation creates an alarm, the alarm state is immediately set to `INSUFFICIENT_DATA`. The alarm is then evaluated and its state is set appropriately. Any actions associated with the new state are then executed. For a composite alarm, this initial time after creation is the only time that the alarm can be in `INSUFFICIENT_DATA` state.

When you update an existing alarm, its state is left unchanged, but the update completely overwrites the previous configuration of the alarm.

To use this operation, you must be signed on with the `cloudwatch:PutCompositeAlarm` permission that is scoped to `*`. You can't create a composite alarms if your `cloudwatch:PutCompositeAlarm` permission has a narrower scope.

If you are an IAM user, you must have `iam:CreateServiceLinkedRole` to create a composite alarm that has Systems Manager OpsItem actions.

Request Parameters

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state of the composite alarm. The default is `TRUE`.

Type: Boolean

Required: No

ActionsSuppressor

Actions will be suppressed if the suppressor alarm is in the `ALARM` state. `ActionsSuppressor` can be an `AlarmName` or an Amazon Resource Name (ARN) from an existing alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

ActionsSuppressorExtensionPeriod

The maximum time in seconds that the composite alarm waits after suppressor alarm goes out of the `ALARM` state. After this time, the composite alarm performs its actions.

⚠ Important

ExtensionPeriod is required only when ActionsSuppressor is specified.

Type: Integer

Required: No

ActionsSuppressorWaitPeriod

The maximum time in seconds that the composite alarm waits for the suppressor alarm to go into the ALARM state. After this time, the composite alarm performs its actions.

⚠ Important

WaitPeriod is required only when ActionsSuppressor is specified.

Type: Integer

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:]

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Systems Manager actions:

`arn:aws:ssm:region:account-id:opsitem:severity`

Start a Amazon Q Developer operational investigation

`arn:aws:aiops:region:account-id:investigation-group:investigation-group-id`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmDescription

The description for the composite alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name for the composite alarm. This name must be unique within the Region.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

AlarmRule

An expression that specifies which other alarms are to be evaluated to determine this composite alarm's state. For each alarm that you reference, you designate a function that specifies whether that alarm needs to be in ALARM state, OK state, or INSUFFICIENT_DATA state. You can use operators (AND, OR and NOT) to combine multiple functions in a single expression. You can use parenthesis to logically group the functions in your expression.

You can use either alarm names or ARNs to reference the other alarms that are to be evaluated.

Functions can include the following:

- `ALARM("alarm-name or alarm-ARN")` is TRUE if the named alarm is in ALARM state.
- `OK("alarm-name or alarm-ARN")` is TRUE if the named alarm is in OK state.
- `INSUFFICIENT_DATA("alarm-name or alarm-ARN")` is TRUE if the named alarm is in INSUFFICIENT_DATA state.
- TRUE always evaluates to TRUE.
- FALSE always evaluates to FALSE.

TRUE and FALSE are useful for testing a complex `AlarmRule` structure, and for testing your alarm actions.

Alarm names specified in `AlarmRule` can be surrounded with double-quotes ("), but do not have to be.

The following are some examples of `AlarmRule`:

- `ALARM(CPUUtilizationTooHigh) AND ALARM(DiskReadOpsTooHigh)` specifies that the composite alarm goes into ALARM state only if both `CPUUtilizationTooHigh` and `DiskReadOpsTooHigh` alarms are in ALARM state.
- `ALARM(CPUUtilizationTooHigh) AND NOT ALARM(DeploymentInProgress)` specifies that the alarm goes to ALARM state if `CPUUtilizationTooHigh` is in ALARM state and `DeploymentInProgress` is not in ALARM state. This example reduces alarm noise during a known deployment window.
- `(ALARM(CPUUtilizationTooHigh) OR ALARM(DiskReadOpsTooHigh)) AND OK(NetworkOutTooHigh)` goes into ALARM state if `CPUUtilizationTooHigh` OR `DiskReadOpsTooHigh` is in ALARM state, and if `NetworkOutTooHigh` is in OK state. This provides another example of using a composite alarm to prevent noise. This rule ensures that you are not notified with an alarm action on high CPU or disk usage if a known network problem is also occurring.

The `AlarmRule` can specify as many as 100 "children" alarms. The `AlarmRule` expression can have as many as 500 elements. Elements are child alarms, TRUE or FALSE statements, and parentheses.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 10240.

Required: Yes

InsufficientDataActions

The actions to execute when this alarm transitions to the `INSUFFICIENT_DATA` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:]

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

OKActions

The actions to execute when this alarm transitions to an `OK` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:]

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`

- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

Tags

A list of key-value pairs to associate with the alarm. You can associate as many as 50 tags with an alarm. To be able to associate tags with the alarm when you create the alarm, you must have the `cloudwatch:TagResource` permission.

Tags can help you organize and categorize your resources. You can also use them to scope user permissions by granting a user permission to access or change only resources with certain tag values.

If you are using this operation to update an existing alarm, any tags you specify in this parameter are ignored. To change the tags of an existing alarm, use [TagResource](#) or [UntagResource](#).

Type: Array of [Tag](#) objects

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

LimitExceeded

The quota for alarms for this customer has already been reached.

message

HTTP Status Code: 400

Examples

Composite alarm example

The following example creates an alarm that notifies an SNS group when either of two specified metric alarms exceeds its threshold.

Sample Request

```
{
  "AlarmDescription": "The host is experiencing problems",
  "AlarmRule": "ALARM(CPUUtilizationTooHigh) OR ALARM(DiskReadOpsTooHigh)",
  "AlarmName": "overall-health-alarm",
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutDashboard

Creates a dashboard if it does not already exist, or updates an existing dashboard. If you update a dashboard, the entire contents are replaced with what you specify here.

All dashboards in your account are global, not region-specific.

A simple way to create a dashboard using PutDashboard is to copy an existing dashboard. To copy an existing dashboard using the console, you can load the dashboard and then use the View/edit source command in the Actions menu to display the JSON block for that dashboard. Another way to copy a dashboard is to use GetDashboard, and then use the data returned within DashboardBody as the template for the new dashboard when you call PutDashboard.

When you create a dashboard with PutDashboard, a good practice is to add a text widget at the top of the dashboard with a message that the dashboard was created by script and should not be changed in the console. This message could also point console users to the location of the DashboardBody script or the CloudFormation template used to create the dashboard.

Request Parameters

DashboardBody

The detailed information about the dashboard in JSON format, including the widgets to include and their location on the dashboard. This parameter is required.

For more information about the syntax, see [Dashboard Body Structure and Syntax](#).

Type: String

Required: Yes

DashboardName

The name of the dashboard. If a dashboard with this name already exists, this call modifies that dashboard, replacing its current contents. Otherwise, a new dashboard is created. The maximum length is 255, and valid characters are A-Z, a-z, 0-9, "-", and "_". This parameter is required.

Type: String

Required: Yes

Tags

A list of key-value pairs to associate with the dashboard. You can associate as many as 50 tags with a dashboard.

Tags can help you organize and categorize your dashboards. You can also use them to scope user permissions by granting a user permission to access or change only dashboards with certain tag values.

You can use this parameter only when creating a new dashboard. If you specify Tags when updating an existing dashboard, the tag updates are ignored. To add or update tags on an existing dashboard, use [TagResource](#). To remove tags, use [UntagResource](#).

Type: Array of [Tag](#) objects

Required: No

Response Elements

The following element is returned by the service.

DashboardValidationMessages

If the input for PutDashboard was correct and the dashboard was successfully created or modified, this result is empty.

If this result includes only warning messages, then the input was valid enough for the dashboard to be created or modified, but some elements of the dashboard might not render.

If this result includes error messages, the input was not valid and the operation failed.

Type: Array of [DashboardValidationMessage](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterInput

Some part of the dashboard data is invalid.

HTTP Status Code: 400

Examples

Example

The following example creates a dashboard with just one text widget.

```
{
  "DashboardName": "Dashboard-with-only-one-text-widget",
  "DashboardBody": {
    "widgets": [
      {
        "type": "text",
        "x": 0,
        "y": 7,
        "width": 3,
        "height": 3,
        "properties": {
          "markdown": "Hello world"
        }
      }
    ]
  }
}
```

Example

The following example modifies an existing dashboard to include one metric widget and one text widget.


```
{
  "DashboardName": "Two-Widget-Dashboard",
  "DashboardBody": {
    "widgets": [
      {
        "type": "metric",
        "x": 0,
        "y": 0,
        "width": 12,
        "height": 6,
        "properties": {
          "metrics": [
            [
              "AWS/EC2",
              "CPUUtilization",
              "InstanceId",
              "i-012345"
            ]
          ],
          "period": 300,
          "stat": "Average",
          "region": "us-east-1",
          "title": "EC2 Instance CPU"
        }
      },
      {
        "type": "text",
        "x": 0,
        "y": 7,
        "width": 3,
        "height": 3,
        "properties": {
          "markdown": "Hello world"
        }
      }
    ]
  }
}
```

Example

The following example creates a dashboard with two metric widgets, side by side.

```
{
  "DashboardName": "Two-metric-widget-Dashboard",
  "DashboardBody": {
    "widgets": [
      {
        "type": "metric",
        "x": 0,
        "y": 0,
        "width": 12,
        "height": 6,
        "properties": {
          "metrics": [
            [
              "AWS/EC2",
              "CPUUtilization",
              "InstanceId",
              "i-012345"
            ]
          ],
          "period": 300,
          "stat": "Average",
          "region": "us-east-1",
          "title": "EC2 Instance CPU"
        }
      },
      {
        "type": "metric",
        "x": 12,
        "y": 0,
        "width": 12,
        "height": 6,
        "properties": {
          "metrics": [
            [
              "AWS/S3",
              "BucketSizeBytes",
              "BucketName",
              "amzn-s3-demo-bucket"
            ]
          ],
          "period": 86400,
          "stat": "Maximum",
          "region": "us-east-1",
```

```

        "title": "amzn-s3-demo-bucket bytes"
      }
    ]
  }
}

```

Example

The following example creates a dashboard with one widget at the top that shows the DiskReadBytes metric for three EC2 instances on one graph, and a separate widget below that, with an alarm.

```

{
  "DashboardName": "Dashboard-with-three-metric-graph-and-alarm",
  "DashboardBody": {
    "widgets": [
      {
        "type": "metric",
        "x": 0,
        "y": 0,
        "width": 12,
        "height": 6,
        "properties": {
          "metrics": [
            [
              "AWS/EC2",
              "DiskReadBytes",
              "InstanceId",
              "i-xyz"
            ],
            [
              ".",
              ".",
              ".",
              "i-abc"
            ],
            [
              ".",
              ".",
              ".",
              "i-123"
            ]
          ]
        }
      }
    ]
  }
}

```

```

        ],
        "period": 300,
        "stat": "Average",
        "region": "us-east-1",
        "title": "EC2 Instance CPU"
    }
},
{
    "type": "metric",
    "x": 0,
    "y": 7,
    "width": 12,
    "height": 12,
    "properties": {
        "annotations": {
            "alarms": [
                "arn:aws:cloudwatch:us-east-1:123456789012:alarm:myalarm"
            ]
        },
        "period": 60,
        "title": "MyAlarm"
    }
}
]
}
}

```

Example

The following example creates a dashboard with one metric widget and one metric math widget.

```

{
  "DashboardName": "One-metric-math-widget-and-One-metric-widget",
  "DashboardBody": {
    "widgets": [
      {
        "type": "metric",
        "x": 0,
        "y": 0,
        "width": 6,
        "height": 6,
        "properties": {
          "metrics": [

```

```
        [
            "AWS/EC2",
            "CPUUtilization",
            "InstanceId",
            "i-012345"
        ]
    ],
    "region": "us-east-1",
    "stat": "Average",
    "period": 300,
    "title": "EC2 Instance CPU"
}
},
{
    "type": "metric",
    "x": 6,
    "y": 0,
    "width": 6,
    "height": 6,
    "properties": {
        "metrics": [
            [
                {
                    "expression": "SUM(METRICS())",
                    "label": "Expression1",
                    "id": "e1",
                    "visible": true
                }
            ],
            [
                "AWS/EC2",
                "CPUUtilization",
                "InstanceId",
                "i-xyz",
                {
                    "id": "m1",
                    "visible": true
                }
            ],
            [
                "...",
                "i-abc",
                {
                    "id": "m2",
```

```
        "visible":true
      }
    ],
    [
      "...",
      "i-123",
      {
        "id":"m3",
        "visible":true
      }
    ],
    [
      "...",
      "i-456",
      {
        "id":"m4",
        "visible":true
      }
    ]
  ],
  "region":"us-east-1",
  "stat":"Average",
  "period":300,
  "title":"Sum of CPUUtilization of four Instances"
}
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)

- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutInsightRule

Creates a Contributor Insights rule. Rules evaluate log events in a CloudWatch Logs log group, enabling you to find contributor data for the log events in that log group. For more information, see [Using Contributor Insights to Analyze High-Cardinality Data](#).

If you create a rule, delete it, and then re-create it with the same name, historical data from the first time the rule was created might not be available.

Request Parameters

ApplyOnTransformedLogs

Specify `true` to have this rule evaluate log events after they have been transformed by [Log transformation](#). If you specify `true`, then the log events in log groups that have transformers will be evaluated by Contributor Insights after being transformed. Log groups that don't have transformers will still have their original log events evaluated by Contributor Insights.

The default is `false`

Note

If a log group has a transformer, and transformation fails for some log events, those log events won't be evaluated by Contributor Insights. For information about investigating log transformation failures, see [Transformation metrics and errors](#).

Type: Boolean

Required: No

RuleDefinition

The definition of the rule, as a JSON object. For details on the valid syntax, see [Contributor Insights Rule Syntax](#).

Type: String

Length Constraints: Minimum length of 1. Maximum length of 8192.

Pattern: `[\x00-\x7F]+`

Required: Yes

RuleName

A unique name for the rule.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

RuleState

The state of the rule. Valid values are ENABLED and DISABLED.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 32.

Pattern: `[\x20-\x7E]+`

Required: No

Tags

A list of key-value pairs to associate with the Contributor Insights rule. You can associate as many as 50 tags with a rule.

Tags can help you organize and categorize your resources. You can also use them to scope user permissions, by granting a user permission to access or change only the resources that have certain tag values.

To be able to associate tags with a rule, you must have the `cloudwatch:TagResource` permission in addition to the `cloudwatch:PutInsightRule` permission.

If you are using this operation to update an existing Contributor Insights rule, any tags you specify in this parameter are ignored. To change the tags of an existing rule, use [TagResource](#).

Type: Array of [Tag](#) objects

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

LimitExceededException

The operation exceeded one or more limits.

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutLogAlarm

Creates or updates a log alarm. A log alarm evaluates the results of a CloudWatch Logs scheduled query against the configured threshold and comparison operator to determine its state.

When you create a log alarm, the operation creates a service-managed CloudWatch Logs scheduled query that runs the query string you provide on the schedule you configure. Each scheduled query execution returns one or more aggregated values determined by the `AggregationExpression`, and each aggregated value is compared against the alarm `Threshold` to determine the alarm state. The alarm uses M-out-of-N evaluation: if `QueryResultsToAlarm` out of the most recent `QueryResultsToEvaluate` query results breach the threshold, the alarm transitions to ALARM.

Log alarms support the alarm states (OK, ALARM, INSUFFICIENT_DATA). Configure transition actions using `OKActions`, `AlarmActions`, and `InsufficientDataActions`.

If you call this operation with the name of an existing log alarm, the operation replaces the previous configuration of that alarm.

Permissions

To create or update a log alarm, you must have the `cloudwatch:PutLogAlarm` permission. The IAM role specified in `ScheduledQueryRoleARN` must grant the CloudWatch Alarms service permission to execute scheduled queries on the specified log groups. If you set `ActionLogLineCount`, the role specified in `ActionLogLineRoleArn` must grant permission to retrieve log events for inclusion in alarm notifications.

Request Parameters

ActionLogLineCount

The number of log lines from the most recent scheduled query execution to include in alarm action notifications. Valid range is 0 through 50. The default is 0, which means no log lines are included.

Type: Integer

Required: No

ActionLogLineRoleArn

The Amazon Resource Name (ARN) of an IAM role that CloudWatch assumes to retrieve log events for inclusion in alarm action notifications. Required when ActionLogLineCount is greater than 0.

Type: String

Required: No

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state. The default is `true`.

Type: Boolean

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Systems Manager actions:

`arn:aws:ssm:region:account-id:opsitem:severity`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmDescription

The description for the alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name for the alarm. This name must be unique within the AWS account and Region.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

ComparisonOperator

The arithmetic operation to use when comparing the aggregated query result and the threshold. The aggregated query result is used as the first operand. Valid values are `GreaterThanThreshold`, `GreaterThanOrEqualToThreshold`, `LessThanThreshold`, and `LessThanOrEqualToThreshold`.

Type: String

Valid Values: `GreaterThanOrEqualToThreshold` | `GreaterThanThreshold` | `LessThanThreshold` | `LessThanOrEqualToThreshold` | `LessThanLowerOrGreaterThanUpperThreshold` | `LessThanLowerThreshold` | `GreaterThanUpperThreshold`

Required: Yes

InsufficientDataActions

The actions to execute when this alarm transitions to the `INSUFFICIENT_DATA` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

OKActions

The actions to execute when this alarm transitions to the `OK` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Valid Values:

Amazon SNS actions:

`arn:aws:sns:region:account-id:sns-topic-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`

- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

QueryResultsToAlarm

The number of query results, out of the most recent QueryResultsToEvaluate results, that must breach the threshold to trigger the alarm to transition to ALARM (the M in M-of-N evaluation). Must be less than or equal to QueryResultsToEvaluate.

Type: Integer

Valid Range: Minimum value of 1.

Required: Yes

QueryResultsToEvaluate

The number of most recent scheduled query results to evaluate against the threshold (the N in M-of-N evaluation). Valid range is 1 through 100.

Type: Integer

Valid Range: Minimum value of 1.

Required: Yes

ScheduledQueryConfiguration

The configuration of the underlying CloudWatch Logs scheduled query that this alarm evaluates, including the query string, log groups, schedule, and aggregation expression.

Type: [ScheduledQueryConfiguration](#) object

Required: Yes

Tags

A list of key-value pairs to associate with the alarm. You can use tags to categorize and manage your alarms.

Type: Array of [Tag](#) objects

Required: No

Threshold

The value to compare with the aggregated query result.

Type: Double

Required: Yes

TreatMissingData

Sets how this alarm is to handle missing data points. Valid values are `breaching`, `notBreaching`, `ignore`, and `missing`. If this parameter is omitted, the default behavior of `missing` is used.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

LimitExceeded

The quota for alarms for this customer has already been reached.

message

HTTP Status Code: 400

ResourceConflict

The operation could not be completed because the request conflicts with the current state of the alarm or its underlying scheduled query resource.

HTTP Status Code: 409

Examples

Create a log alarm that monitors error count

Create a log alarm that runs a CloudWatch Logs Insights query every 5 minutes counting log entries that contain "ERROR", and triggers when the count exceeds 10 in 3 of the last 5 evaluations.

Sample Request

```
aws cloudwatch put-log-alarm \  
  --alarm-name "HighErrorRateAlarm" \  
  --alarm-description "Alarm when error count exceeds threshold" \  
  --scheduled-query-configuration '{  
    "QueryString": "fields @timestamp, @message | filter @message like /ERROR/",  
    "LogGroupIdentifiers": ["arn:aws:logs:us-east-1:111122223333:log-group:/aws/  
lambda/my-function"],  
    "ScheduledQueryRoleARN": "arn:aws:iam::111122223333:role/ScheduledQueryRole",  
    "AggregationExpression": "count(*)",  
    "ScheduleConfiguration": {  
      "ScheduleExpression": "rate(5 minutes)",  
      "StartTimeOffset": 360  
    }  
  }' \  
  --query-results-to-alarm 3 \  
  --query-results-to-evaluate 5 \  
  --threshold 10.0 \  
  --comparison-operator "GreaterThanThreshold" \  
  --alarm-actions "arn:aws:sns:us-east-1:111122223333:my-topic" \  
  --treat-missing-data "notBreaching"
```

Sample Response

```
{  
  "AlarmArn": "arn:aws:cloudwatch:us-east-1:111122223333:alarm:HighErrorRateAlarm"  
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutManagedInsightRules

Creates a managed Contributor Insights rule for a specified AWS resource. When you enable a managed rule, you create a Contributor Insights rule that collects data from AWS services. You cannot edit these rules with `PutInsightRule`. The rules can be enabled, disabled, and deleted using `EnableInsightRules`, `DisableInsightRules`, and `DeleteInsightRules`. If a previously created managed rule is currently disabled, a subsequent call to this API will re-enable it. Use `ListManagedInsightRules` to describe all available rules.

Request Parameters

ManagedRules

A list of `ManagedRules` to enable.

Type: Array of [ManagedRule](#) objects

Required: Yes

Response Elements

The following element is returned by the service.

Failures

An array that lists the rules that could not be enabled.

Type: Array of [PartialFailure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutMetricAlarm

Creates or updates an alarm and associates it with the specified metric, metric math expression, anomaly detection model, Metrics Insights query, or PromQL query. For more information about using a Metrics Insights query for an alarm, see [Create alarms on Metrics Insights queries](#).

Alarms based on anomaly detection models cannot have Auto Scaling actions.

When this operation creates an alarm, the alarm state is immediately set to `INSUFFICIENT_DATA`. For PromQL alarms, the alarm state is instead immediately set to `OK`. The alarm is then evaluated and its state is set appropriately. Any actions associated with the new state are then executed.

When you update an existing alarm, its state is left unchanged, but the update completely overwrites the previous configuration of the alarm.

If you are an IAM user, you must have Amazon EC2 permissions for some alarm operations:

- The `iam:CreateServiceLinkedRole` permission for all alarms with EC2 actions
- The `iam:CreateServiceLinkedRole` permissions to create an alarm with Systems Manager OpsItem or response plan actions.

The first time you create an alarm in the AWS Management Console, the AWS CLI, or by using the `PutMetricAlarm` API, CloudWatch creates the necessary service-linked role for you. The service-linked roles are called `AWSServiceRoleForCloudWatchEvents` and `AWSServiceRoleForCloudWatchAlarms_ActionSSM`. For more information, see [AWS service-linked role](#).

Each `PutMetricAlarm` action has a maximum uncompressed payload of 120 KB.

Cross-account alarms

You can set an alarm on metrics in the current account, or in another account. To create a cross-account alarm that watches a metric in a different account, you must have completed the following pre-requisites:

- The account where the metrics are located (the *sharing account*) must already have a sharing role named **CloudWatch-CrossAccountSharingRole**. If it does not already have this role, you must create it using the instructions in **Set up a sharing account** in [Cross-account cross-Region CloudWatch console](#). The policy for that role must grant access to the ID of the account where you are creating the alarm.

- The account where you are creating the alarm (the *monitoring account*) must already have a service-linked role named **AWSServiceRoleForCloudWatchCrossAccount** to allow CloudWatch to assume the sharing role in the sharing account. If it does not, you must create it following the directions in **Set up a monitoring account** in [Cross-account cross-Region CloudWatch console](#).

Request Parameters

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state. The default is TRUE.

Type: Boolean

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN). Valid values:

EC2 actions:

- `arn:aws:automate:region:ec2:stop`
- `arn:aws:automate:region:ec2:terminate`
- `arn:aws:automate:region:ec2:reboot`
- `arn:aws:automate:region:ec2:recover`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Stop/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Terminate/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Reboot/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Recover/1.0`

Autoscaling action:

- `arn:aws:autoscaling:region:account-id:scalingPolicy:policy-id:autoScalingGroupName/group-friendly-name:policyName/policy-friendly-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

SNS notification action:

- `arn:aws:sns:region:account-id:sns-topic-name`

SSM integration actions:

- `arn:aws:ssm:region:account-id:opsitem:severity#CATEGORY=category-name`
- `arn:aws:ssm-incidents::account-id:responseplan/response-plan-name`

Start a Amazon Q Developer operational investigation

`arn:aws:aiops:region:account-id:investigation-group:investigation-group-id`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmDescription

The description for the alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name for the alarm. This name must be unique within the Region.

The name must contain only UTF-8 characters, and can't contain ASCII control characters

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

ComparisonOperator

The arithmetic operation to use when comparing the specified statistic and threshold. The specified statistic value is used as the first operand.

The values `LessThanLowerOrGreaterThanUpperThreshold`, `LessThanLowerThreshold`, and `GreaterThanUpperThreshold` are used only for alarms based on anomaly detection models.

Type: String

Valid Values: `GreaterThanOrEqualToThreshold` | `GreaterThanThreshold` | `LessThanThreshold` | `LessThanOrEqualToThreshold` | `LessThanLowerOrGreaterThanUpperThreshold` | `LessThanLowerThreshold` | `GreaterThanUpperThreshold`

Required: No

DatapointsToAlarm

The number of data points that must be breaching to trigger the alarm. This is used only if you are setting an "M out of N" alarm. In that case, this value is the M. For more information, see [Evaluating an Alarm](#) in the *Amazon CloudWatch User Guide*.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

Dimensions

The dimensions for the metric specified in `MetricName`.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

EvaluateLowSampleCountPercentile

Used only for alarms based on percentiles. If you specify `ignore`, the alarm state does not change during periods with too few data points to be statistically significant. If you specify `evaluate` or omit this parameter, the alarm is always evaluated and possibly changes state no matter how many data points are available. For more information, see [Percentile-Based CloudWatch Alarms and Low Data Samples](#).

Valid Values: `evaluate` | `ignore`

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

EvaluationCriteria

The evaluation criteria for the alarm. For each `PutMetricAlarm` operation, you must specify either `MetricName`, a `Metrics` array, or an `EvaluationCriteria`.

If you use the `EvaluationCriteria` parameter, you cannot include the `Namespace`, `MetricName`, `Dimensions`, `Period`, `Unit`, `Statistic`, `ExtendedStatistic`, `Metrics`, `Threshold`, `ComparisonOperator`, `ThresholdMetricId`, `EvaluationPeriods`, or `DatapointsToAlarm` parameters of `PutMetricAlarm` in the same operation. Instead, all evaluation parameters are defined within this structure.

For an example of how to use this parameter, see the **PromQL alarm** example on this page.

Type: [EvaluationCriteria](#) object

Note: This object is a Union. Only one member of this object can be specified or returned.

Required: No

EvaluationInterval

The frequency, in seconds, at which the alarm is evaluated. Valid values are 10, 20, 30, and any multiple of 60.

This parameter is required for alarms that use `EvaluationCriteria`, and cannot be specified for alarms configured with `MetricName` or `Metrics`.

Type: Integer

Valid Range: Minimum value of 10. Maximum value of 3600.

Required: No

EvaluationPeriods

The number of periods over which data is compared to the specified threshold. If you are setting an alarm that requires that a number of consecutive data points be breaching to trigger the alarm, this value specifies that number. If you are setting an "M out of N" alarm, this value is the N.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

EvaluationWindow

The evaluation window that the alarm uses to select the range of metric data that it evaluates. Specify either a sliding window or a wall clock window. If you omit this parameter, the alarm uses a sliding window.

A sliding window advances each time the alarm is evaluated, forming a rolling time window. A wall clock window aligns the evaluated range to fixed clock boundaries, such as the top of the hour or the start of the day.

You can use `EvaluationWindow` with any type of metric alarm except alarms that are based on a PromQL query.

For more information, see [Alarm evaluation windows](#) in the *CloudWatch User Guide*.

Type: [EvaluationWindow](#) object

Note: This object is a Union. Only one member of this object can be specified or returned.

Required: No

ExtendedStatistic

The extended statistic for the metric specified in `MetricName`. When you call `PutMetricAlarm` and specify a `MetricName`, you must specify either `Statistic` or `ExtendedStatistic` but not both.

If you specify `ExtendedStatistic`, the following are valid values:

- `p90`
- `tm90`
- `tc90`
- `ts90`
- `wm90`
- `IQM`
- `PR(n:m)` where *n* and *m* are values of the metric
- `TC(X%:X%)` where *X* is between 10 and 90 inclusive.
- `TM(X%:X%)` where *X* is between 10 and 90 inclusive.
- `TS(X%:X%)` where *X* is between 10 and 90 inclusive.
- `WM(X%:X%)` where *X* is between 10 and 90 inclusive.

For more information about these extended statistics, see [CloudWatch statistics definitions](#).

Type: String

Required: No

InsufficientDataActions

The actions to execute when this alarm transitions to the `INSUFFICIENT_DATA` state from any other state. Each action is specified as an Amazon Resource Name (ARN). Valid values:

EC2 actions:

- `arn:aws:automate:region:ec2:stop`
- `arn:aws:automate:region:ec2:terminate`
- `arn:aws:automate:region:ec2:reboot`

- `arn:aws:automate:region:ec2:recover`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Stop/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Terminate/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Reboot/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Recover/1.0`

Autoscaling action:

- `arn:aws:autoscaling:region:account-id:scalingPolicy:policy-id:autoScalingGroupName/group-friendly-name:policyName/policy-friendly-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

SNS notification action:

- `arn:aws:sns:region:account-id:sns-topic-name`

SSM integration actions:

- `arn:aws:ssm:region:account-id:opsitem:severity#CATEGORY=category-name`
- `arn:aws:ssm-incidents::account-id:responseplan/response-plan-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

MetricName

The name for the metric associated with the alarm. For each PutMetricAlarm operation, you must specify either MetricName, a Metrics array, or an EvaluationCriteria.

If you are creating an alarm based on a math expression, you cannot specify this parameter, or any of the Namespace, Dimensions, Period, Unit, Statistic, or ExtendedStatistic parameters. Instead, you specify all this information in the Metrics array.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Metrics

An array of MetricDataQuery structures that enable you to create an alarm based on the result of a metric math expression. For each PutMetricAlarm operation, you must specify either MetricName, a Metrics array, or an EvaluationCriteria.

Each item in the Metrics array either retrieves a metric or performs a math expression.

One item in the Metrics array is the expression that the alarm watches. You designate this expression by setting ReturnData to true for this object in the array. For more information, see [MetricDataQuery](#).

If you use the Metrics parameter, you cannot include the Namespace, MetricName, Dimensions, Period, Unit, Statistic, or ExtendedStatistic parameters of PutMetricAlarm in the same operation. Instead, you retrieve the metrics you are using in your math expression as part of the Metrics array.

Type: Array of [MetricDataQuery](#) objects

Required: No

Namespace

The namespace for the metric associated specified in MetricName.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*.*`

Required: No

OKActions

The actions to execute when this alarm transitions to an OK state from any other state. Each action is specified as an Amazon Resource Name (ARN). Valid values:

EC2 actions:

- `arn:aws:automate:region:ec2:stop`
- `arn:aws:automate:region:ec2:terminate`
- `arn:aws:automate:region:ec2:reboot`
- `arn:aws:automate:region:ec2:recover`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Stop/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Terminate/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Reboot/1.0`
- `arn:aws:swf:region:account-id:action/actions/AWS_EC2.InstanceId.Recover/1.0`

Autoscaling action:

- `arn:aws:autoscaling:region:account-id:scalingPolicy:policy-id:autoScalingGroupName/group-friendly-name:policyName/policy-friendly-name`

Lambda actions:

- Invoke the latest version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name`
- Invoke a specific version of a Lambda function: `arn:aws:lambda:region:account-id:function:function-name:version-number`
- Invoke a function by using an alias Lambda function: `arn:aws:lambda:region:account-id:function:function-name:alias-name`

SNS notification action:

- `arn:aws:sns:region:account-id:sns-topic-name`

SSM integration actions:

- `arn:aws:ssm:region:account-id:opsitem:severity#CATEGORY=category-name`
- `arn:aws:ssm-incidents::account-id:responseplan/response-plan-name`

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

Period

The length, in seconds, used each time the metric specified in `MetricName` is evaluated. Valid values are 10, 20, 30, and any multiple of 60.

Period is required for alarms based on static thresholds. If you are creating an alarm based on a metric math expression, you specify the period for each metric within the objects in the `Metrics` array.

Be sure to specify 10, 20, or 30 only for metrics that are stored by a `PutMetricData` call with a `StorageResolution` of 1. If you specify a period of 10, 20, or 30 for a metric that does not have sub-minute resolution, the alarm still attempts to gather data at the period rate that you specify. In this case, it does not receive data for the attempts that do not correspond to a one-minute data resolution, and the alarm might often lapse into `INSUFFICIENT_DATA` status. Specifying 10, 20, or 30 also sets this alarm as a high-resolution alarm, which has a higher charge than other alarms. For more information about pricing, see [Amazon CloudWatch Pricing](#).

An alarm's total current evaluation period can be no longer than seven days, so `Period` multiplied by `EvaluationPeriods` can't be more than 604,800 seconds. For alarms with a period of less than one hour (3,600 seconds), the total evaluation period can't be longer than one day (86,400 seconds).

Type: Integer

Valid Range: Minimum value of 10.

Required: No

Statistic

The statistic for the metric specified in `MetricName`, other than percentile. For percentile statistics, use `ExtendedStatistic`. When you call `PutMetricAlarm` and specify a `MetricName`, you must specify either `Statistic` or `ExtendedStatistic`, but not both.

Type: String

Valid Values: `SampleCount` | `Average` | `Sum` | `Minimum` | `Maximum`

Required: No

Tags

A list of key-value pairs to associate with the alarm. You can associate as many as 50 tags with an alarm. To be able to associate tags with the alarm when you create the alarm, you must have the `cloudwatch:TagResource` permission.

Tags can help you organize and categorize your resources. You can also use them to scope user permissions by granting a user permission to access or change only resources with certain tag values.

If you are using this operation to update an existing alarm, any tags you specify in this parameter are ignored. To change the tags of an existing alarm, use [TagResource](#) or [UntagResource](#).

To use this field to set tags for an alarm when you create it, you must be signed on with both the `cloudwatch:PutMetricAlarm` and `cloudwatch:TagResource` permissions.

Type: Array of [Tag](#) objects

Required: No

Threshold

The value against which the specified statistic is compared.

This parameter is required for alarms based on static thresholds, but should not be used for alarms based on anomaly detection models.

Type: Double

Required: No

ThresholdMetricId

If this is an alarm based on an anomaly detection model, make this value match the ID of the `ANOMALY_DETECTION_BAND` function.

For an example of how to use this parameter, see the **Anomaly Detection Model Alarm** example on this page.

If your alarm uses this parameter, it cannot have Auto Scaling actions.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

TreatMissingData

Sets how this alarm is to handle missing data points. If `TreatMissingData` is omitted, the default behavior of missing is used. For more information, see [Configuring How CloudWatch Alarms Treats Missing Data](#).

Valid Values: `breaching` | `notBreaching` | `ignore` | `missing`

Note

Alarms that evaluate metrics in the AWS/DynamoDB namespace always ignore missing data even if you choose a different option for `TreatMissingData`. When an AWS/DynamoDB metric has missing data, alarms that evaluate that metric remain in their current state.

Note

This parameter is not applicable to PromQL alarms.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Unit

The unit of measure for the statistic. For example, the units for the Amazon EC2 NetworkIn metric are Bytes because NetworkIn tracks the number of bytes that an instance receives on all network interfaces. You can also specify a unit when you create a custom metric. Units help provide conceptual meaning to your data. Metric data points that specify a unit of measure, such as Percent, are aggregated separately. If you are creating an alarm based on a metric math expression, you can specify the unit for each metric (if needed) within the objects in the `Metrics` array.

If you don't specify `Unit`, CloudWatch retrieves all unit types that have been published for the metric and attempts to evaluate the alarm. Usually, metrics are published with only one unit, so the alarm works as intended.

However, if the metric is published with multiple types of units and you don't specify a unit, the alarm's behavior is not defined and it behaves unpredictably.

We recommend omitting `Unit` so that you don't inadvertently specify an incorrect unit that is not published for this metric. Doing so causes the alarm to be stuck in the `INSUFFICIENT DATA` state.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

LimitExceeded

The quota for alarms for this customer has already been reached.

message

HTTP Status Code: 400

Examples

Static Threshold Alarm

The following example creates an alarm that notifies an SNS group when the CPUUtilization of a certain instance goes over 40% for three out of four periods.

Sample Request

```
{
  "Namespace": "AWS/EC2",
  "MetricName": "CPUUtilization",
  "Dimensions": [
    {
      "Name": "InstanceId",
      "Value": "i-1234567890abcdef0"
    }
  ],
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic"
  ],
  "ComparisonOperator": "GreaterThanThreshold",
  "DatapointsToAlarm": 3,
  "EvaluationPeriods": 4,
  "Period": 60,
  "Statistic": "Average",
  "Threshold": 40,
  "AlarmDescription": "CPU Utilization of i-1234567890abcdef0 with 40% as threshold",
  "AlarmName": "Instance i-1234567890abcdef0 CPU Utilization"
}
```

Metric Math Function Alarm

The following example retrieves three metrics that each track a different type of connection error to a custom service. These error counts are first summed in one expression, then divided by total connection attempts in another expression. The alarm goes to the ALARM state if the error rate is over 40% for three consecutive periods, and notifies two different SNS groups.

Sample Request

```
{
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic",
    "arn:aws:sns:us-west-1:123456789012:my_other_sns_topic"
  ],
  "ComparisonOperator": "GreaterThanThreshold",
  "EvaluationPeriods": 3,
  "Threshold": 40,
  "AlarmDescription": "MyService Aggregate Connection Error Rate (Alarm at 40%)",
  "AlarmName": "MyService Connection Error Rate",
  "Metrics": [
    {
      "MetricStat": {
        "Metric": {
          "MetricName": "ConnectionsFailed",
          "Namespace": "MyService"
        },
        "Period": 60,
        "Stat": "Sum"
      },
      "Id": "m1",
      "ReturnData": "False"
    },
    {
      "MetricStat": {
        "Metric": {
          "MetricName": "ConnectionsDropped",
          "Namespace": "MyService"
        },
        "Period": 60,
        "Stat": "Sum"
      },
      "Id": "m2",
      "ReturnData": "False"
    },
    {
      "MetricStat": {
        "Metric": {
          "MetricName": "RequestsThrottled",
          "Namespace": "MyService"
        },
        "Period": 60,
```

```

        "Stat": "Sum"
    },
    "Id": "m3",
    "ReturnData": "False"
},
{
    "MetricStat": {
        "Metric": {
            "MetricName": "ConnectionAttempts",
            "Namespace": "MyService"
        },
        "Period": 60,
        "Stat": "Sum"
    },
    "Id": "m4",
    "ReturnData": "False"
},
{
    "Id": "error_total",
    "Expression": "m1+m2+m3",
    "ReturnData": "False"
},
{
    "Id": "error_rate",
    "Expression": "(error_total/m4)*100",
    "ReturnData": "true",
    "Label": "Total Connection Error Rate"
}
]
}

```

Anomaly Detection Model Alarm

The following example sets an alarm on an anomaly detection model. The `Id` of `m1` is assigned to the `CPUUtilization` metric of an instance. `t1` is the anomaly detection model function for that metric, and uses 3 standard deviations to set the width of the band. The setting of `ThresholdMetricId` is `t1` and the `ComparisonOperator` is `LessThanLowerOrGreaterThanUpperThreshold`, specifying that the alarm goes to alarm state when the metric value is outside the anomaly model band in either direction for two consecutive evaluation periods.

Sample Request

```
{
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic",
    "arn:aws:sns:us-west-1:123456789012:my_other_sns_topic"
  ],
  "AlarmName": "MyAlarmName",
  "AlarmDescription": "This alarm uses an anomaly detection model",
  "Metrics": [
    {
      "Id": "m1",
      "ReturnData": true,
      "MetricStat": {
        "Metric": {
          "MetricName": "CPUUtilization",
          "Namespace": "AWS/EC2",
          "Dimensions": [
            {
              "Name": "instanceId",
              "Value": "i-1234567890abcdef0"
            }
          ]
        },
        "Stat": "Average",
        "Period": 60
      }
    },
    {
      "Id": "t1",
      "Expression": "ANOMALY_DETECTION_BAND(m1, 3)"
    }
  ],
  "EvaluationPeriods": 2,
  "ThresholdMetricId": "t1",
  "ComparisonOperator": "LessThanLowerOrGreaterThanUpperThreshold"
}
```

Metrics Insights query alarm

The following example sets an alarm on an Metrics Insights query.

Sample Request

```
{
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic",
    "arn:aws:sns:us-west-1:123456789012:my_other_sns_topic"
  ],
  "AlarmName": "MetricsInsightsAlarm",
  "AlarmDescription": "This alarm uses a Metrics Insights query",
  "Metrics": [
    {
      "Id": "m1",
      "Expression": "SELECT AVG(CPUUtilization) FROM SCHEMA('AWS/EC2',
InstanceId)",
      "Period": 60,
      "Label": "Average CPUUtilization query"
    }
  ],
  "EvaluationPeriods": 1,
  "Threshold": 65,
  "ComparisonOperator": "GreaterThanThreshold"
}
```

PromQL alarm

The following example creates an alarm based on a PromQL query. The alarm evaluates the PromQL query every 30 seconds and its contributors (matching series) transition to the ALARM state if they are continuously breaching for 300 seconds (pending period). Contributors transition back to OK after they are no longer breaching for 120 seconds (recovery period).

Sample Request

```
{
  "AlarmName": "HighCPUPromQLAlarm",
  "AlarmDescription": "Alarm when average CPU exceeds 80% using PromQL",
  "AlarmActions": [
    "arn:aws:sns:us-west-1:123456789012:my_sns_topic"
  ],
  "EvaluationCriteria": {
    "PromQLCriteria": {
      "Query": "avg(cpu_utilization_percent) > 80",
      "PendingPeriod": 300,

```



```
        "RecoveryPeriod": 120
    },
    "EvaluationInterval": 30
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutMetricData

Publishes metric data to Amazon CloudWatch. CloudWatch associates the data with the specified metric. If the specified metric does not exist, CloudWatch creates the metric. When CloudWatch creates a metric, it can take up to fifteen minutes for the metric to appear in calls to [ListMetrics](#).

You can publish metrics with associated entity data (so that related telemetry can be found and viewed together), or publish metric data by itself. To send entity data with your metrics, use the `EntityMetricData` parameter. To send metrics without entity data, use the `MetricData` parameter. The `EntityMetricData` structure includes `MetricData` structures for the metric data.

You can publish either individual values in the `Value` field, or arrays of values and the number of times each value occurred during the period by using the `Values` and `Counts` fields in the `MetricData` structure. Using the `Values` and `Counts` method enables you to publish up to 150 values per metric with one `PutMetricData` request, and supports retrieving percentile statistics on this data.

Each `PutMetricData` request is limited to 1 MB in size for HTTP POST requests. You can send a payload compressed by gzip. Each request is also limited to no more than 1000 different metrics (across both the `MetricData` and `EntityMetricData` properties).

Although the `Value` parameter accepts numbers of type `Double`, CloudWatch rejects values that are either too small or too large. Values must be in the range of -2^{360} to 2^{360} . In addition, special values (for example, NaN, +Infinity, -Infinity) are not supported.

You can use up to 30 dimensions per metric to further clarify what data the metric collects. Each dimension consists of a `Name` and `Value` pair. For more information about specifying dimensions, see [Publishing Metrics](#) in the *Amazon CloudWatch User Guide*.

You specify the time stamp to be associated with each data point. You can specify time stamps that are as much as two weeks before the current date, and as much as 2 hours after the current day and time.

Data points with time stamps from 24 hours ago or longer can take at least 48 hours to become available for [GetMetricData](#) or [GetMetricStatistics](#) from the time they are submitted. Data points with time stamps between 3 and 24 hours ago can take as much as 2 hours to become available for [GetMetricData](#) or [GetMetricStatistics](#).

CloudWatch needs raw data points to calculate percentile statistics. If you publish data using a statistic set instead, you can only retrieve percentile statistics for this data if one of the following conditions is true:

- The `SampleCount` value of the statistic set is 1 and `Min`, `Max`, and `Sum` are all equal.
- The `Min` and `Max` are equal, and `Sum` is equal to `Min` multiplied by `SampleCount`.

Request Parameters

EntityMetricData

Data for metrics that contain associated entity information. You can include up to two `EntityMetricData` objects, each of which can contain a single `Entity` and associated metrics.

The limit of metrics allowed, 1000, is the sum of both `EntityMetricData` and `MetricData` metrics.

Type: Array of [EntityMetricData](#) objects

Required: No

MetricData

The data for the metrics. Use this parameter if your metrics do not contain associated entities. The array can include no more than 1000 metrics per call.

The limit of metrics allowed, 1000, is the sum of both `EntityMetricData` and `MetricData` metrics.

Type: Array of [MetricDatum](#) objects

Required: No

Namespace

The namespace for the metric data. You can use ASCII characters for the namespace, except for control characters which are not supported.

To avoid conflicts with AWS service namespaces, you should not specify a namespace that begins with `AWS/`

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: Yes

StrictEntityValidation

Whether to accept valid metric data when an invalid entity is sent.

- When set to `true`: Any validation error (for entity or metric data) will fail the entire request, and no data will be ingested. The failed operation will return a 400 result with the error.
- When set to `false`: Validation errors in the entity will not associate the metric with the entity, but the metric data will still be accepted and ingested. Validation errors in the metric data will fail the entire request, and no data will be ingested.

In the case of an invalid entity, the operation will return a 200 status, but an additional response header will contain information about the validation errors. The new header, `X-Amzn-Failure-Message` is an enumeration of the following values:

- `InvalidEntity` - The provided entity is invalid.
- `InvalidKeyAttributes` - The provided `KeyAttributes` of an entity is invalid.
- `InvalidAttributes` - The provided `Attributes` of an entity is invalid.
- `InvalidTypeValue` - The provided `Type` in the `KeyAttributes` of an entity is invalid.
- `EntitySizeTooLarge` - The number of `EntityMetricData` objects allowed is 2.
- `MissingRequiredFields` - There are missing required fields in the `KeyAttributes` for the provided `Type`.

For details of the requirements for specifying an entity, see [How to add related information to telemetry](#) in the *CloudWatch User Guide*.

This parameter is *required* when `EntityMetricData` is included.

Type: Boolean

Required: No

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

Examples

Example

The following example puts data for a single metric containing one dimension:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
```

```
&Namespace=TestNamespace
&MetricData.member.1.MetricName=buffers
&MetricData.member.1.Unit=Bytes
&MetricData.member.1.Value=231434333
&MetricData.member.1.Dimensions.member.1.Name=InstanceType
&MetricData.member.1.Dimensions.member.1.Value=m1.small
&AUTHPARAMS
```

Example

The following example puts data for a single metric containing two dimensions:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&Namespace=TestNamespace
&MetricData.member.1.MetricName=buffers
&MetricData.member.1.Unit=Bytes
&MetricData.member.1.Value=231434333
&MetricData.member.1.Dimensions.member.1.Name=InstanceID
&MetricData.member.1.Dimensions.member.1.Value=i-aaba32d4
&MetricData.member.1.Dimensions.member.2.Name=InstanceType
&MetricData.member.1.Dimensions.member.2.Value=m1.small
&AUTHPARAMS
```

Example

The following example puts data for two metrics, each with two dimensions:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&Namespace=TestNamespace
&MetricData.member.1.MetricName=buffers
&MetricData.member.1.Unit=Bytes
&MetricData.member.1.Value=231434333
&MetricData.member.1.Dimensions.member.1.Name=InstanceID
&MetricData.member.1.Dimensions.member.1.Value=i-aaba32d4
```

```
&MetricData.member.1.Dimensions.member.2.Name=InstanceType
&MetricData.member.1.Dimensions.member.2.Value=m1.small
&MetricData.member.2.MetricName=latency
&MetricData.member.2.Unit=Milliseconds
&MetricData.member.2.Value=23
&MetricData.member.2.Dimensions.member.1.Name=InstanceID
&MetricData.member.2.Dimensions.member.1.Value=i-aaba32d4
&MetricData.member.2.Dimensions.member.2.Name=InstanceType
&MetricData.member.2.Dimensions.member.2.Value=m1.small
&AUTHPARAMS
```

Example

The following example puts data for a high-resolution metric:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&Namespace=HighResolutionMetric
&MetricData.member.1.MetricName=HighResdata
&MetricData.member.1.Unit=Bytes
&MetricData.member.1.Value=542868
&MetricData.member.1.StorageResolution=1
&AUTHPARAMS
```

Example

The following example puts multiple values for each of two metrics, using Values and Counts arrays:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&Namespace=TestNamespace
&MetricData.member.1.MetricName=Reads
&MetricData.member.1.Unit=Count
&MetricData.member.1.Values.member.1=5
&MetricData.member.1.Values.member.2=8
```

```
&MetricData.member.1.Values.member.3=10
&MetricData.member.1.Values.member.4=9
&MetricData.member.1.Counts.member.1=1
&MetricData.member.1.Counts.member.2=5
&MetricData.member.1.Counts.member.3=6
&MetricData.member.1.Counts.member.4=5
&MetricData.member.1.Dimensions.member.1.Name=InstanceId
&MetricData.member.1.Dimensions.member.1.Value=i-aaba32d4
&MetricData.member.2.MetricName=Writes
&MetricData.member.2.Unit=Count
&MetricData.member.2.Values.member.1=2
&MetricData.member.2.Values.member.2=3
&MetricData.member.2.Values.member.3=0
&MetricData.member.2.Counts.member.1=2
&MetricData.member.2.Counts.member.2=2
&MetricData.member.2.Counts.member.3=1
&MetricData.member.2.Dimensions.member.1.Name=InstanceId
&MetricData.member.2.Dimensions.member.1.Value=i-aaba32d4
&AUTHPARAMS
```

Example

The following example uses `EntityMetricData` to put a metric with entity data for a service running in Amazon EC2:

Sample Request

```
https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&StrictEntityValidation=true
&Namespace=TestNamespace
&EntityMetricData.member.1.Entity.KeyAttributes.entry.1.key=Type
&EntityMetricData.member.1.Entity.KeyAttributes.entry.1.value=Service
&EntityMetricData.member.1.Entity.KeyAttributes.entry.2.key=Name
&EntityMetricData.member.1.Entity.KeyAttributes.entry.2.value=MyTestService
&EntityMetricData.member.1.Entity.KeyAttributes.entry.3.key=Environment
&EntityMetricData.member.1.Entity.KeyAttributes.entry.3.value=MyTestEnvironment
&EntityMetricData.member.1.Entity.Attributes.entry.1.key=PlatformType
&EntityMetricData.member.1.Entity.Attributes.entry.1.value=AWS::EC2
&EntityMetricData.member.1.Entity.Attributes.entry.2.key=EC2.InstanceId
&EntityMetricData.member.1.Entity.Attributes.entry.2.value=i-1234567890abcdef0
&EntityMetricData.member.1.MetricData.member.1.MetricName=buffers
```



```

&EntityMetricData.member.1.MetricData.member.1.Timestamp=2024-11-06T02:16:28Z
&EntityMetricData.member.1.MetricData.member.1.Unit=Count
&EntityMetricData.member.1.MetricData.member.1.Values.member.1=2
&EntityMetricData.member.1.MetricData.member.1.Values.member.2=3
&EntityMetricData.member.1.MetricData.member.1.Values.member.3=0
&EntityMetricData.member.1.MetricData.member.1.Counts.member.1=2
&EntityMetricData.member.1.MetricData.member.1.Counts.member.2=2
&EntityMetricData.member.1.MetricData.member.1.Counts.member.3=1
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.1.Name=InstanceID
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.1.Value=i-aaba32d4
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.2.Name=InstanceType
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.2.Value=m1.small
&AUTHPARAMS

```

Example

The following example uses `EntityMetricData` to put a metric with entity data for a service running in Lambda:

Sample Request

```

https://monitoring.us-west-1.amazonaws.com
?Action=PutMetricData
&Version=2010-08-01
&StrictEntityValidation=true
&Namespace=TestNamespace
&EntityMetricData.member.1.Entity.KeyAttributes.entry.1.key=Type
&EntityMetricData.member.1.Entity.KeyAttributes.entry.1.value=Service
&EntityMetricData.member.1.Entity.KeyAttributes.entry.2.key=Name
&EntityMetricData.member.1.Entity.KeyAttributes.entry.2.value=MyTestService
&EntityMetricData.member.1.Entity.KeyAttributes.entry.3.key=Environment
&EntityMetricData.member.1.Entity.KeyAttributes.entry.3.value=MyTestEnvironment
&EntityMetricData.member.1.Entity.Attributes.entry.1.key=PlatformType
&EntityMetricData.member.1.Entity.Attributes.entry.1.value=AWS::Lambda
&EntityMetricData.member.1.Entity.Attributes.entry.2.key=Lambda.Function
&EntityMetricData.member.1.Entity.Attributes.entry.2.value=MyTestFunction
&EntityMetricData.member.1.MetricData.member.1.MetricName=faults
&EntityMetricData.member.1.MetricData.member.1.Timestamp=2024-11-06T02:16:28Z
&EntityMetricData.member.1.MetricData.member.1.Unit=Count
&EntityMetricData.member.1.MetricData.member.1.Values.member.1=2
&EntityMetricData.member.1.MetricData.member.1.Values.member.2=3
&EntityMetricData.member.1.MetricData.member.1.Values.member.3=0
&EntityMetricData.member.1.MetricData.member.1.Counts.member.1=2

```

```
&EntityMetricData.member.1.MetricData.member.1.Counts.member.2=2
&EntityMetricData.member.1.MetricData.member.1.Counts.member.3=1
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.1.Name=InstanceID
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.1.Value=i-aaba32d4
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.2.Name=InstanceType
&EntityMetricData.member.1.MetricData.member.1.Dimensions.member.2.Value=m1.small
&AUTHPARAMS
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutMetricStream

Creates or updates a metric stream. Metric streams can automatically stream CloudWatch metrics to AWS destinations, including Amazon S3, and to many third-party solutions.

For more information, see [Using Metric Streams](#).

To create a metric stream, you must be signed in to an account that has the `iam:PassRole` permission and either the `CloudWatchFullAccess` policy or the `cloudwatch:PutMetricStream` permission.

When you create or update a metric stream, you choose one of the following:

- Stream metrics from all metric namespaces in the account.
- Stream metrics from all metric namespaces in the account, except for the namespaces that you list in `ExcludeFilters`.
- Stream metrics from only the metric namespaces that you list in `IncludeFilters`.

By default, a metric stream always sends the MAX, MIN, SUM, and SAMPLECOUNT statistics for each metric that is streamed. You can use the `StatisticsConfigurations` parameter to have the metric stream send additional statistics in the stream. Streaming additional statistics incurs additional costs. For more information, see [Amazon CloudWatch Pricing](#).

When you use `PutMetricStream` to create a new metric stream, the stream is created in the running state. If you use it to update an existing stream, the state of the stream is not changed.

If you are using CloudWatch cross-account observability and you create a metric stream in a monitoring account, you can choose whether to include metrics from source accounts in the stream. For more information, see [CloudWatch cross-account observability](#).

Request Parameters

ExcludeFilters

If you specify this parameter, the stream sends metrics from all metric namespaces except for the namespaces that you specify here.

You cannot include `ExcludeFilters` and `IncludeFilters` in the same operation.

Type: Array of [MetricStreamFilter](#) objects

Required: No

FirehoseArn

The ARN of the Amazon Kinesis Data Firehose delivery stream to use for this metric stream. This Amazon Kinesis Data Firehose delivery stream must already exist and must be in the same account as the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

IncludeFilters

If you specify this parameter, the stream sends only the metrics from the metric namespaces that you specify here.

You cannot include `IncludeFilters` and `ExcludeFilters` in the same operation.

Type: Array of [MetricStreamFilter](#) objects

Required: No

IncludeLinkedAccountsMetrics

If you are creating a metric stream in a monitoring account, specify `true` to include metrics from source accounts in the metric stream.

Type: Boolean

Required: No

Name

If you are creating a new metric stream, this is the name for the new stream. The name must be different than the names of other metric streams in this account and Region.

If you are updating a metric stream, specify the name of that stream here.

Valid characters are A-Z, a-z, 0-9, "-" and "_".

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

OutputFormat

The output format for the stream. Valid values are `json`, `opentelemetry1.0`, and `opentelemetry0.7`. For more information about metric stream output formats, see [Metric streams output formats](#).

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Valid Values: `json` | `opentelemetry0.7` | `opentelemetry1.0`

Required: Yes

RoleArn

The ARN of an IAM role that this metric stream will use to access Amazon Kinesis Data Firehose resources. This IAM role must already exist and must be in the same account as the metric stream. This IAM role must include the following permissions:

- `firehose:PutRecord`
- `firehose:PutRecordBatch`

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

StatisticsConfigurations

By default, a metric stream always sends the MAX, MIN, SUM, and SAMPLECOUNT statistics for each metric that is streamed. You can use this parameter to have the metric stream also send additional statistics in the stream. This array can have up to 100 members.

For each entry in this array, you specify one or more metrics and the list of additional statistics to stream for those metrics. The additional statistics that you can stream depend on the stream's `OutputFormat`. If the `OutputFormat` is `json`, you can stream any additional statistic that is supported by CloudWatch, listed in [CloudWatch statistics definitions](#). If the `OutputFormat` is `opentelemetry1.0` or `opentelemetry0.7`, you can stream percentile statistics such as `p95`, `p99.9`, and so on.

Type: Array of [MetricStreamStatisticsConfiguration](#) objects

Required: No

Tags

A list of key-value pairs to associate with the metric stream. You can associate as many as 50 tags with a metric stream.

Tags can help you organize and categorize your resources. You can also use them to scope user permissions by granting a user permission to access or change only resources with certain tag values.

You can use this parameter only when you are creating a new metric stream. If you are using this operation to update an existing metric stream, any tags you specify in this parameter are ignored. To change the tags of an existing metric stream, use [TagResource](#) or [UntagResource](#).

Type: Array of [Tag](#) objects

Required: No

Response Elements

The following element is returned by the service.

Arn

The ARN of the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConcurrentModificationException

More than one process tried to modify a resource at the same time.

HTTP Status Code: 429

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterCombination

Parameters were used together that cannot be used together.

message

HTTP Status Code: 400

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

Examples

Stream two namespaces with the default statistics

The following example creates a metric stream that streams all the metrics from the AWS/EC2 and AWS/ELB namespaces, with only the default statistics.

Sample Request

```
{
  "Name": "MyMetricStream",
  "FirehoseArn": "arn:aws:firehose:us-east-1:123456789098:stream/MyFirehose",
```

```
"RoleArn": "arn:aws:iam::123456789098:role/MyFirehoseWriteAccessRole",
"IncludeFilters": [
  {
    "Namespace": "AWS/EC2"
  },
  {
    "Namespace": "AWS/ELB"
  }
],
"OutputFormat": "opentelemetry1.0"
}
```

In a monitoring account, stream two namespaces with default statistics from all source accounts

The following example creates a metric stream that streams all the metrics from the AWS/EC2 and AWS/ELB namespaces from this monitoring account and from all source accounts that it is linked to.

Sample Request

```
{
  "Name": "MyMetricStream",
  "FirehoseArn": "arn:aws:firehose:us-east-1:123456789098:stream/MyFirehose",
  "RoleArn": "arn:aws:iam::123456789098:role/MyFirehoseWriteAccessRole",
  "IncludeLinkedAccountsMetrics": "true",
  "IncludeFilters": [
    {
      "Namespace": "AWS/EC2"
    },
    {
      "Namespace": "AWS/ELB"
    }
  ],
  "OutputFormat": "opentelemetry1.0"
}
```

Stream additional statistics

The following example creates a metric stream that streams all metrics from the AWS/EC2 namespace with only the default statistics, and also streams two other metrics with the default statistics and some additional statistics.

Sample Request

```
{
  "Name": "MyMetricStream",
  "FirehoseArn": "arn:aws:firehose:us-east-1:123456789098:stream/MyFirehose",
  "RoleArn": "arn:aws:iam::123456789098:role/MyFirehoseWriteAccessRole",
  "IncludeFilters": [
    {
      "Namespace": "AWS/EC2"
    }
  ],
  "OutputFormat": "json",
  "StatisticsConfigurations": [
    {
      "IncludeMetrics": [
        {
          "Namespace": "AWS/ApplicationELB",
          "MetricName": "TargetResponseTime"
        },
        {
          "Namespace": "AWS/ELB",
          "MetricName": "Latency"
        }
      ],
      "AdditionalStatistics": [
        "tm90",
        "p90",
        "p99",
        "p99.9"
      ]
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)

- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

SetAlarmState

Temporarily sets the state of an alarm for testing purposes. When the updated state differs from the previous value, the action configured for the appropriate state is invoked. For example, if your alarm is configured to send an Amazon SNS message when an alarm is triggered, temporarily changing the alarm state to ALARM sends an SNS message.

Metric alarms returns to their actual state quickly, often within seconds. Because the metric alarm state change happens quickly, it is typically only visible in the alarm's **History** tab in the Amazon CloudWatch console or through [DescribeAlarmHistory](#).

If you use SetAlarmState on a composite alarm, the composite alarm is not guaranteed to return to its actual state. It returns to its actual state only once any of its children alarms change state. It is also reevaluated if you update its configuration.

If an alarm triggers EC2 Auto Scaling policies or application Auto Scaling policies, you must include information in the StateReasonData parameter to enable the policy to take the correct action.

Request Parameters

AlarmName

The name of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

StateReason

The reason that this alarm is set to this specific state, in text format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1023.

Required: Yes

StateReasonData

The reason that this alarm is set to this specific state, in JSON format.

For SNS or EC2 alarm actions, this is just informational. But for EC2 Auto Scaling or application Auto Scaling alarm actions, the Auto Scaling policy uses the information in this field to take the correct action.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 4000.

Required: No

StateValue

The value of the state.

Type: String

Valid Values: OK | ALARM | INSUFFICIENT_DATA

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InvalidFormat

Data was not syntactically valid JSON.

message

HTTP Status Code: 400

ResourceNotFound

The named resource does not exist.

message

HTTP Status Code: 404

Examples

Example

The following example sets the alarm state to ALARM, and provides information inside of StateReasonData so that Auto Scaling actions can be performed correctly according to your Auto Scaling policies.

```
{
  "AlarmName": "ExampleAlarmName",
  "StateValue": "ALARM",
  "StateReason": "Testing Alarm State",
  "StateReasonData": {
    "Version": "1.0",
    "QueryDate": "2018-10-31T14:32:52.031+0000",
    "StartDate": "2018-10-31T14:31:00.000+0000",
    "Statistic": "Average",
    "Period": 60,
    "RecentDatapoints": [
      100
    ],
    "Threshold": 50
  }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StartMetricStreams

Starts the streaming of metrics for one or more of your metric streams.

Request Parameters

Names

The array of the names of metric streams to start streaming.

This is an "all or nothing" operation. If you do not have permission to access all of the metric streams that you list here, then none of the streams that you list in the operation will start streaming.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StartOTelEnrichment

Enables enrichment and PromQL access for CloudWatch vended metrics for [supported AWS resources](#) in the account. Once enabled, metrics that contain a resource identifier dimension (for example, EC2 CPUUtilization with an InstanceId dimension) are enriched with resource ARN and resource tag labels and become queryable using PromQL.

Before calling this operation, you must enable resource tags on telemetry for your account. For more information, see [Enable resource tags on telemetry](#).

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StopMetricStreams

Stops the streaming of metrics for one or more of your metric streams.

Request Parameters

Names

The array of the names of metric streams to stop streaming.

This is an "all or nothing" operation. If you do not have permission to access all of the metric streams that you list here, then none of the streams that you list in the operation will stop streaming.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

MissingParameter

An input parameter that is required is missing.

message

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StopOTelEnrichment

Disables enrichment and PromQL access for CloudWatch vended metrics for [supported AWS resources](#) in the account. After disabling, these metrics are no longer enriched with resource ARN and resource tag labels, and cannot be queried using PromQL.

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

TagResource

Assigns one or more tags (key-value pairs) to the specified CloudWatch resource. Currently, the only CloudWatch resources that can be tagged are alarms, dashboards, metric streams and Contributor Insights rules.

Tags can help you organize and categorize your resources. You can also use them to scope user permissions by granting a user permission to access or change only resources with certain tag values.

Tags don't have any semantic meaning to AWS and are interpreted strictly as strings of characters.

You can use the `TagResource` action with an alarm that already has tags. If you specify a new tag key for the alarm, this tag is appended to the list of tags associated with the alarm. If you specify a tag key that is already associated with the alarm, the new tag value that you specify replaces the previous value for that tag.

You can associate as many as 50 tags with a CloudWatch resource.

Request Parameters

ResourceARN

The ARN of the CloudWatch resource that you're adding tags to.

The ARN format of an alarm is `arn:aws:cloudwatch:Region:account-id:alarm:alarm-name`

The ARN format of a Contributor Insights rule is `arn:aws:cloudwatch:Region:account-id:insight-rule/insight-rule-name`

The ARN format of a dashboard is `arn:aws:cloudwatch::account-id:dashboard/dashboard-name`

The ARN format of a metric stream is `arn:aws:cloudwatch:Region:account-id:metric-stream/metric-stream-name`

For more information about ARN format, see [Resource Types Defined by Amazon CloudWatch](#) in the *Amazon Web Services General Reference*.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

Tags

The list of key-value pairs to associate with the alarm.

Type: Array of [Tag](#) objects

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConcurrentModificationException

More than one process tried to modify a resource at the same time.

HTTP Status Code: 429

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UntagResource

Removes one or more tags from the specified resource. Currently, alarms, dashboards, metric streams and Contributor Insights rules support tagging.

Request Parameters

ResourceARN

The ARN of the CloudWatch resource that you're removing tags from.

The ARN format of an alarm is `arn:aws:cloudwatch:Region:account-id:alarm:alarm-name`

The ARN format of a Contributor Insights rule is `arn:aws:cloudwatch:Region:account-id:insight-rule/insight-rule-name`

The ARN format of a dashboard is `arn:aws:cloudwatch::account-id:dashboard/dashboard-name`

The ARN format of a metric stream is `arn:aws:cloudwatch:Region:account-id:metric-stream/metric-stream-name`

For more information about ARN format, see [Resource Types Defined by Amazon CloudWatch](#) in the *Amazon Web Services General Reference*.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

TagKeys

The list of tag keys to remove from the resource.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 128.

Required: Yes

Errors

For information about the errors that are common to all actions, see [Common Error Types](#).

ConcurrentModificationException

More than one process tried to modify a resource at the same time.

HTTP Status Code: 429

ConflictException

This operation attempted to create a resource that already exists.

HTTP Status Code: 409

InternalServerError

Request processing has failed due to some unknown error, exception, or failure.

Message

HTTP Status Code: 500

InvalidParameterValue

The value of an input parameter is bad or out-of-range.

message

HTTP Status Code: 400

ResourceNotFoundException

The named resource does not exist.

HTTP Status Code: 404

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)

- [AWS SDK for .NET V4](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

Data Types

The Amazon CloudWatch API contains several data types that various actions use. This section describes each data type in detail.

Note

The order of each element in a data type structure is not guaranteed. Applications should not assume a particular order.

The following data types are supported:

- [AlarmContributor](#)
- [AlarmHistoryItem](#)
- [AlarmMuteRuleSummary](#)
- [AlarmPromQLCriteria](#)
- [AnomalyDetector](#)
- [AnomalyDetectorConfiguration](#)
- [CompositeAlarm](#)
- [DashboardEntry](#)
- [DashboardValidationMessage](#)
- [Datapoint](#)
- [Dimension](#)
- [DimensionFilter](#)
- [Entity](#)
- [EntityMetricData](#)
- [EvaluationCriteria](#)
- [EvaluationWindow](#)
- [InsightRule](#)
- [InsightRuleContributor](#)
- [InsightRuleContributorDatapoint](#)
- [InsightRuleMetricDatapoint](#)

- [LabelOptions](#)
- [LogAlarm](#)
- [ManagedRule](#)
- [ManagedRuleDescription](#)
- [ManagedRuleState](#)
- [MessageData](#)
- [Metric](#)
- [MetricAlarm](#)
- [MetricCharacteristics](#)
- [MetricDataQuery](#)
- [MetricDataResult](#)
- [MetricDatum](#)
- [MetricMathAnomalyDetector](#)
- [MetricStat](#)
- [MetricStreamEntry](#)
- [MetricStreamFilter](#)
- [MetricStreamStatisticsConfiguration](#)
- [MetricStreamStatisticsMetric](#)
- [MuteTargets](#)
- [PartialFailure](#)
- [Range](#)
- [Rule](#)
- [Schedule](#)
- [ScheduleConfiguration](#)
- [ScheduledQueryConfiguration](#)
- [SingleMetricAnomalyDetector](#)
- [SlidingWindow](#)
- [StatisticSet](#)
- [Tag](#)
- [WallClockWindow](#)

AlarmContributor

Represents an individual contributor to a multi-timeseries alarm, containing information about a specific time series and its contribution to the alarm's state.

Contents

ContributorAttributes

A map of attributes that describe the contributor, such as metric dimensions and other identifying characteristics.

Type: String to string map

Map Entries: Minimum number of 0 items. Maximum number of 150 items.

Key Length Constraints: Minimum length of 1. Maximum length of 255.

Value Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

ContributorId

The unique identifier for this alarm contributor.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 16.

Required: Yes

StateReason

An explanation for the contributor's current state, providing context about why it is in its current condition.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1023.

Required: Yes

StateTransitionedTimestamp

The timestamp when the contributor last transitioned to its current state.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AlarmHistoryItem

Represents the history of a specific alarm.

Contents

AlarmContributorAttributes

A map of attributes that describe the alarm contributor associated with this history item, providing context about the contributor's characteristics at the time of the event.

Type: String to string map

Map Entries: Minimum number of 0 items. Maximum number of 150 items.

Key Length Constraints: Minimum length of 1. Maximum length of 255.

Value Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmContributorId

The unique identifier of the alarm contributor associated with this history item, if applicable.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 16.

Required: No

AlarmName

The descriptive name for the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

AlarmType

The type of alarm, either metric alarm or composite alarm.

Type: String

Valid Values: CompositeAlarm | MetricAlarm | LogAlarm

Required: No

HistoryData

Data about the alarm, in JSON format.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 4095.

Required: No

HistoryItemType

The type of alarm history item.

Type: String

Valid Values: ConfigurationUpdate | StateUpdate | Action | AlarmContributorStateUpdate | AlarmContributorAction

Required: No

HistorySummary

A summary of the alarm history, in text format.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Timestamp

The time stamp for the alarm history item.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AlarmMuteRuleSummary

Summary information about an alarm mute rule, including its name, status, and configuration details.

Contents

AlarmMuteRuleArn

The Amazon Resource Name (ARN) of the alarm mute rule.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

ExpireDate

The date and time when the mute rule expires and is no longer evaluated. This field is only present if an expiration date was configured.

Type: Timestamp

Required: No

LastUpdatedTimestamp

The date and time when the mute rule was last updated.

Type: Timestamp

Required: No

MuteType

Indicates whether the mute rule is one-time or recurring. Valid values are ONE_TIME or RECURRING.

Type: String

Required: No

Status

The current status of the alarm mute rule. Valid values are SCHEDULED, ACTIVE, or EXPIRED.

Type: String

Valid Values: SCHEDULED | ACTIVE | EXPIRED

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AlarmPromQLCriteria

Contains the configuration that determines how a PromQL alarm evaluates its contributors, including the query to run and the durations that define when contributors transition between states.

Contents

Query

The PromQL query that the alarm evaluates. The query must return a result of vector type. Each entry in the vector result represents an alarm contributor.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 10000.

Required: Yes

PendingPeriod

The duration, in seconds, that a contributor must be continuously breaching before it transitions to the ALARM state.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 86400.

Required: No

RecoveryPeriod

The duration, in seconds, that a contributor must continuously not be breaching before it transitions back to the OK state.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 86400.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AnomalyDetector

An anomaly detection model associated with a particular CloudWatch metric, statistic, or metric math expression. You can use the model to display a band of expected, normal values when the metric is graphed.

If you have enabled unified cross-account observability, and this account is a monitoring account, the metric can be in the same account or a source account.

Contents

Configuration

The configuration specifies details about how the anomaly detection model is to be trained, including time ranges to exclude from use for training the model, and the time zone to use for the metric.

Type: [AnomalyDetectorConfiguration](#) object

Required: No

Dimensions

This member has been deprecated.

The metric dimensions associated with the anomaly detection model.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MetricCharacteristics

This object includes parameters that you can use to provide information about your metric to CloudWatch to help it build more accurate anomaly detection models. Currently, it includes the `PeriodicSpikes` parameter.

Type: [MetricCharacteristics](#) object

Required: No

MetricMathAnomalyDetector

The CloudWatch metric math expression for this anomaly detector.

Type: [MetricMathAnomalyDetector](#) object

Required: No

MetricName

This member has been deprecated.

The name of the metric associated with the anomaly detection model.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

This member has been deprecated.

The namespace of the metric associated with the anomaly detection model.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: No

SingleMetricAnomalyDetector

The CloudWatch metric and statistic for this anomaly detector.

Type: [SingleMetricAnomalyDetector](#) object

Required: No

Stat

This member has been deprecated.

The statistic associated with the anomaly detection model.

Type: String

Length Constraints: Maximum length of 50.

Pattern: (SampleCount|Average|Sum|Minimum|Maximum|IQM|(p|tc|tm|ts|wm)(\d{1,2}(\.\d{0,10})?|100)|[ou]\d+(\.\d*)?)(_E|_L|_H)?|(TM|TC|TS|WM)\((((((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%)?:((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%|((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):(((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%)?)\)|((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]? \d+)):(\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]? \d+)))?|((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]? \d+)))?:(\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]? \d+)))\)

Required: No

StateValue

The current status of the anomaly detector's training.

Type: String

Valid Values: PENDING_TRAINING | TRAINED_INSUFFICIENT_DATA | TRAINED

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AnomalyDetectorConfiguration

The configuration specifies details about how the anomaly detection model is to be trained, including time ranges to exclude from use for training the model and the time zone to use for the metric.

Contents

ExcludedTimeRanges

An array of time ranges to exclude from use when the anomaly detection model is trained. Use this to make sure that events that could cause unusual values for the metric, such as deployments, aren't used when CloudWatch creates the model.

Type: Array of [Range](#) objects

Required: No

MetricTimezone

The time zone to use for the metric. This is useful to enable the model to automatically account for daylight savings time changes if the metric is sensitive to such time changes.

To specify a time zone, use the name of the time zone as specified in the standard tz database. For more information, see [tz database](#).

Type: String

Length Constraints: Maximum length of 50.

Pattern: .*

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

CompositeAlarm

The details about a composite alarm.

Contents

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state.

Type: Boolean

Required: No

ActionsSuppressedBy

When the value is `ALARM`, it means that the actions are suppressed because the suppressor alarm is in `ALARM`. When the value is `WaitPeriod`, it means that the actions are suppressed because the composite alarm is waiting for the suppressor alarm to go into the `ALARM` state. The maximum waiting time is as specified in `ActionsSuppressorWaitPeriod`. After this time, the composite alarm performs its actions. When the value is `ExtensionPeriod`, it means that the actions are suppressed because the composite alarm is waiting after the suppressor alarm went out of the `ALARM` state. The maximum waiting time is as specified in `ActionsSuppressorExtensionPeriod`. After this time, the composite alarm performs its actions.

Type: String

Valid Values: `WaitPeriod` | `ExtensionPeriod` | `Alarm`

Required: No

ActionsSuppressedReason

Captures the reason for action suppression.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

ActionsSuppressor

Actions will be suppressed if the suppressor alarm is in the ALARM state. `ActionsSuppressor` can be an `AlarmName` or an Amazon Resource Name (ARN) from an existing alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

ActionsSuppressorExtensionPeriod

The maximum time in seconds that the composite alarm waits after suppressor alarm goes out of the ALARM state. After this time, the composite alarm performs its actions.

Important

`ExtensionPeriod` is required only when `ActionsSuppressor` is specified.

Type: Integer

Required: No

ActionsSuppressorWaitPeriod

The maximum time in seconds that the composite alarm waits for the suppressor alarm to go into the ALARM state. After this time, the composite alarm performs its actions.

Important

`WaitPeriod` is required only when `ActionsSuppressor` is specified.

Type: Integer

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmArn

The Amazon Resource Name (ARN) of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

AlarmConfigurationUpdatedTimestamp

The time stamp of the last update to the alarm configuration.

Type: Timestamp

Required: No

AlarmDescription

The description of the alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

AlarmRule

The rule that this alarm uses to evaluate its alarm state.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 10240.

Required: No

InsufficientDataActions

The actions to execute when this alarm transitions to the INSUFFICIENT_DATA state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

OKActions

The actions to execute when this alarm transitions to the OK state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

StateReason

An explanation for the alarm state, in text format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1023.

Required: No

StateReasonData

An explanation for the alarm state, in JSON format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 4000.

Required: No

StateTransitionedTimestamp

The timestamp of the last change to the alarm's StateValue.

Type: Timestamp

Required: No

StateUpdatedTimestamp

Tracks the timestamp of any state update, even if StateValue doesn't change.

Type: Timestamp

Required: No

StateValue

The state value for the alarm.

Type: String

Valid Values: OK | ALARM | INSUFFICIENT_DATA

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DashboardEntry

Represents a specific dashboard.

Contents

DashboardArn

The Amazon Resource Name (ARN) of the dashboard.

Type: String

Required: No

DashboardName

The name of the dashboard.

Type: String

Required: No

LastModified

The time stamp of when the dashboard was last modified, either by an API call or through the console. This number is expressed as the number of milliseconds since Jan 1, 1970 00:00:00 UTC.

Type: Timestamp

Required: No

Size

The size of the dashboard, in bytes.

Type: Long

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DashboardValidationMessage

An error or warning for the operation.

Contents

DataPath

The data path related to the message.

Type: String

Required: No

Message

A message describing the error or warning.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Datapoint

Encapsulates the statistical data that CloudWatch computes from metric data.

Contents

Average

The average of the metric values that correspond to the data point.

Type: Double

Required: No

ExtendedStatistics

The percentile statistic for the data point.

Type: String to double map

Required: No

Maximum

The maximum metric value for the data point.

Type: Double

Required: No

Minimum

The minimum metric value for the data point.

Type: Double

Required: No

SampleCount

The number of metric values that contributed to the aggregate value of this data point.

Type: Double

Required: No

Sum

The sum of the metric values for the data point.

Type: Double

Required: No

Timestamp

The time stamp used for the data point.

Type: Timestamp

Required: No

Unit

The standard unit for the data point.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Dimension

A dimension is a name/value pair that is part of the identity of a metric. Because dimensions are part of the unique identifier for a metric, whenever you add a unique name/value pair to one of your metrics, you are creating a new variation of that metric. For example, many Amazon EC2 metrics publish InstanceId as a dimension name, and the actual instance ID as the value for that dimension.

You can assign up to 30 dimensions to a metric.

Contents

Name

The name of the dimension. Dimension names must contain only ASCII characters, must include at least one non-whitespace character, and cannot start with a colon (:). ASCII control characters are not supported as part of dimension names.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Value

The value of the dimension. Dimension values must contain only ASCII characters and must include at least one non-whitespace character. ASCII control characters are not supported as part of dimension values.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DimensionFilter

Represents filters for a dimension.

Contents

Name

The dimension name to be matched.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Value

The value of the dimension to be matched.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Entity

An entity associated with metrics, to allow for finding related telemetry. An entity is typically a resource or service within your system. For example, metrics from an Amazon EC2 instance could be associated with that instance as the entity. Similarly, metrics from a service that you own could be associated with that service as the entity.

Contents

Attributes

Additional attributes of the entity that are not used to specify the identity of the entity. A list of key-value pairs.

For details about how to use the attributes, see [How to add related information to telemetry](#) in the *CloudWatch User Guide*.

Type: String to string map

Map Entries: Minimum number of 0 items. Maximum number of 10 items.

Key Length Constraints: Minimum length of 1. Maximum length of 256.

Value Length Constraints: Minimum length of 1. Maximum length of 2048.

Required: No

KeyAttributes

The attributes of the entity which identify the specific entity, as a list of key-value pairs. Entities with the same KeyAttributes are considered to be the same entity. For an entity to be valid, the KeyAttributes must exist and be formatted correctly.

There are five allowed attributes (key names): Type, ResourceType, Identifier, Name, and Environment.

For details about how to use the key attributes to specify an entity, see [How to add related information to telemetry](#) in the *CloudWatch User Guide*.

Type: String to string map

Map Entries: Minimum number of 2 items. Maximum number of 4 items.

Key Length Constraints: Minimum length of 1. Maximum length of 32.

Value Length Constraints: Minimum length of 1. Maximum length of 2048.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EntityMetricData

A set of metrics that are associated with an entity, such as a specific service or resource. Contains the entity and the list of metric data associated with it.

Contents

Entity

The entity associated with the metrics.

Type: [Entity](#) object

Required: No

MetricData

The metric data.

Type: Array of [MetricDatum](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EvaluationCriteria

The evaluation criteria for an alarm. This is a union type that currently supports PromQLCriteria.

Contents

Important

This data type is a UNION, so only one of the following members can be specified when used or returned.

PromQLCriteria

The PromQL criteria for the alarm evaluation.

Type: [AlarmPromQLCriteria](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EvaluationWindow

The evaluation window that an alarm uses to select the range of metric data that it evaluates each time it runs. This is a union type. Set exactly one of its members, `SlidingWindow` or `WallClockWindow`. If you don't set `EvaluationWindow`, the alarm uses a `SlidingWindow` by default.

For more information, see [Alarm evaluation windows](#) in the *CloudWatch User Guide*.

Contents

Important

This data type is a UNION, so only one of the following members can be specified when used or returned.

SlidingWindow

A sliding window, which advances each time the alarm is evaluated, forming a rolling time window. This is the default evaluation window.

Type: [SlidingWindow](#) object

Required: No

WallClockWindow

A wall clock window, which aligns the evaluated range to fixed clock boundaries that match the alarm's period, such as the top of the hour, midnight, or the start of the calendar week.

Type: [WallClockWindow](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InsightRule

This structure contains the definition for a Contributor Insights rule. For more information about this rule, see [Using Contributor Insights to analyze high-cardinality data](#) in the *Amazon CloudWatch User Guide*.

Contents

Definition

The definition of the rule, as a JSON object. The definition contains the keywords used to define contributors, the value to aggregate on if this rule returns a sum instead of a count, and the filters. For details on the valid syntax, see [Contributor Insights Rule Syntax](#).

Type: String

Length Constraints: Minimum length of 1. Maximum length of 8192.

Pattern: `[\x00-\x7F]+`

Required: Yes

Name

The name of the rule.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

Schema

For rules that you create, this is always `{"Name": "CloudWatchLogRule", "Version": 1}`. For managed rules, this is `{"Name": "ServiceLogRule", "Version": 1}`

Type: String

Required: Yes

State

Indicates whether the rule is enabled or disabled.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 32.

Pattern: `[\x20-\x7E]+`

Required: Yes

ApplyOnTransformedLogs

Displays whether the rule is evaluated on the transformed versions of logs, for log groups that have [Log transformation](#) enabled. If this is false, log events are evaluated before they are transformed.

Type: Boolean

Required: No

ManagedRule

An optional built-in rule that AWS manages.

Type: Boolean

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InsightRuleContributor

One of the unique contributors found by a Contributor Insights rule. If the rule contains multiple keys, then a unique contributor is a unique combination of values from all the keys in the rule.

If the rule contains a single key, then each unique contributor is each unique value for this key.

For more information, see [GetInsightRuleReport](#).

Contents

ApproximateAggregateValue

An approximation of the aggregate value that comes from this contributor.

Type: Double

Required: Yes

Datapoints

An array of the data points where this contributor is present. Only the data points when this contributor appeared are included in the array.

Type: Array of [InsightRuleContributorDatapoint](#) objects

Required: Yes

Keys

One of the log entry field keywords that is used to define contributors for this rule.

Type: Array of strings

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InsightRuleContributorDatapoint

One data point related to one contributor.

For more information, see [GetInsightRuleReport](#) and [InsightRuleContributor](#).

Contents

ApproximateValue

The approximate value that this contributor added during this timestamp.

Type: Double

Required: Yes

Timestamp

The timestamp of the data point.

Type: Timestamp

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InsightRuleMetricDatapoint

One data point from the metric time series returned in a Contributor Insights rule report.

For more information, see [GetInsightRuleReport](#).

Contents

Timestamp

The timestamp of the data point.

Type: Timestamp

Required: Yes

Average

The average value from all contributors during the time period represented by that data point.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

MaxContributorValue

The maximum value provided by one contributor during this timestamp. Each timestamp is evaluated separately, so the identity of the max contributor could be different for each timestamp.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

Maximum

The maximum value from a single occurrence from a single contributor during the time period represented by that data point.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

Minimum

The minimum value from a single contributor during the time period represented by that data point.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

SampleCount

The number of occurrences that matched the rule during this data point.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

Sum

The sum of the values from all contributors during the time period represented by that data point.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

UniqueContributors

The number of unique contributors who published data during this timestamp.

This statistic is returned only if you included it in the `Metrics` array in your request.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LabelOptions

This structure includes the `Timezone` parameter, which you can use to specify your time zone so that the labels that are associated with returned metrics display the correct time for your time zone.

The `Timezone` value affects a label only if you have a time-based dynamic expression in the label. For more information about dynamic expressions in labels, see [Using Dynamic Labels](#).

Contents

Timezone

The time zone to use for metric data return in this operation. The format is + or - followed by four digits. The first two digits indicate the number of hours ahead or behind of UTC, and the final two digits are the number of minutes. For example, +0130 indicates a time zone that is 1 hour and 30 minutes ahead of UTC. The default is +0000.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LogAlarm

The details about a log alarm.

Contents

ActionLogLineCount

The number of log lines from the most recent scheduled query execution that are included in alarm action notifications. Valid range is 0 through 50. A value of 0 means no log lines are included.

Type: Integer

Required: No

ActionLogLineRoleArn

The Amazon Resource Name (ARN) of the IAM role that CloudWatch assumes to retrieve log events for inclusion in alarm action notifications. Set when ActionLogLineCount is greater than 0.

Type: String

Required: No

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state.

Type: Boolean

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmArn

The Amazon Resource Name (ARN) of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

AlarmConfigurationUpdatedTimestamp

The time stamp of the last update to the alarm configuration.

Type: Timestamp

Required: No

AlarmDescription

The description of the alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

ComparisonOperator

The arithmetic operation to use when comparing the aggregated query result and the threshold. The aggregated query result is used as the first operand.

Type: String

Valid Values: `GreaterThanOrEqualToThreshold` | `GreaterThanThreshold` | `LessThanThreshold` | `LessThanOrEqualToThreshold` | `LessThanLowerOrGreaterThanUpperThreshold` | `LessThanLowerThreshold` | `GreaterThanUpperThreshold`

Required: No

EvaluationState

If the value of this field is `EVALUATION_ERROR`, it indicates configuration errors in the alarm setup that require review and correction. Refer to the `StateReason` field of the alarm for more details.

If the value of this field is `EVALUATION_FAILURE`, it indicates temporary CloudWatch issues. We recommend manual monitoring until the issue is resolved.

If the value of this field is `PARTIAL_DATA`, it indicates that the query returned the maximum 500 contributor groups but more matched. The alarm evaluates the available contributors, but results might be incomplete.

Type: String

Valid Values: `PARTIAL_DATA` | `EVALUATION_FAILURE` | `EVALUATION_ERROR`

Required: No

InsufficientDataActions

The actions to execute when this alarm transitions to the `INSUFFICIENT_DATA` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

OKActions

The actions to execute when this alarm transitions to the `OK` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

QueryResultsToAlarm

The number of query results, out of the most recent `QueryResultsToEvaluate` results, that must breach the threshold to trigger the alarm to transition to ALARM (the M in M-of-N evaluation).

Type: Integer

Valid Range: Minimum value of 1.

Required: No

QueryResultsToEvaluate

The number of most recent scheduled query results that the alarm evaluates against the threshold (the N in M-of-N evaluation).

Type: Integer

Valid Range: Minimum value of 1.

Required: No

ScheduledQueryConfiguration

The configuration of the underlying CloudWatch Logs scheduled query, including the query string, log groups, schedule, aggregation expression, and the ARN of the managed scheduled query.

Type: [ScheduledQueryConfiguration](#) object

Required: No

StateReason

An explanation for the alarm state, in text format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1023.

Required: No

StateReasonData

An explanation for the alarm state, in JSON format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 4000.

Required: No

StateTransitionedTimestamp

The date and time that the alarm's StateValue most recently changed.

Type: Timestamp

Required: No

StateUpdatedTimestamp

The time stamp of the last update to the value of either the StateValue or EvaluationState parameters.

Type: Timestamp

Required: No

StateValue

The state value for the alarm.

Type: String

Valid Values: OK | ALARM | INSUFFICIENT_DATA

Required: No

Threshold

The value to compare with the aggregated query result.

Type: Double

Required: No

TreatMissingData

How this alarm handles missing data points. Valid values are `breaching`, `notBreaching`, `ignore`, and `missing`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedRule

Contains the information that's required to enable a managed Contributor Insights rule for an AWS resource.

Contents

ResourceARN

The ARN of an AWS resource that has managed Contributor Insights rules.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

TemplateName

The template name for the managed Contributor Insights rule, as returned by `ListManagedInsightRules`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[0-9A-Za-z][\-\.\_0-9A-Za-z]{0,126}[0-9A-Za-z]`

Required: Yes

Tags

A list of key-value pairs that you can associate with a managed Contributor Insights rule. You can associate as many as 50 tags with a rule. Tags can help you organize and categorize your resources. You also can use them to scope user permissions by granting a user permission to access or change only the resources that have certain tag values. To associate tags with a rule, you must have the `cloudwatch:TagResource` permission in addition to the `cloudwatch:PutInsightRule` permission. If you are using this operation to update an existing Contributor Insights rule, any tags that you specify in this parameter are ignored. To change the tags of an existing rule, use `TagResource`.

Type: Array of [Tag](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedRuleDescription

Contains information about managed Contributor Insights rules, as returned by `ListManagedInsightRules`.

Contents

ResourceARN

If a managed rule is enabled, this is the ARN for the related AWS resource.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

RuleState

Describes the state of a managed rule. If present, it contains information about the Contributor Insights rule that contains information about the related AWS resource.

Type: [ManagedRuleState](#) object

Required: No

TemplateName

The template name for the managed rule. Used to enable managed rules using `PutManagedInsightRules`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[0-9A-Za-z][\-\.\_0-9A-Za-z]{0,126}[0-9A-Za-z]`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedRuleState

The status of a managed Contributor Insights rule.

Contents

RuleName

The name of the Contributor Insights rule that contains data for the specified AWS resource.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `[\x20-\x7E]+`

Required: Yes

State

Indicates whether the rule is enabled or disabled.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 32.

Pattern: `[\x20-\x7E]+`

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MessageData

A message returned by the `GetMetricData` API, including a code and a description.

If a cross-Region `GetMetricData` operation fails with a code of `Forbidden` and a value of `Authentication too complex to retrieve cross region data`, you can correct the problem by running the `GetMetricData` operation in the same Region where the metric data is.

Contents

Code

The error code or status code associated with the message.

Type: String

Required: No

Value

The message text.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Metric

Represents a specific metric.

Contents

Dimensions

The dimensions for the metric.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MetricName

The name of the metric. This is a required field.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

The namespace of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `^[^:]*`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricAlarm

The details about a metric alarm.

Contents

ActionsEnabled

Indicates whether actions should be executed during any changes to the alarm state.

Type: Boolean

Required: No

AlarmActions

The actions to execute when this alarm transitions to the ALARM state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

AlarmArn

The Amazon Resource Name (ARN) of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1600.

Required: No

AlarmConfigurationUpdatedTimestamp

The time stamp of the last update to the alarm configuration.

Type: Timestamp

Required: No

AlarmDescription

The description of the alarm.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1024.

Required: No

AlarmName

The name of the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

ComparisonOperator

The arithmetic operation to use when comparing the specified statistic and threshold. The specified statistic value is used as the first operand.

Type: String

Valid Values: `GreaterThanOrEqualToThreshold` | `GreaterThanThreshold` | `LessThanThreshold` | `LessThanOrEqualToThreshold` | `LessThanLowerOrGreaterThanUpperThreshold` | `LessThanLowerThreshold` | `GreaterThanUpperThreshold`

Required: No

DatapointsToAlarm

The number of data points that must be breaching to trigger the alarm.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

Dimensions

The dimensions for the metric associated with the alarm.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

EvaluateLowSampleCountPercentile

Used only for alarms based on percentiles. If `ignore`, the alarm state does not change during periods with too few data points to be statistically significant. If `evaluate` or this parameter is not used, the alarm is always evaluated and possibly changes state no matter how many data points are available.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

EvaluationCriteria

The evaluation criteria for the alarm.

Type: [EvaluationCriteria](#) object

Note: This object is a Union. Only one member of this object can be specified or returned.

Required: No

EvaluationInterval

The frequency, in seconds, at which the alarm is evaluated.

Type: Integer

Valid Range: Minimum value of 10. Maximum value of 3600.

Required: No

EvaluationPeriods

The number of periods over which data is compared to the specified threshold.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

EvaluationState

If the value of this field is `PARTIAL_DATA`, it indicates that not all the available data was able to be retrieved due to quota limitations. For more information, see [Create alarms on Metrics Insights queries](#).

If the value of this field is `EVALUATION_ERROR`, it indicates configuration errors in alarm setup that require review and correction. Refer to `StateReason` field of the alarm for more details.

If the value of this field is `EVALUATION_FAILURE`, it indicates temporary CloudWatch issues. We recommend manual monitoring until the issue is resolved

Type: String

Valid Values: `PARTIAL_DATA` | `EVALUATION_FAILURE` | `EVALUATION_ERROR`

Required: No

EvaluationWindow

The evaluation window that the alarm uses to select the range of metric data that it evaluates. This is either a sliding window or a wall clock window. For more information, see [Alarm evaluation windows](#) in the *CloudWatch User Guide*.

Type: [EvaluationWindow](#) object

Note: This object is a Union. Only one member of this object can be specified or returned.

Required: No

ExtendedStatistic

The percentile statistic for the metric associated with the alarm. Specify a value between `p0.0` and `p100`.

Type: String

Required: No

InsufficientDataActions

The actions to execute when this alarm transitions to the `INSUFFICIENT_DATA` state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

MetricName

The name of the metric associated with the alarm, if this is an alarm based on a single metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Metrics

An array of `MetricDataQuery` structures, used in an alarm based on a metric math expression. Each structure either retrieves a metric or performs a math expression. One item in the `Metrics` array is the math expression that the alarm watches. This expression is designated by having `ReturnData` set to true.

Type: Array of [MetricDataQuery](#) objects

Required: No

Namespace

The namespace of the metric associated with the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `^[^:]*`

Required: No

OKActions

The actions to execute when this alarm transitions to the OK state from any other state. Each action is specified as an Amazon Resource Name (ARN).

Type: Array of strings

Array Members: Maximum number of 5 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

Period

The period, in seconds, over which the statistic is applied.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

StateReason

An explanation for the alarm state, in text format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 1023.

Required: No

StateReasonData

An explanation for the alarm state, in JSON format.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 4000.

Required: No

StateTransitionedTimestamp

The date and time that the alarm's StateValue most recently changed.

Type: Timestamp

Required: No

StateUpdatedTimestamp

The time stamp of the last update to the value of either the StateValue or EvaluationState parameters.

Type: Timestamp

Required: No

StateValue

The state value for the alarm.

Type: String

Valid Values: OK | ALARM | INSUFFICIENT_DATA

Required: No

Statistic

The statistic for the metric associated with the alarm, other than percentile. For percentile statistics, use ExtendedStatistic.

Type: String

Valid Values: SampleCount | Average | Sum | Minimum | Maximum

Required: No

Threshold

The value to compare with the specified statistic.

Type: Double

Required: No

ThresholdMetricId

In an alarm based on an anomaly detection model, this is the ID of the ANOMALY_DETECTION_BAND function used as the threshold for the alarm.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

TreatMissingData

Sets how this alarm is to handle missing data points. The valid values are breaching, notBreaching, ignore, and missing. For more information, see [Configuring how CloudWatch alarms treat missing data](#).

If this parameter is omitted, the default behavior of missing is used.

Note

This parameter is not applicable to PromQL alarms.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Unit

The unit of the metric associated with the alarm.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricCharacteristics

This object includes parameters that you can use to provide information to CloudWatch to help it build more accurate anomaly detection models.

Contents

PeriodicSpikes

Set this parameter to `true` if values for this metric consistently include spikes that should not be considered to be anomalies. With this set to `true`, CloudWatch will expect to see spikes that occurred consistently during the model training period, and won't flag future similar spikes as anomalies.

Type: Boolean

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricDataQuery

This structure is used in both `GetMetricData` and `PutMetricAlarm`. The supported use of this structure is different for those two operations.

When used in `GetMetricData`, it indicates the metric data to return, and whether this call is just retrieving a batch set of data for one metric, or is performing a Metrics Insights query or a math expression. A single `GetMetricData` call can include up to 500 `MetricDataQuery` structures.

When used in `PutMetricAlarm`, it enables you to create an alarm based on a metric math expression. Each `MetricDataQuery` in the array specifies either a metric to retrieve, or a math expression to be performed on retrieved metrics. A single `PutMetricAlarm` call can include up to 20 `MetricDataQuery` structures in the array. The 20 structures can include as many as 10 structures that contain a `MetricStat` parameter to retrieve a metric, and as many as 10 structures that contain the `Expression` parameter to perform a math expression. Of those `Expression` structures, one must have `true` as the value for `ReturnData`. The result of this expression is the value the alarm watches.

Any expression used in a `PutMetricAlarm` operation must return a single time series. For more information, see [Metric Math Syntax and Functions](#) in the *Amazon CloudWatch User Guide*.

Some of the parameters of this structure also have different uses whether you are using this structure in a `GetMetricData` operation or a `PutMetricAlarm` operation. These differences are explained in the following parameter list.

Contents

Id

A short name used to tie this object to the results in the response. This name must be unique within a single call to `GetMetricData`. If you are performing math expressions on this set of data, this name represents that data and can serve as a variable in the mathematical expression. The valid characters are letters, numbers, and underscore. The first character must be a lowercase letter.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

AccountId

The ID of the account where the metrics are located.

If you are performing a `GetMetricData` operation in a monitoring account, use this to specify which account to retrieve this metric from.

If you are performing a `PutMetricAlarm` operation, use this to specify which account contains the metric that the alarm is watching.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Expression

This field can contain either a Metrics Insights query, or a metric math expression to be performed on the returned data. For more information about Metrics Insights queries, see [Metrics Insights query components and syntax](#) in the *Amazon CloudWatch User Guide*.

A math expression can use the `Id` of the other metrics or queries to refer to those metrics, and can also use the `Id` of other expressions to use the result of those expressions. For more information about metric math expressions, see [Metric Math Syntax and Functions](#) in the *Amazon CloudWatch User Guide*.

Within each `MetricDataQuery` object, you must specify either `Expression` or `MetricStat` but not both.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 2048.

Required: No

Label

A human-readable label for this metric or expression. This is especially useful if this is an expression, so that you know what the value represents. If the metric or expression is shown in a CloudWatch dashboard widget, the label is shown. If `Label` is omitted, CloudWatch generates a default.

You can put dynamic expressions into a label, so that it is more descriptive. For more information, see [Using Dynamic Labels](#).

Type: String

Required: No

MetricStat

The metric to be returned, along with statistics, period, and units. Use this parameter only if this object is retrieving a metric and not performing a math expression on returned data.

Within one `MetricDataQuery` object, you must specify either `Expression` or `MetricStat` but not both.

Type: [MetricStat](#) object

Required: No

Period

The granularity, in seconds, of the returned data points. For metrics with regular resolution, a period can be as short as one minute (60 seconds) and must be a multiple of 60. For high-resolution metrics that are collected at intervals of less than one minute, the period can be 1, 5, 10, 20, 30, 60, or any multiple of 60. High-resolution metrics are those metrics stored by a `PutMetricData` operation that includes a `StorageResolution` of 1 second.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

ReturnData

When used in `GetMetricData`, this option indicates whether to return the timestamps and raw data values of this metric. If you are performing this call just to do math expressions and do not also need the raw data returned, you can specify `false`. If you omit this, the default of `true` is used.

When used in `PutMetricAlarm`, specify `true` for the one expression result to use as the alarm. For all other metrics and expressions in the same `PutMetricAlarm` operation, specify `ReturnData` as `False`.

Type: Boolean

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricDataResult

A `GetMetricData` call returns an array of `MetricDataResult` structures. Each of these structures includes the data points for that metric, along with the timestamps of those data points and other identifying information.

Contents

Id

The short name you specified to represent this metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Label

The human-readable label associated with the data.

Type: String

Required: No

Messages

A list of messages with additional information about the data returned.

Type: Array of [MessageData](#) objects

Required: No

StatusCode

The status of the returned data. `Complete` indicates that all data points in the requested time range were returned. `PartialData` means that an incomplete set of data points were returned. You can use the `NextToken` value that was returned and repeat your request to get more data points. `NextToken` is not returned if you are performing a math expression. `InternalError` indicates that an error occurred. Retry your request using `NextToken`, if present.

Type: String

Valid Values: Complete | InternalError | PartialData | Forbidden

Required: No

Timestamps

The timestamps for the data points, formatted in Unix timestamp format. The number of timestamps always matches the number of values and the value for Timestamps[x] is Values[x].

Type: Array of timestamps

Required: No

Values

The data points for the metric corresponding to Timestamps. The number of values always matches the number of timestamps and the timestamp for Values[x] is Timestamps[x].

Type: Array of doubles

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricDatum

Encapsulates the information sent to either create a metric or add new values to be aggregated into an existing metric.

Contents

MetricName

The name of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Counts

Array of numbers that is used along with the `Values` array. Each number in the `Count` array is the number of times the corresponding value in the `Values` array occurred during the period.

If you omit the `Counts` array, the default of 1 is used as the value for each count. If you include a `Counts` array, it must include the same amount of values as the `Values` array.

Type: Array of doubles

Required: No

Dimensions

The dimensions associated with the metric.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

StatisticValues

The statistical values for the metric.

Type: [StatisticSet](#) object

Required: No

StorageResolution

Valid values are 1 and 60. Setting this to 1 specifies this metric as a high-resolution metric, so that CloudWatch stores the metric with sub-minute resolution down to one second. Setting this to 60 specifies this metric as a regular-resolution metric, which CloudWatch stores at 1-minute resolution. Currently, high resolution is available only for custom metrics. For more information about high-resolution metrics, see [High-Resolution Metrics](#) in the *Amazon CloudWatch User Guide*.

This field is optional, if you do not specify it the default of 60 is used.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

Timestamp

The time the metric data was received, expressed as the number of milliseconds since Jan 1, 1970 00:00:00 UTC.

Type: Timestamp

Required: No

Unit

When you are using a Put operation, this defines what unit you want to use when storing the metric.

In a Get operation, this displays the unit that is used for the metric.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

Value

The value for the metric.

Although the parameter accepts numbers of type `Double`, CloudWatch rejects values that are either too small or too large. Values must be in the range of -2^{360} to 2^{360} . In addition, special values (for example, NaN, +Infinity, -Infinity) are not supported.

Type: `Double`

Required: No

Values

Array of numbers representing the values for the metric during the period. Each unique value is listed just once in this array, and the corresponding number in the `Counts` array specifies the number of times that value occurred during the period. You can include up to 150 unique values in each `PutMetricData` action that specifies a `Values` array.

Although the `Values` array accepts numbers of type `Double`, CloudWatch rejects values that are either too small or too large. Values must be in the range of -2^{360} to 2^{360} . In addition, special values (for example, NaN, +Infinity, -Infinity) are not supported.

Type: Array of doubles

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricMathAnomalyDetector

Indicates the CloudWatch math expression that provides the time series the anomaly detector uses as input. The designated math expression must return a single time series.

Contents

MetricDataQueries

An array of metric data query structures that enables you to create an anomaly detector based on the result of a metric math expression. Each item in `MetricDataQueries` gets a metric or performs a math expression. One item in `MetricDataQueries` is the expression that provides the time series that the anomaly detector uses as input. Designate the expression by setting `ReturnData` to `true` for this object in the array. For all other expressions and metrics, set `ReturnData` to `false`. The designated expression must return a single time series.

Type: Array of [MetricDataQuery](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricStat

This structure defines the metric to be returned, along with the statistics, period, and units.

Contents

Metric

The metric to return, including the metric name, namespace, and dimensions.

Type: [Metric](#) object

Required: Yes

Period

The granularity, in seconds, of the returned data points. For metrics with regular resolution, a period can be as short as one minute (60 seconds) and must be a multiple of 60. For high-resolution metrics that are collected at intervals of less than one minute, the period can be 1, 5, 10, 20, 30, 60, or any multiple of 60. High-resolution metrics are those metrics stored by a `PutMetricData` call that includes a `StorageResolution` of 1 second.

If the `StartTime` parameter specifies a time stamp that is greater than 3 hours ago, you must specify the period as follows or no data points in that time range is returned:

- Start time between 3 hours and 15 days ago - Use a multiple of 60 seconds (1 minute).
- Start time between 15 and 63 days ago - Use a multiple of 300 seconds (5 minutes).
- Start time greater than 63 days ago - Use a multiple of 3600 seconds (1 hour).

Type: Integer

Valid Range: Minimum value of 1.

Required: Yes

Stat

The statistic to return. It can include any CloudWatch statistic or extended statistic.

Type: String

Required: Yes

Unit

When you are using a Put operation, this defines what unit you want to use when storing the metric.

In a Get operation, if you omit Unit then all data that was collected with any unit is returned, along with the corresponding units that were specified when the data was reported to CloudWatch. If you specify a unit, the operation returns only data that was collected with that unit specified. If you specify a unit that does not match the data collected, the results of the operation are null. CloudWatch does not perform unit conversions.

Type: String

Valid Values: Seconds | Microseconds | Milliseconds | Bytes | Kilobytes | Megabytes | Gigabytes | Terabytes | Bits | Kilobits | Megabits | Gigabits | Terabits | Percent | Count | Bytes/Second | Kilobytes/Second | Megabytes/Second | Gigabytes/Second | Terabytes/Second | Bits/Second | Kilobits/Second | Megabits/Second | Gigabits/Second | Terabits/Second | Count/Second | None

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricStreamEntry

This structure contains the configuration information about one metric stream.

Contents

Arn

The ARN of the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

CreationDate

The date that the metric stream was originally created.

Type: Timestamp

Required: No

FirehoseArn

The ARN of the Kinesis Firehose devlivery stream that is used for this metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

LastUpdateDate

The date that the configuration of this metric stream was most recently updated.

Type: Timestamp

Required: No

Name

The name of the metric stream.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

OutputFormat

The output format of this metric stream. Valid values are `json`, `opentelemetry1.0`, and `opentelemetry0.7`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Valid Values: `json` | `opentelemetry0.7` | `opentelemetry1.0`

Required: No

State

The current state of this stream. Valid values are `running` and `stopped`.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricStreamFilter

This structure contains a metric namespace and optionally, a list of metric names, to either include in a metric stream or exclude from a metric stream.

A metric stream's filters can include up to 1000 total names. This limit applies to the sum of namespace names and metric names in the filters. For example, this could include 10 metric namespace filters with 99 metrics each, or 20 namespace filters with 49 metrics specified in each filter.

Contents

MetricNames

The names of the metrics to either include or exclude from the metric stream.

If you omit this parameter, all metrics in the namespace are included or excluded, depending on whether this filter is specified as an exclude filter or an include filter.

Each metric name can contain only ASCII printable characters (ASCII range 32 through 126). Each metric name must contain at least one non-whitespace character.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

The name of the metric namespace for this filter.

The namespace can contain only ASCII printable characters (ASCII range 32 through 126). It must contain at least one non-whitespace character.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*.*`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricStreamStatisticsConfiguration

By default, a metric stream always sends the MAX, MIN, SUM, and SAMPLECOUNT statistics for each metric that is streamed. This structure contains information for one metric that includes additional statistics in the stream. For more information about statistics, see CloudWatch, listed in [CloudWatch statistics definitions](#).

Contents

AdditionalStatistics

The list of additional statistics that are to be streamed for the metrics listed in the `IncludeMetrics` array in this structure. This list can include as many as 20 statistics.

If the `OutputFormat` for the stream is `opentelemetry1.0` or `opentelemetry0.7`, the only valid values are `p??` percentile statistics such as `p90`, `p99` and so on.

If the `OutputFormat` for the stream is `json`, the valid values include the abbreviations for all of the statistics listed in [CloudWatch statistics definitions](#). For example, this includes `tm98`, `wm90`, `PR(:300)`, and so on.

Type: Array of strings

Required: Yes

IncludeMetrics

An array of metric name and namespace pairs that stream the additional statistics listed in the value of the `AdditionalStatistics` parameter. There can be as many as 100 pairs in the array.

All metrics that match the combination of metric name and namespace will be streamed with the additional statistics, no matter their dimensions.

Type: Array of [MetricStreamStatisticsMetric](#) objects

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MetricStreamStatisticsMetric

This object contains the information for one metric that is to be streamed with additional statistics.

Contents

MetricName

The name of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

Namespace

The namespace of the metric.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `^[^:]*`

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MuteTargets

Specifies which alarms an alarm mute rule applies to.

You can target up to 100 specific alarms by name. When a mute rule is active, the targeted alarms continue to evaluate metrics and transition between states, but their configured actions are muted.

Contents

AlarmNames

The list of alarm names that this mute rule targets. You can specify up to 100 alarm names.

Each alarm name must be between 1 and 255 characters in length. The alarm names must match existing alarms in your AWS account and region.

Type: Array of strings

Array Members: Maximum number of 100 items.

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

PartialFailure

This array is empty if the API operation was successful for all the rules specified in the request. If the operation could not process one of the rules, the following data is returned for each of those rules.

Contents

ExceptionType

The type of error.

Type: String

Required: No

FailureCode

The code of the error.

Type: String

Required: No

FailureDescription

A description of the error.

Type: String

Required: No

FailureResource

The specified rule that could not be deleted.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Range

Specifies one range of days or times to exclude from use for training an anomaly detection model.

Contents

EndTime

The end time of the range to exclude. The format is yyyy-MM-dd'T'HH:mm:ss. For example, 2019-07-01T23:59:59.

Type: Timestamp

Required: Yes

StartTime

The start time of the range to exclude. The format is yyyy-MM-dd'T'HH:mm:ss. For example, 2019-07-01T23:59:59.

Type: Timestamp

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Rule

Defines the schedule configuration for an alarm mute rule.

The rule contains a schedule that specifies when and how long alarms should be muted. The schedule can be a recurring pattern using cron expressions or a one-time mute window using at expressions.

Contents

Schedule

The schedule configuration that defines when the mute rule activates and how long it remains active.

Type: [Schedule](#) object

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Schedule

Specifies when and how long an alarm mute rule is active.

The schedule uses either a cron expression for recurring mute windows or an at expression for one-time mute windows. When the schedule activates, the mute rule mutes alarm actions for the specified duration.

Contents

Duration

The length of time that alarms remain muted when the schedule activates. The duration must be between 1 and 50 characters in length.

Specify the duration using ISO 8601 duration format with a minimum of 1 minute (PT1M) and maximum of 15 days (P15D).

Examples:

- PT4H - 4 hours for weekly system maintenance
- P2DT12H - 2 days and 12 hours for weekend muting from Friday 6:00 PM to Monday 6:00 AM
- PT6H - 6 hours for monthly database maintenance
- PT2H - 2 hours for nightly backup operations
- P7D - 7 days for annual company shutdown

The duration begins when the schedule expression time is reached. For recurring schedules, the duration applies to each occurrence.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 50.

Required: Yes

Expression

The schedule expression that defines when the mute rule activates. The expression must be between 1 and 256 characters in length.

You can use one of two expression formats:

- **Cron expressions** - For recurring mute windows. Format: `cron(Minutes Hours Day-of-month Month Day-of-week)`

Examples:

- `cron(0 2 * * *)` - Activates daily at 2:00 AM
- `cron(0 2 * * SUN)` - Activates every Sunday at 2:00 AM for weekly system maintenance
- `cron(0 1 1 * *)` - Activates on the first day of each month at 1:00 AM for monthly database maintenance
- `cron(0 18 * * FRI)` - Activates every Friday at 6:00 PM
- `cron(0 23 * * *)` - Activates every day at 11:00 PM during nightly backup operations

The characters `*`, `-`, and `,` are supported in all fields. English names can be used for the month (JAN-DEC) and day of week (SUN-SAT) fields.

- **At expressions** - For one-time mute windows. Format: `at(yyyy-MM-ddThh:mm)`

Examples:

- `at(2024-05-10T14:00)` - Activates once on May 10, 2024 at 2:00 PM during an active incident response session
- `at(2024-12-23T00:00)` - Activates once on December 23, 2024 at midnight during annual company shutdown

Type: String

Length Constraints: Minimum length of 1. Maximum length of 256.

Required: Yes

Timezone

The time zone to use when evaluating the schedule expression. The time zone must be between 1 and 50 characters in length.

Specify the time zone using standard timezone identifiers (for example, `America/New_York`, `Europe/London`, or `Asia/Tokyo`).

If you don't specify a time zone, UTC is used by default. The time zone affects how cron and at expressions are interpreted, as well as start and expire dates you specify

Examples:

- America/New_York - Eastern Time (US)
- America/Los_Angeles - Pacific Time (US)
- Europe/London - British Time
- Asia/Tokyo - Japan Standard Time
- UTC - Coordinated Universal Time

Type: String

Length Constraints: Minimum length of 1. Maximum length of 50.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ScheduleConfiguration

Contains the schedule expression and time-range offsets that define when a scheduled query runs and what time range each execution covers.

Contents

ScheduleExpression

The schedule expression that defines how often the underlying CloudWatch Logs scheduled query runs. Specify a `rate()` expression, for example `rate(5 minutes)`.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 256.

Required: Yes

EndTimeOffset

The offset, in seconds, before the scheduled execution time at which the query time range ends. Must be non-negative and less than `StartTimeOffset`. The default is 0.

Type: Long

Required: No

StartTimeOffset

The offset, in seconds, before the scheduled execution time at which the query time range begins. For example, an offset of 360 (6 minutes) on a query running at 12:05:00 starts the query time range at 11:59:00.

Type: Long

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ScheduledQueryConfiguration

The configuration of the CloudWatch Logs scheduled query that backs a log alarm.

Contents

AggregationExpression

The expression that defines how to aggregate query results into one or more scalar values for alarm evaluation. For example, `count(*)` or `avg(latency) by host | sort desc`. Length constraints: minimum 1 character, maximum 2048 characters.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 256.

Required: Yes

QueryString

The CloudWatch Logs query to execute on each scheduled run. Length constraints: maximum of 10,000 characters.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 10000.

Required: Yes

ScheduleConfiguration

The schedule and time-range offset configuration for the underlying scheduled query.

Type: [ScheduleConfiguration](#) object

Required: Yes

ScheduledQueryRoleARN

The Amazon Resource Name (ARN) of the IAM role that CloudWatch assumes when executing the scheduled query against the configured log groups.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: Yes

LogGroupIdentifiers

The log groups to query. Each entry can be a log group name or ARN. Use the ARN form when querying log groups in a different account (for example, when running cross-account queries from a monitoring account). The list must contain between 1 and 50 entries.

Type: Array of strings

Array Members: Minimum number of 1 item. Maximum number of 50 items.

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

QueryARN

The Amazon Resource Name (ARN) of the CloudWatch Logs scheduled query that the alarm uses. This field is populated in `DescribeAlarms` responses.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 1024.

Required: No

Tags

A list of key-value pairs to associate with the underlying scheduled query resource.

Type: Array of [Tag](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

SingleMetricAnomalyDetector

Designates the CloudWatch metric and statistic that provides the time series the anomaly detector uses as input. If you have enabled unified cross-account observability, and this account is a monitoring account, the metric can be in the same account or a source account.

Contents

AccountId

If the CloudWatch metric that provides the time series that the anomaly detector uses as input is in another account, specify that account ID here. If you omit this parameter, the current account is used.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Dimensions

The metric dimensions to create the anomaly detection model for.

Type: Array of [Dimension](#) objects

Array Members: Maximum number of 30 items.

Required: No

MetricName

The name of the metric to create the anomaly detection model for.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Required: No

Namespace

The namespace of the metric to create the anomaly detection model for.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 255.

Pattern: `[^:]*`

Required: No

Stat

The statistic to use for the metric and anomaly detection model.

Type: String

Length Constraints: Maximum length of 50.

Pattern: `(SampleCount|Average|Sum|Minimum|Maximum|IQM|(p|tc|tm|ts|wm)(\d{1,2}(\.\d{0,10})?|100)|[ou]\d+(\.\d*)?)(_E|_L|_H)?|(TM|TC|TS|WM)\((((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%|((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%):((\d{1,2})(\.\d{0,10})?|100(\.\d{0,10})?)%))\)| (TM|TC|TS|WM|PR)\((((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)):(\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))?|((\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))?:(\d+(\.\d{0,10})?|(\d+(\.\d{0,10})?[Ee][+-]?\d+)))\)`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

SlidingWindow

An evaluation window that advances each time the alarm is evaluated, forming a rolling time window. This is the default evaluation window. A sliding window has no additional configuration options.

Choose a sliding window when you need the fastest detection and the calendar boundaries of the data don't matter, such as for continuous performance, latency, or resource-exhaustion monitoring.

Contents

The members of this exception structure are context-dependent.

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

StatisticSet

Represents a set of statistics that describes a specific metric.

Contents

Maximum

The maximum value of the sample set.

Type: Double

Required: Yes

Minimum

The minimum value of the sample set.

Type: Double

Required: Yes

SampleCount

The number of samples used for the statistic set.

Type: Double

Required: Yes

Sum

The sum of values for the sample set.

Type: Double

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Tag

A key-value pair associated with a CloudWatch resource.

Contents

Key

A string that you can use to assign a value. The combination of tag keys and values can help you organize and categorize your resources.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Required: Yes

Value

The value for the specified tag key.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 256.

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

WallClockWindow

An evaluation window that aligns the evaluated range to fixed clock boundaries that match the alarm's period, such as the top of the hour, midnight, or the start of the calendar week, optionally in a specific time zone.

When you use a wall clock window, the alarm's period must be 1 minute (60 seconds), 5 minutes (300 seconds), 1 hour (3,600 seconds), 1 day (86,400 seconds), or 1 week (604,800 seconds). Other period values aren't supported with a wall clock window.

Choose a wall clock window when your monitoring is tied to a business or calendar period, such as daily reports, batch jobs, or backups, or when you want alarm evaluations to match the periods shown on a metric dashboard.

Contents

Timezone

The time zone to use when the alarm aligns the evaluation window to clock boundaries. You can specify an IANA time zone name (for example, `America/New_York`), a fixed UTC offset (for example, `+05:30`), or an offset-prefixed identifier (for example, `UTC+05:30`). The offset must be aligned to a multiple of 5 minutes. If you don't specify a time zone, CloudWatch uses UTC.

The time zone affects window alignment for all periods, including periods of one hour or shorter.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 50.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

Common Parameters

The following list contains the parameters that all actions use for signing Signature Version 4 requests with a query string. Any action-specific parameters are listed in the topic for that action. For more information about Signature Version 4, see [Signing AWS API requests](#) in the *IAM User Guide*.

X-Amz-Algorithm

The hash algorithm that you used to create the request signature.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Valid Values: AWS4-HMAC-SHA256

Required: Conditional

X-Amz-Credential

The credential scope value, which is a string that includes your access key, the date, the region you are targeting, the service you are requesting, and a termination string ("aws4_request"). The value is expressed in the following format: *access_key/YYYYMMDD/region/service/aws4_request*.

For more information, see [Create a signed AWS API request](#) in the *IAM User Guide*.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

X-Amz-Date

The date that is used to create the signature. The format must be ISO 8601 basic format (YYYYMMDD'T'HHMMSS'Z'). For example, the following date time is a valid X-Amz-Date value: 20120325T120000Z.

Condition: X-Amz-Date is optional for all requests; it can be used to override the date used for signing requests. If the Date header is specified in the ISO 8601 basic format, X-Amz-Date is not required. When X-Amz-Date is used, it always overrides the value of the Date header. For more information, see [Elements of an AWS API request signature](#) in the *IAM User Guide*.

Type: string

Required: Conditional

X-Amz-Security-Token

The temporary security token that was obtained through a call to AWS Security Token Service (AWS STS). For a list of services that support temporary security credentials from AWS STS, see [AWS services that work with IAM](#) in the *IAM User Guide*.

Condition: If you're using temporary security credentials from AWS STS, you must include the security token.

Type: string

Required: Conditional

X-Amz-Signature

Specifies the hex-encoded signature that was calculated from the string to sign and the derived signing key.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

X-Amz-SignedHeaders

Specifies all the HTTP headers that were included as part of the canonical request. For more information about specifying signed headers, see [Create a signed AWS API request](#) in the *IAM User Guide*.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

Common Error Types

This section lists common error types that this AWS service may return. Not all services return all error types listed here. For errors specific to an API action for this service, see the topic for that API action.

AccessDeniedException

You don't have permission to perform this action. Verify that your IAM policy includes the required permissions.

HTTP Status Code: 403

ExpiredTokenException

The security token included in the request has expired. Request a new security token and try again.

HTTP Status Code: 403

IncompleteSignature

The request signature doesn't conform to AWS standards. Verify that you're using valid AWS credentials and that your request is properly formatted. If you're using an SDK, ensure it's up to date.

HTTP Status Code: 403

InternalFailure

The request can't be processed right now because of an internal server issue. Try again later. If the problem persists, contact AWS Support.

HTTP Status Code: 500

MalformedHttpRequestException

The request body can't be processed. This typically happens when the request body can't be decompressed using the specified content encoding algorithm. Verify that the content encoding header matches the compression format used.

HTTP Status Code: 400

NotAuthorized

You don't have permissions to perform this action. Verify that your IAM policy includes the required permissions.

HTTP Status Code: 401

OptInRequired

Your AWS account needs a subscription for this service. Verify that you've enabled the service in your account.

HTTP Status Code: 403

RequestAbortedException

The request was aborted before a response could be returned. This typically happens when the client closes the connection.

HTTP Status Code: 400

RequestEntityTooLargeException

The request entity is too large. Reduce the size of the request body and try again.

HTTP Status Code: 413

RequestTimeoutException

The request timed out. The server didn't receive the complete request within the expected time frame. Try again.

HTTP Status Code: 408

ServiceUnavailable

The service is temporarily unavailable. Try again later.

HTTP Status Code: 503

ThrottlingException

Your request rate is too high. The AWS SDKs automatically retry requests that receive this exception. Reduce the frequency of requests.

HTTP Status Code: 400

UnknownOperationException

The action or operation isn't recognized. Verify that the action name is spelled correctly and that it's supported by the API version you're using.

HTTP Status Code: 404

UnrecognizedClientException

The X.509 certificate or AWS access key ID you provided doesn't exist in our records. Verify that you're using valid credentials and that they haven't expired.

HTTP Status Code: 403

ValidationError

The input doesn't meet the required format or constraints. Check that all required parameters are included and that values are valid.

HTTP Status Code: 400